1 Within Plot Level

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Okay, the goal of this section is all about understanding how a much plant variation there can be within a 2x2 m quadrat. The goal is to compare the focal communities to their immediate surroundings. If they are tightly conserved, that would suggest a niche-y assembly which will then lead to an examination of which environmental variables matter. If no environmental variables are good, then we either did not measure the correct ones or we can blame spatial auto-correlation (See: 3_WithinSite/BetweenPointsLevel). If it is spatial auto-correlation and/or random, then we are thinking about neutral. I expect the relative importance of the environemnt to vary with environmental stress.

Species Composition

Ten Mile

This is a section for analyzing data only from ten mile. I will repeat same process for each location.

```
#I make unique identifiers for each plot
TeMi_Names<- colnames(TeMi_Cover)
TeMi_Names<-gsub('.{1}$', '', TeMi_Names)
TeMi_Names<-unique(gsub('TeMi.', '', TeMi_Names))
TeMi_Diss<-data.frame(Plot = rep("", length(TeMi_Names)), Dissimilarity= rep(0, length(TeMi_Names)))


for (i in 1:length(TeMi_Names)) {
  Temp_Cover<-TeMi_Cover %>% select(starts_with(paste0("TeMi.", TeMi_Names[i])))
  #Temp_Cover<- cbind(Temp_Cover[1], rowSums(Temp_Cover[2:5], na.rm = TRUE)/4)
  Temp_Diss<-vegdist(t(Temp_Cover), na.rm = TRUE)
  Temp_Av_Dis<-sum(as.matrix(Temp_Diss)[,1])/4
  TeMi_Diss$Plot[i]<- TeMi_Names[i]
  TeMi_Diss$Dissimilarity[i]<- ifelse(is.na(Temp_Av_Dis) == TRUE, 0, ifelse(Temp_Av_Dis == "NaN", 0, Ter

}
```

```
## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): missing values in results

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): missing values in results

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"
## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"
```

```
## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): missing values in results
## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                   meaningless in method "bray"
## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): missing values in results
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##                   meaningless in method "bray"
## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                   meaningless in method "bray"
```
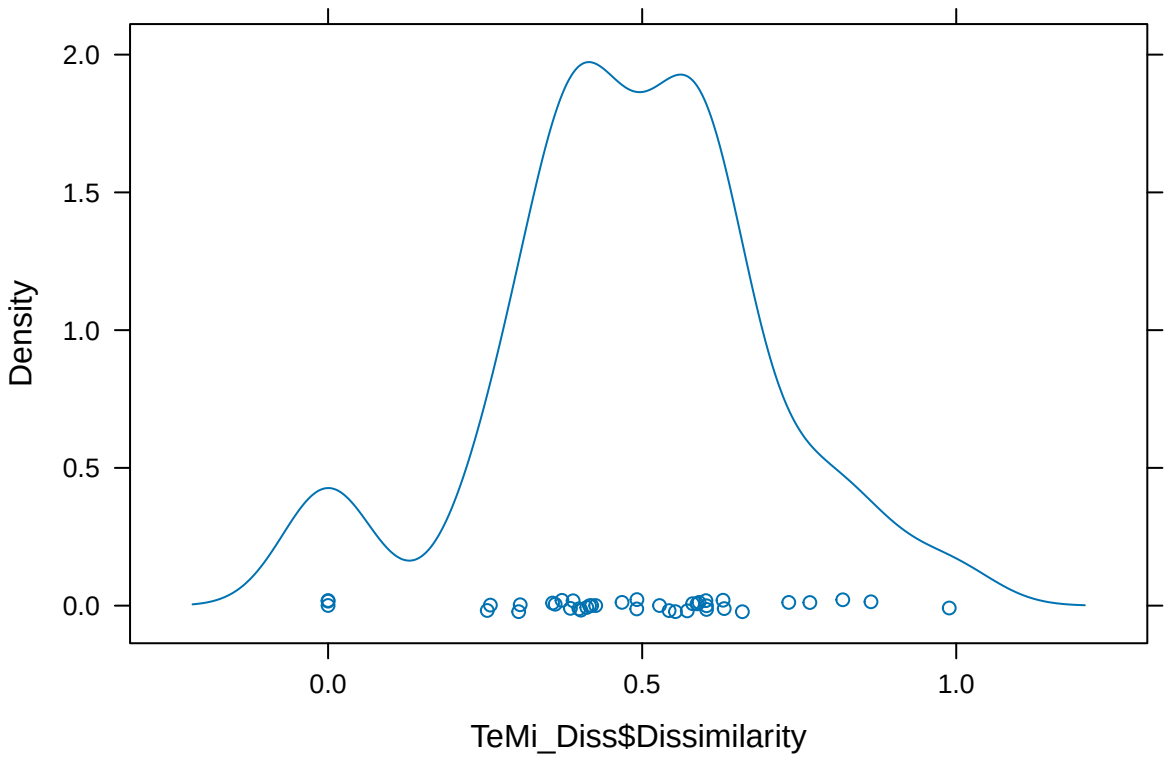
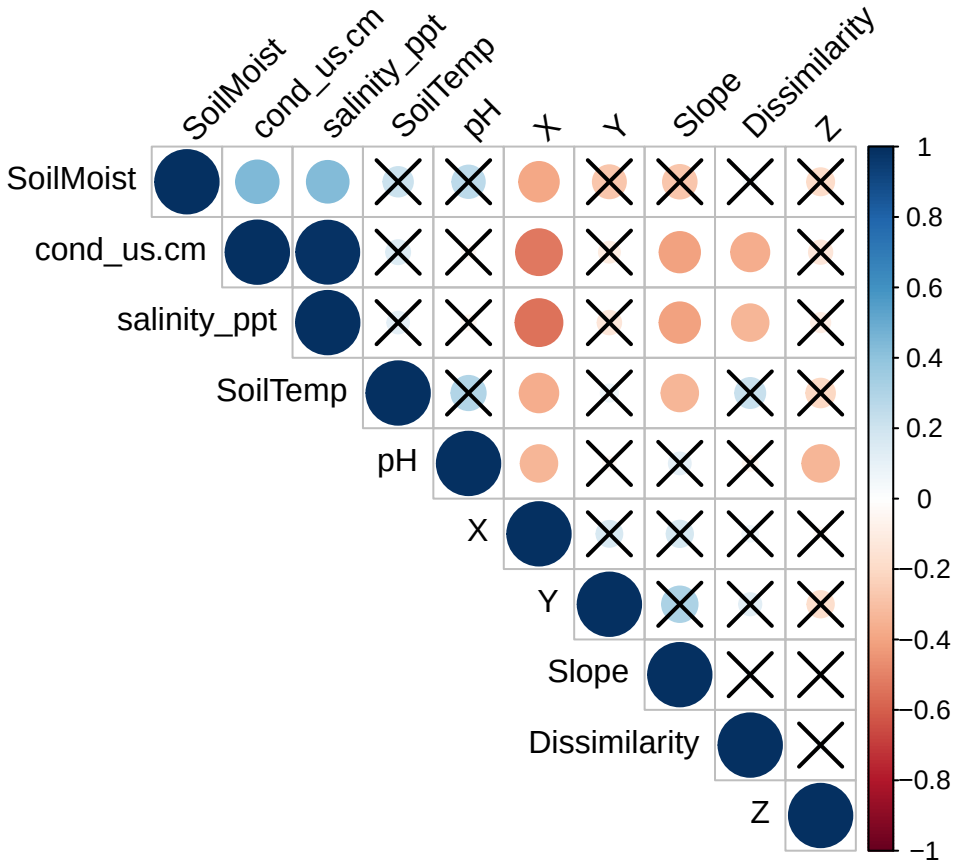

```r
densityplot(TeMi_Diss$Dissimilarity)
```

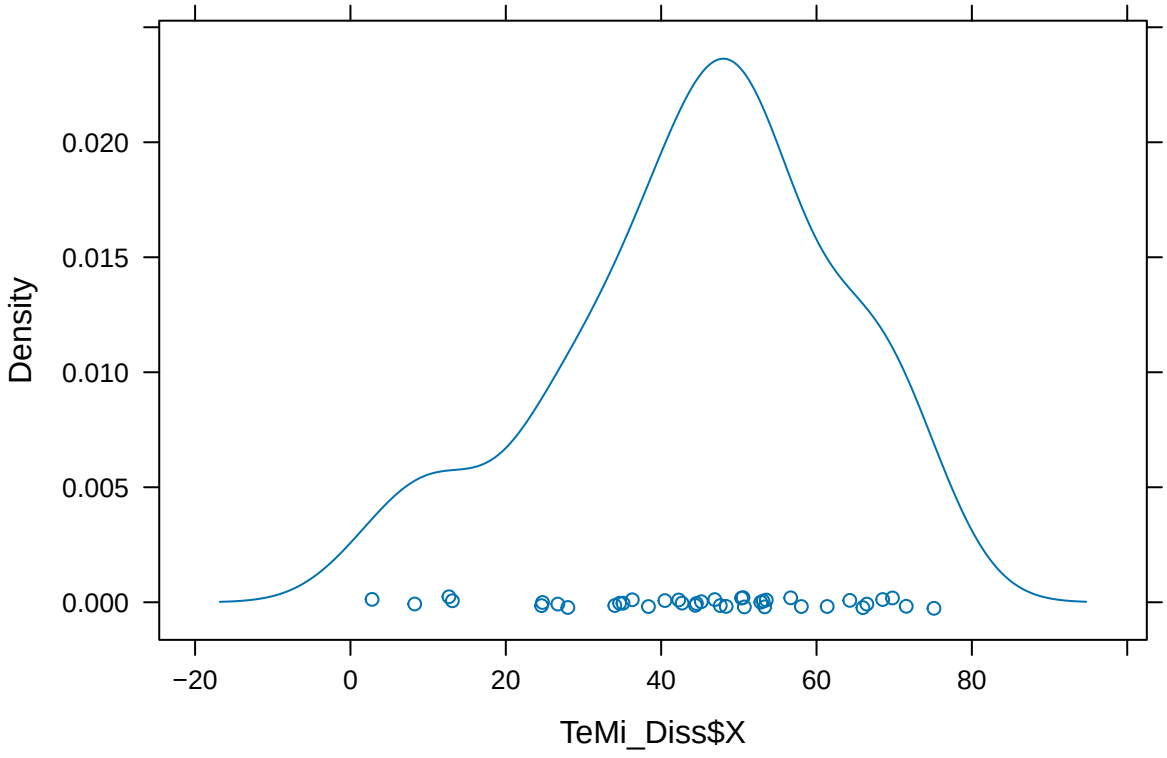



```r
TeMi_Diss$Plot<- as.numeric(TeMi_Diss$Plot)

TeMi_Diss<-merge(TeMi_Diss, TeMi_RelCords)

TeMi_Diss<-merge(TeMi_Diss, subset(DuneEnv_scaled, Site == "TeMi"))


TeMi_pcor<- TeMi_Diss %>%
  select(-c(Site, Plot, Aspect, Date, Collectors, dry_weight, Notes))%>%
  as.matrix()%>%
  cor.mtest(use =  "complete")

TeMi_Diss %>%
  select(-c(Site, Plot, Aspect, Date, Collectors, dry_weight, Notes))%>%
  as.matrix()%>%
  cor(use =  "complete")%>%
  corrplot( type = "upper", order = "hclust",  tl.col = "black", tl.srt = 45, p.mat = TeMi_pcor$p)
```



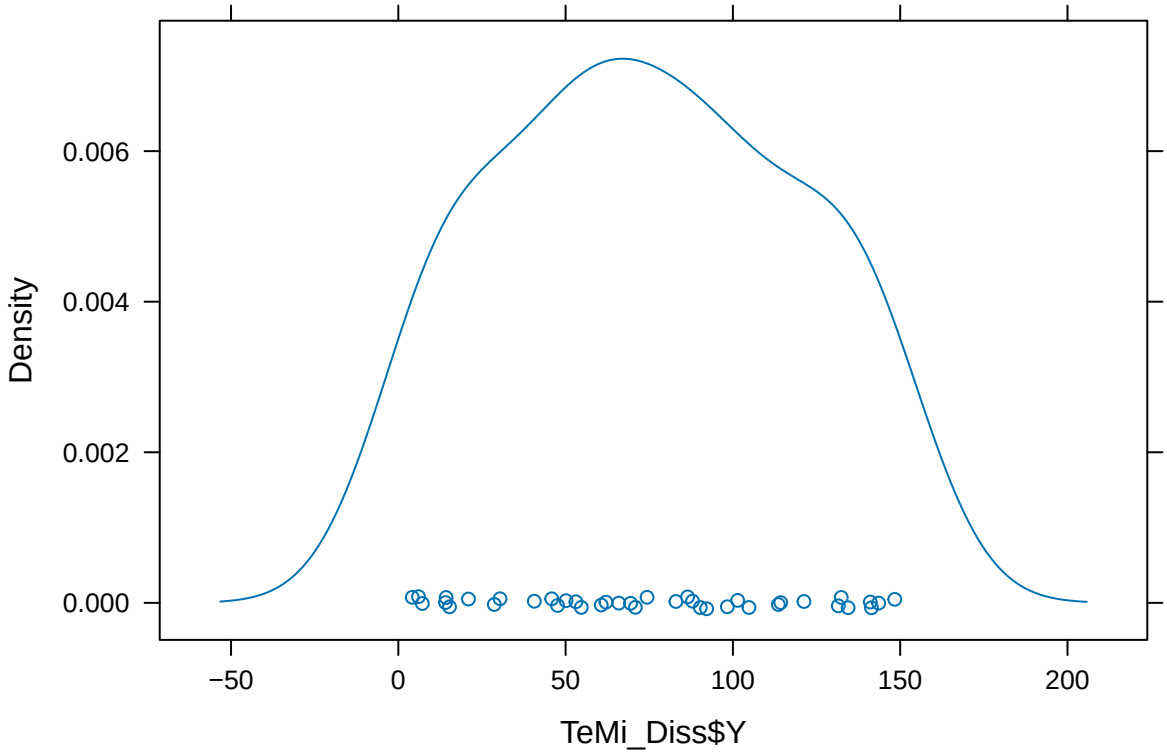
Graphs and models

```
densityplot(TeMi_Diss$X)
```

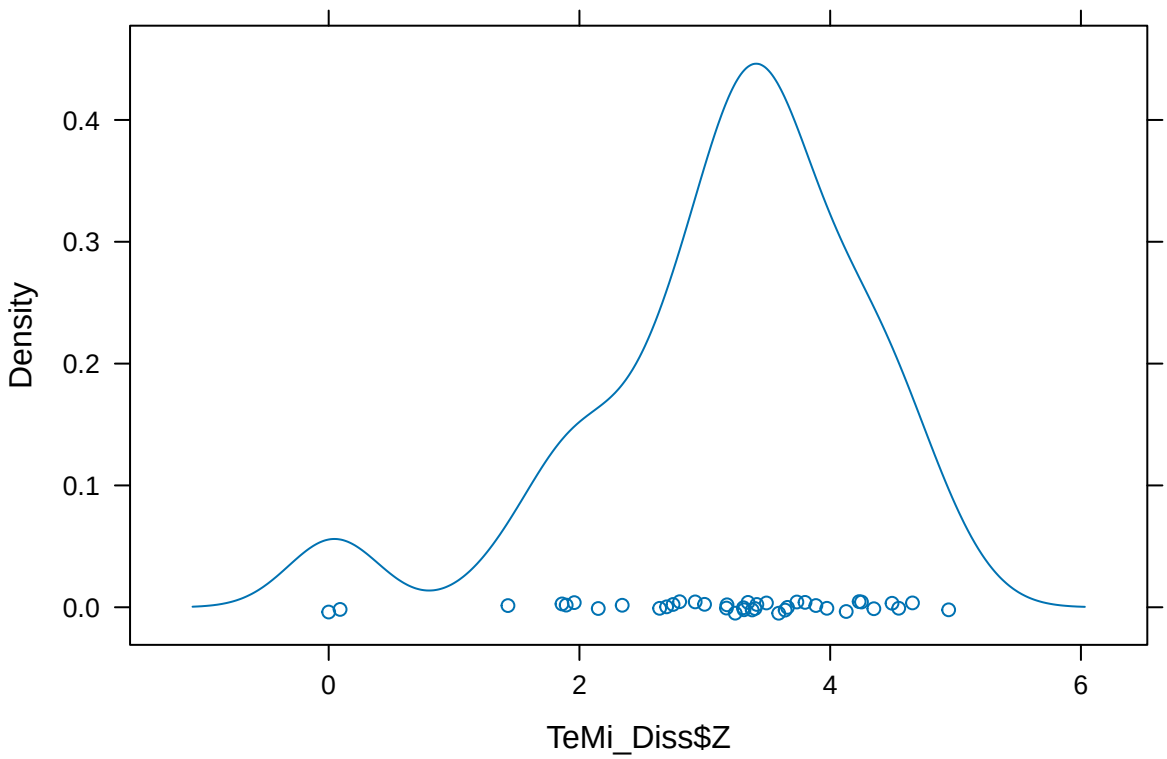


Distance

```
densityplot(TeMi_Diss$Y)
```



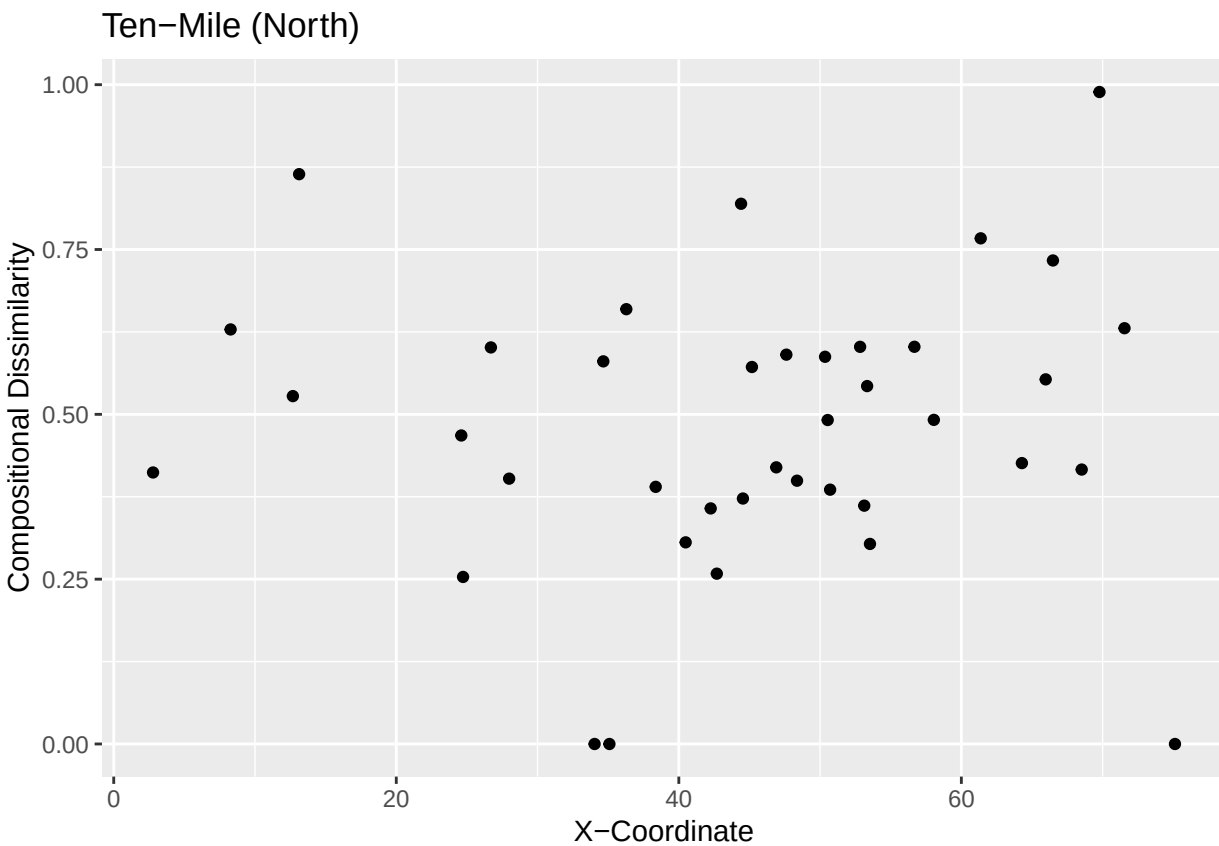

```
densityplot(TeMi_Diss$Z)
```




```r
summary(lm(Dissimilarity ~ X, data = TeMi_Diss)) # Distance from waterline not sig
```

```
##
## Call:
## lm(formula = Dissimilarity ~ X, data = TeMi_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.50365 -0.10430  0.00164  0.11383  0.48919
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.4481275  0.0956314   4.686 3.71e-05 ***
## X           0.0007391  0.0019894   0.372    0.712
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2191 on 37 degrees of freedom
## Multiple R-squared:  0.003717,   Adjusted R-squared:  -0.02321
## F-statistic: 0.138 on 1 and 37 DF,  p-value: 0.7124
```

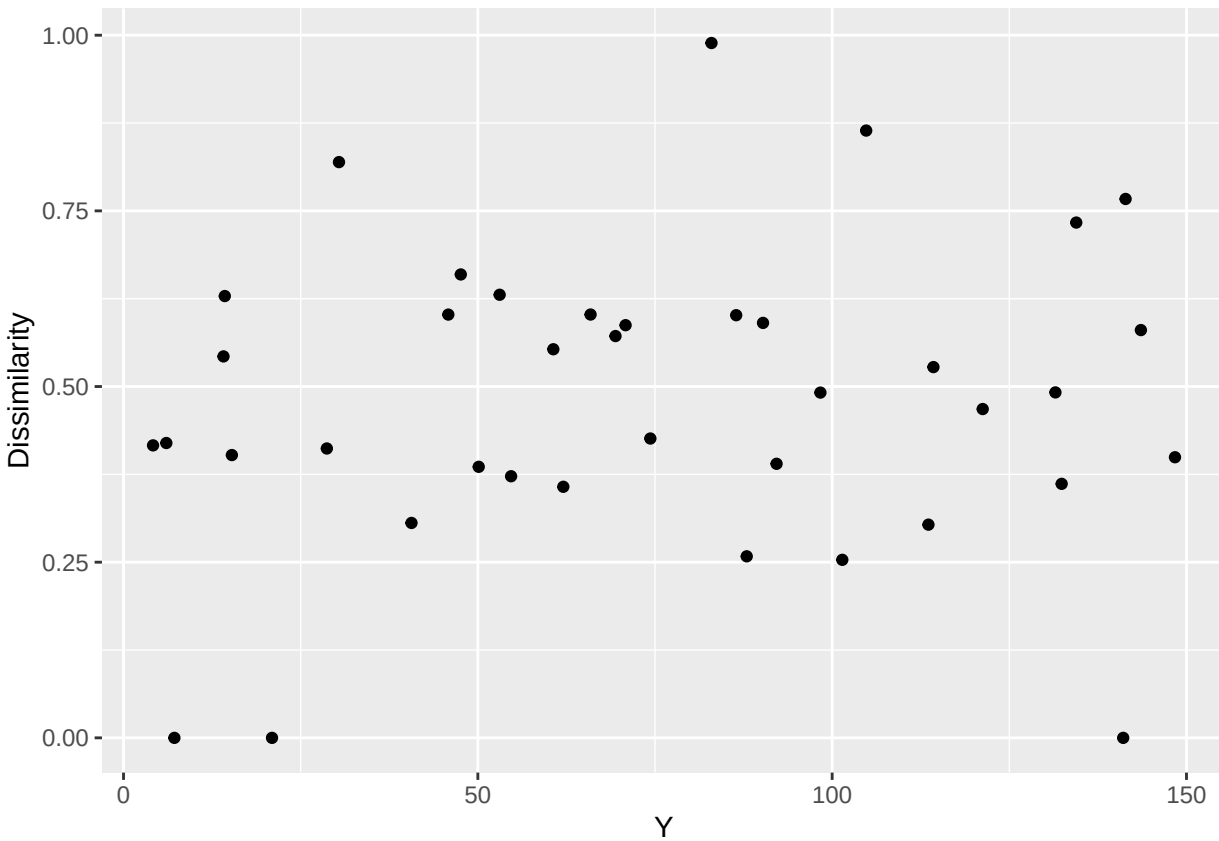
```r
ggplot(aes(x = X, y = Dissimilarity), data = TeMi_Diss)+
  geom_point()+
  labs(x = "X-Coordinate", y = "Compositional Dissimilarity", title = "Ten-Mile (North)")
```




```r
summary(lm(Dissimilarity ~ Y, data = TeMi_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ Y, data = TeMi_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.52103 -0.10911 -0.02069  0.11974  0.50260
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.4366741  0.0689841   6.330 2.25e-07 ***
## Y           0.0005980  0.0007996   0.748    0.459
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2179 on 37 degrees of freedom
## Multiple R-squared:  0.01489,    Adjusted R-squared:  -0.01173
## F-statistic: 0.5593 on 1 and 37 DF,  p-value: 0.4593
```

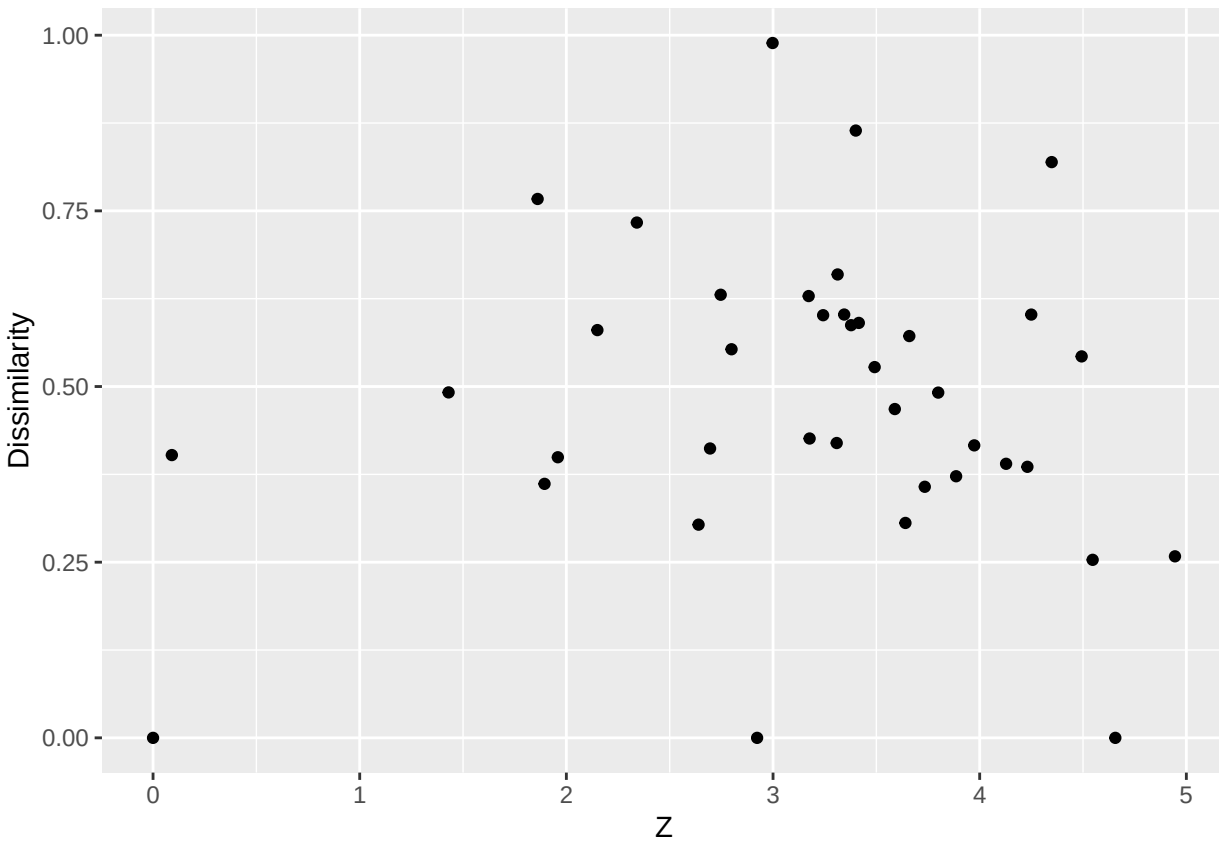
```r
ggplot(aes(x = Y, y = Dissimilarity), data = TeMi_Diss)+
  geom_point()
```




```r
summary(lm(Dissimilarity ~ Z, data = TeMi_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ Z, data = TeMi_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.49223 -0.10675  0.00552  0.11649  0.50899
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.457596   0.107918   4.240 0.000143 ***
## Z           0.007438   0.032184   0.231 0.818501
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2194 on 37 degrees of freedom
## Multiple R-squared:  0.001442,   Adjusted R-squared:  -0.02555
## F-statistic: 0.05341 on 1 and 37 DF,  p-value: 0.8185
```

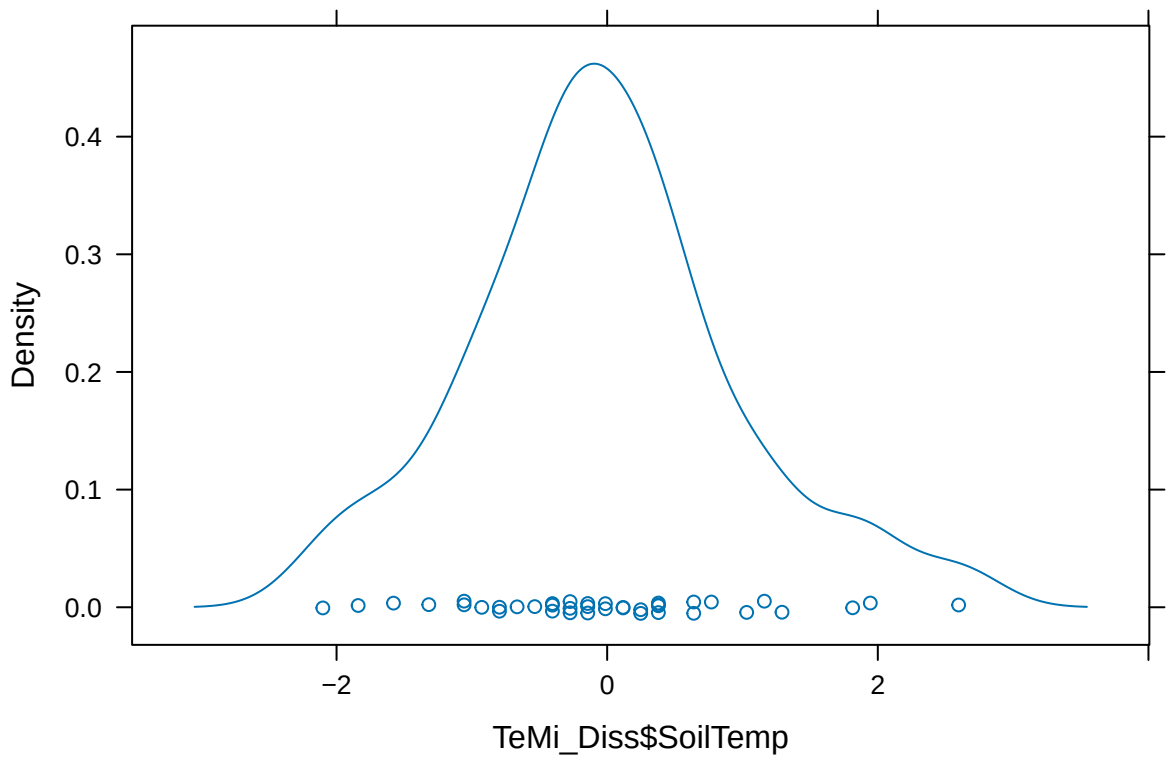
```r
ggplot(aes(x = Z, y = Dissimilarity), data = TeMi_Diss)+
  geom_point()
```




```r
summary(lm(Dissimilarity ~ X+Y+Z, data = TeMi_Diss))
```

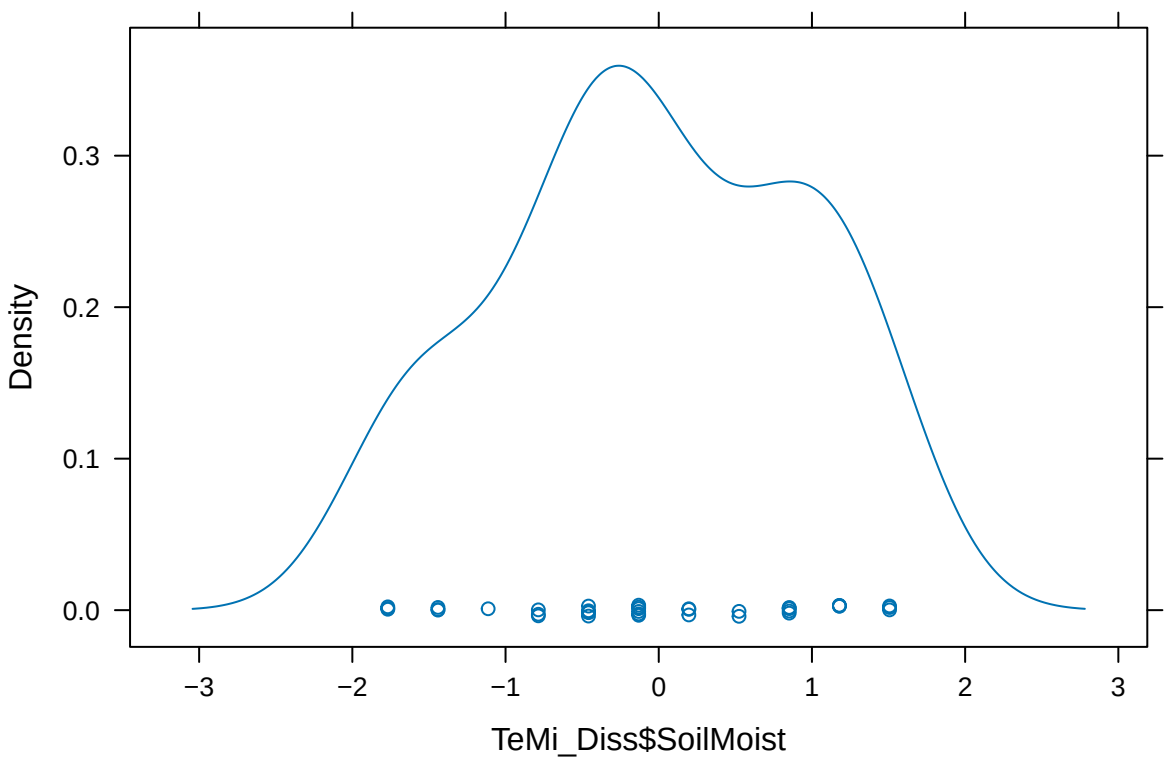
```
##
## Call:
## lm(formula = Dissimilarity ~ X + Y + Z, data = TeMi_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.53486 -0.11246 -0.01063  0.11980  0.49162
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.3743341  0.1592676   2.350   0.0245 *
## X           0.0005148  0.0020571   0.250   0.8039
## Y           0.0006152  0.0008435   0.729   0.4706
## Z           0.0119969  0.0332784   0.361   0.7206
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2234 on 35 degrees of freedom
## Multiple R-squared:  0.02023,    Adjusted R-squared:  -0.06375
## F-statistic: 0.2409 on 3 and 35 DF,  p-value: 0.8672
```

```r
densityplot(TeMi_Diss$SoilTemp)
```

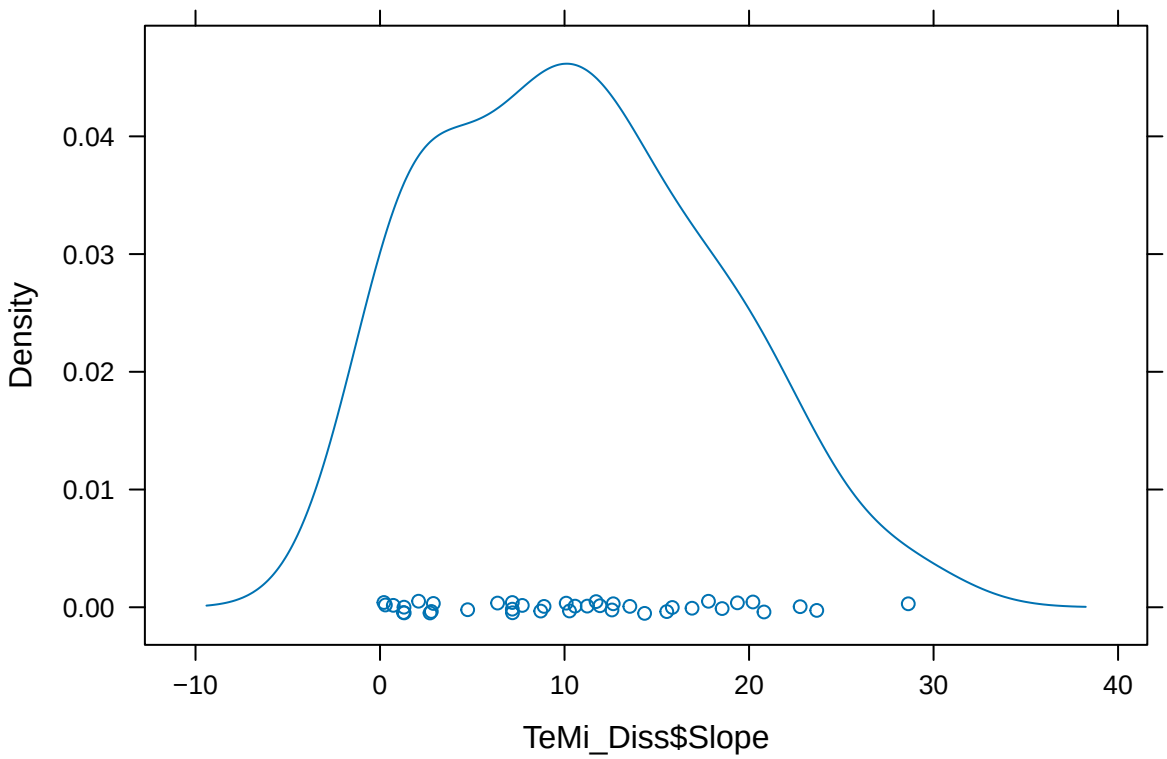


Environmental

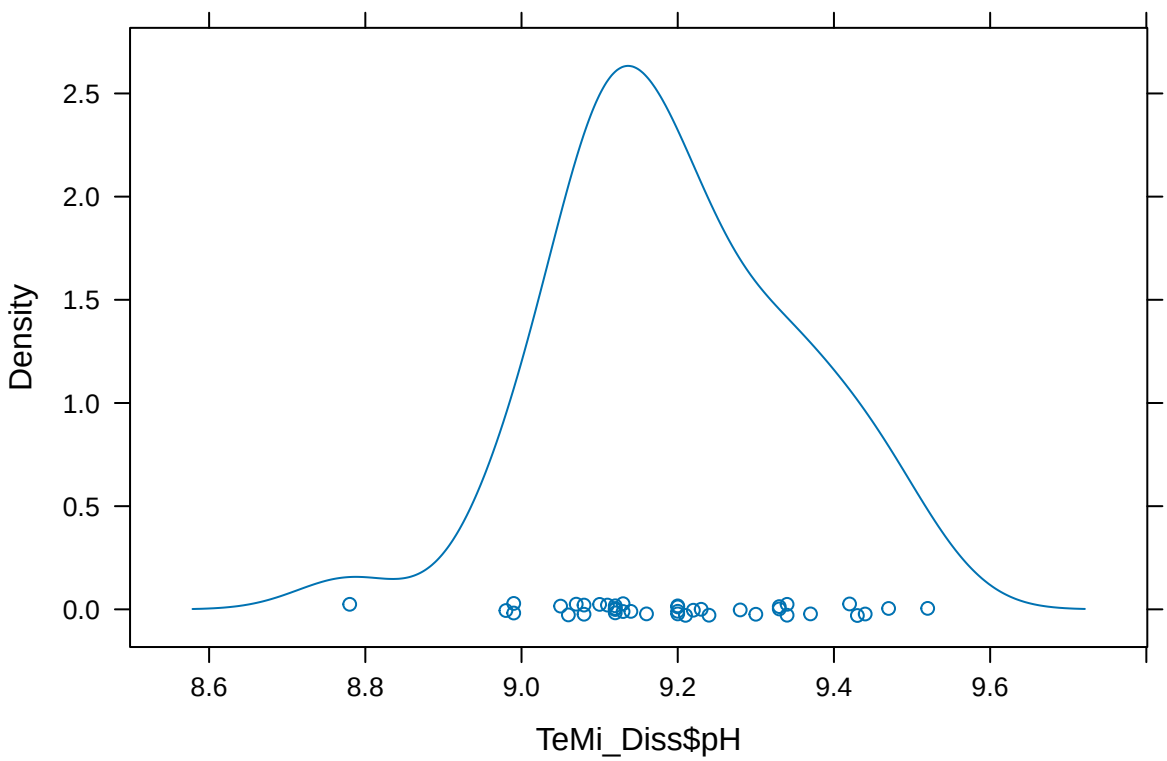
```
densityplot(TeMi_Diss$SoilMoist)
```



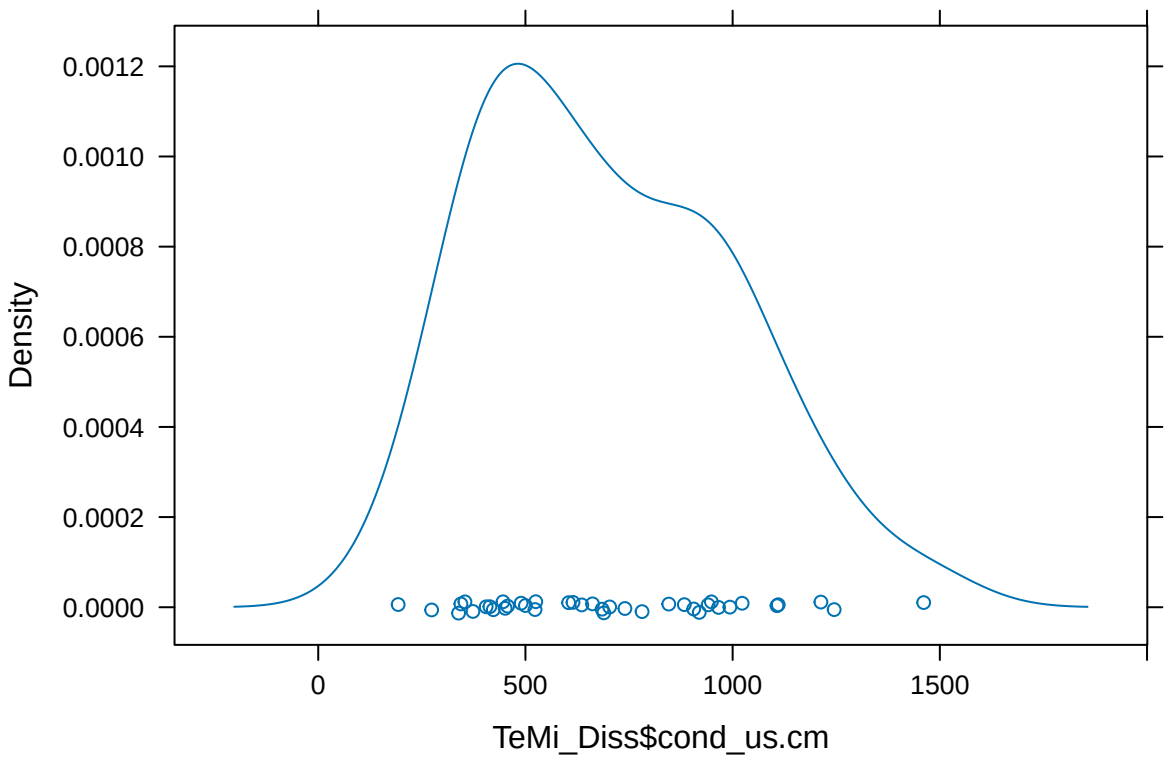

```
densityplot(TeMi_Diss$Slope)
```



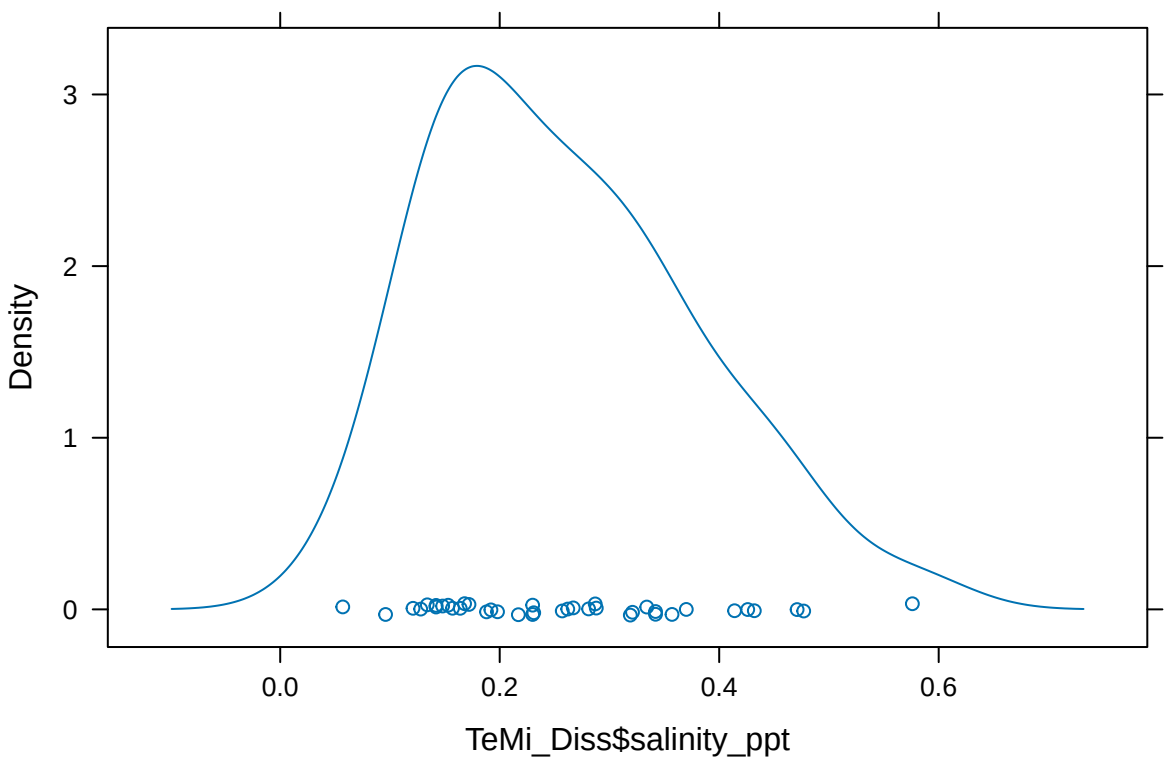

```
densityplot(TeMi_Diss$pH)
```




```
densityplot(TeMi_Diss$cond_us.cm)
```




```
densityplot(TeMi_Diss$salinity_ppt)
```




```
summary(lm(Dissimilarity ~ SoilTemp+SoilMoist+Slope+Aspect+pH, data = TeMi_Diss))
```

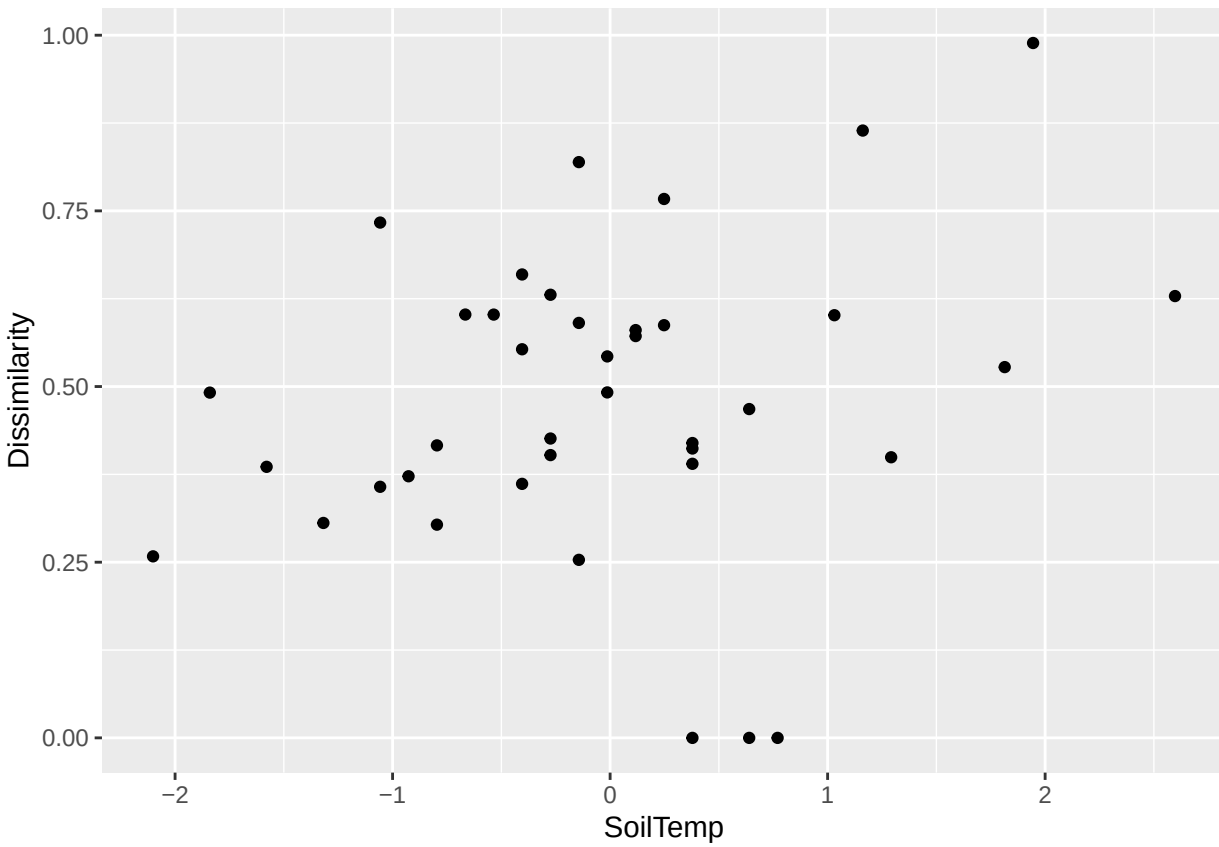
```
##
## Call:
```

```
## lm(formula = Dissimilarity ~ SoilTemp + SoilMoist + Slope + Aspect +
##     pH, data = TeMi_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.44742 -0.07485  0.04279  0.11765  0.41705
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.299387   2.568834   1.284   0.2099
## SoilTemp     0.083259   0.042761   1.947   0.0620 .
## SoilMoist    0.007286   0.045226   0.161   0.8732
## Slope        0.005069   0.005600   0.905   0.3734
## AspectN      0.056043   0.124394   0.451   0.6559
## AspectNE     0.108816   0.142961   0.761   0.4532
## AspectNW     0.017896   0.188791   0.095   0.9252
## AspectS     -0.043995   0.132690  -0.332   0.7428
## AspectSE     0.295925   0.163518   1.810   0.0815 .
## AspectSW     0.201576   0.195685   1.030   0.3121
## AspectW      0.026679   0.155897   0.171   0.8654
## pH          -0.318728   0.281098  -1.134   0.2668
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2197 on 27 degrees of freedom
## Multiple R-squared:  0.2688, Adjusted R-squared:  -0.02909
## F-statistic: 0.9024 on 11 and 27 DF,  p-value: 0.5506
```

```r
summary(lm(Dissimilarity ~ SoilTemp, data = TeMi_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ SoilTemp, data = TeMi_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.51982 -0.09517 -0.02008  0.10663  0.41207
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.48246    0.03427  14.077   <2e-16 ***
## SoilTemp     0.04852    0.03475   1.396    0.171
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.214 on 37 degrees of freedom
## Multiple R-squared:  0.05006,    Adjusted R-squared:  0.02438
## F-statistic:  1.95 on 1 and 37 DF,  p-value: 0.1709
```

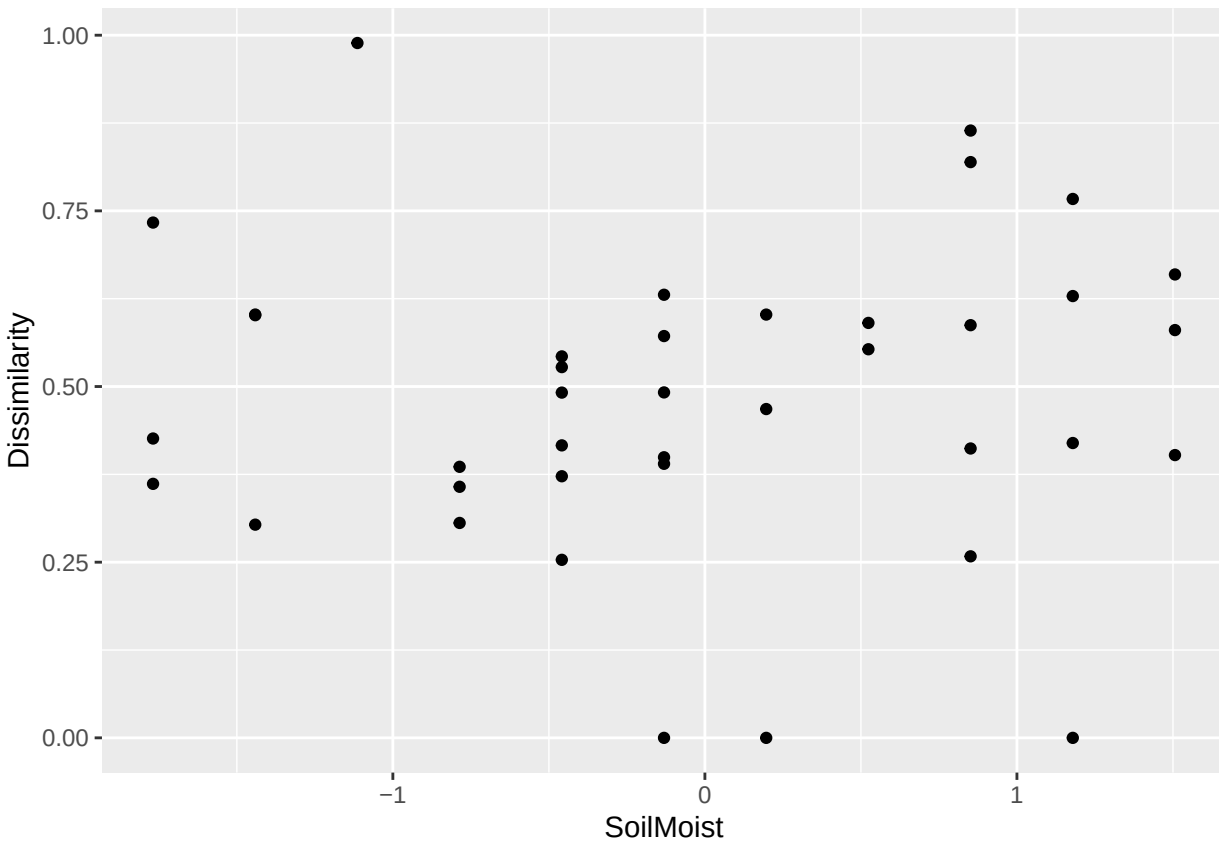
```r
ggplot(aes(x = SoilTemp, y = Dissimilarity), data = TeMi_Diss)+
  geom_point()
```




```r
summary(lm(Dissimilarity ~ SoilMoist, data = TeMi_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ SoilMoist, data = TeMi_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.49269 -0.09833  0.01134  0.12628  0.51787
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.481544   0.035147  13.701 4.55e-16 ***
## SoilMoist   0.009452   0.036216   0.261    0.796
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2193 on 37 degrees of freedom
## Multiple R-squared:  0.001838,   Adjusted R-squared:  -0.02514
## F-statistic: 0.06812 on 1 and 37 DF,  p-value: 0.7955
```

```r
ggplot(aes(x = SoilMoist, y = Dissimilarity), data = TeMi_Diss)+
  geom_point()
```




```r
summary(lm(Dissimilarity ~ SoilTemp*SoilMoist, data = TeMi_Diss))
```

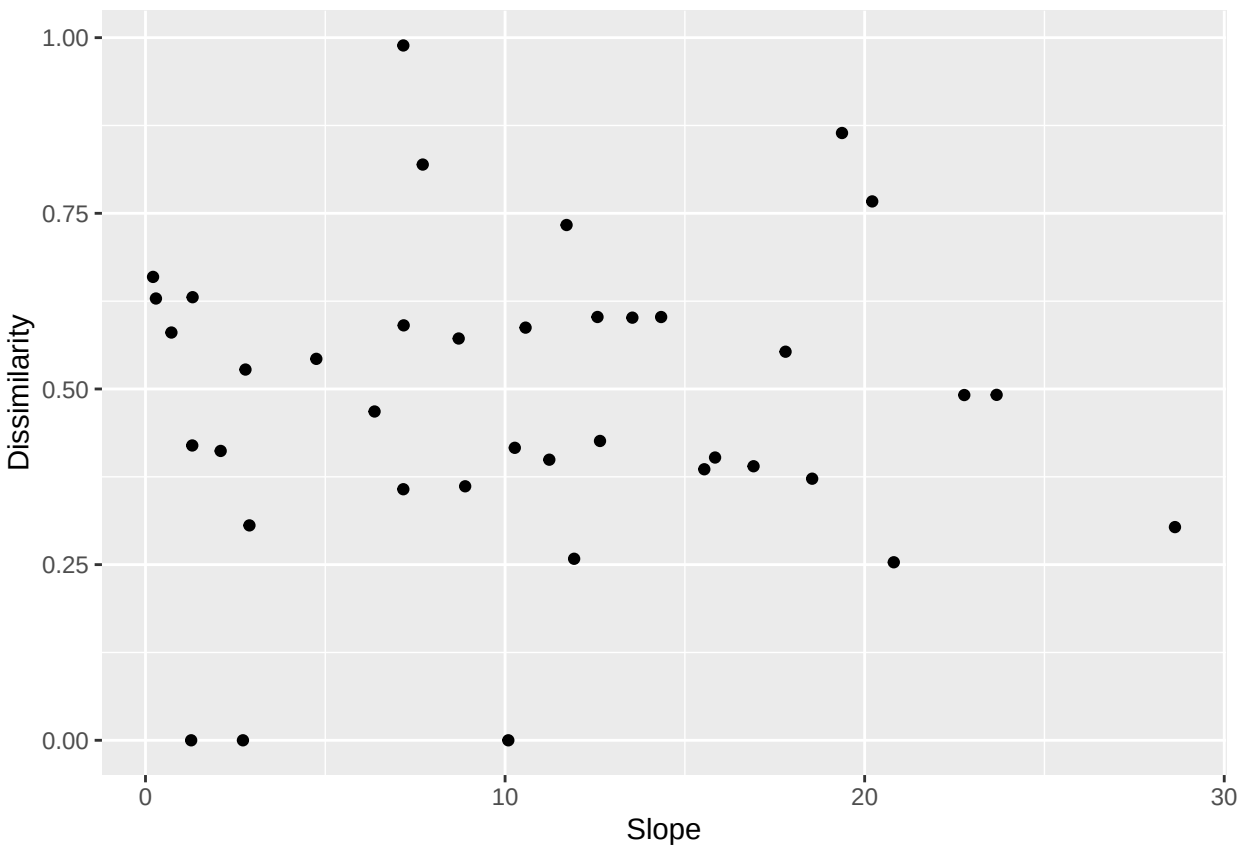
```
##
## Call:
## lm(formula = Dissimilarity ~ SoilTemp * SoilMoist, data = TeMi_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.52343 -0.09197 -0.00934  0.10833  0.38021
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)         0.48508    0.03609  13.439 2.21e-15 ***
## SoilTemp            0.04772    0.03667   1.301    0.202
## SoilMoist          -0.00213    0.03724  -0.057    0.955
## SoilTemp:SoilMoist -0.01312    0.03936  -0.333    0.741
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2196 on 35 degrees of freedom
## Multiple R-squared:  0.0531, Adjusted R-squared:  -0.02807
## F-statistic: 0.6542 on 3 and 35 DF,  p-value: 0.5857
```

```r
summary(lm(Dissimilarity ~ Slope, data = TeMi_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ Slope, data = TeMi_Diss)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.48073 -0.10809 -0.00156  0.11768  0.51084
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.4714657  0.0617574   7.634 4.12e-09 ***
## Slope       0.0009183  0.0048021   0.191    0.849
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2194 on 37 degrees of freedom
## Multiple R-squared:  0.0009875,  Adjusted R-squared:  -0.02601
## F-statistic: 0.03657 on 1 and 37 DF,  p-value: 0.8494
```

```r
ggplot(aes(x = Slope, y = Dissimilarity), data = TeMi_Diss)+
  geom_point()
```

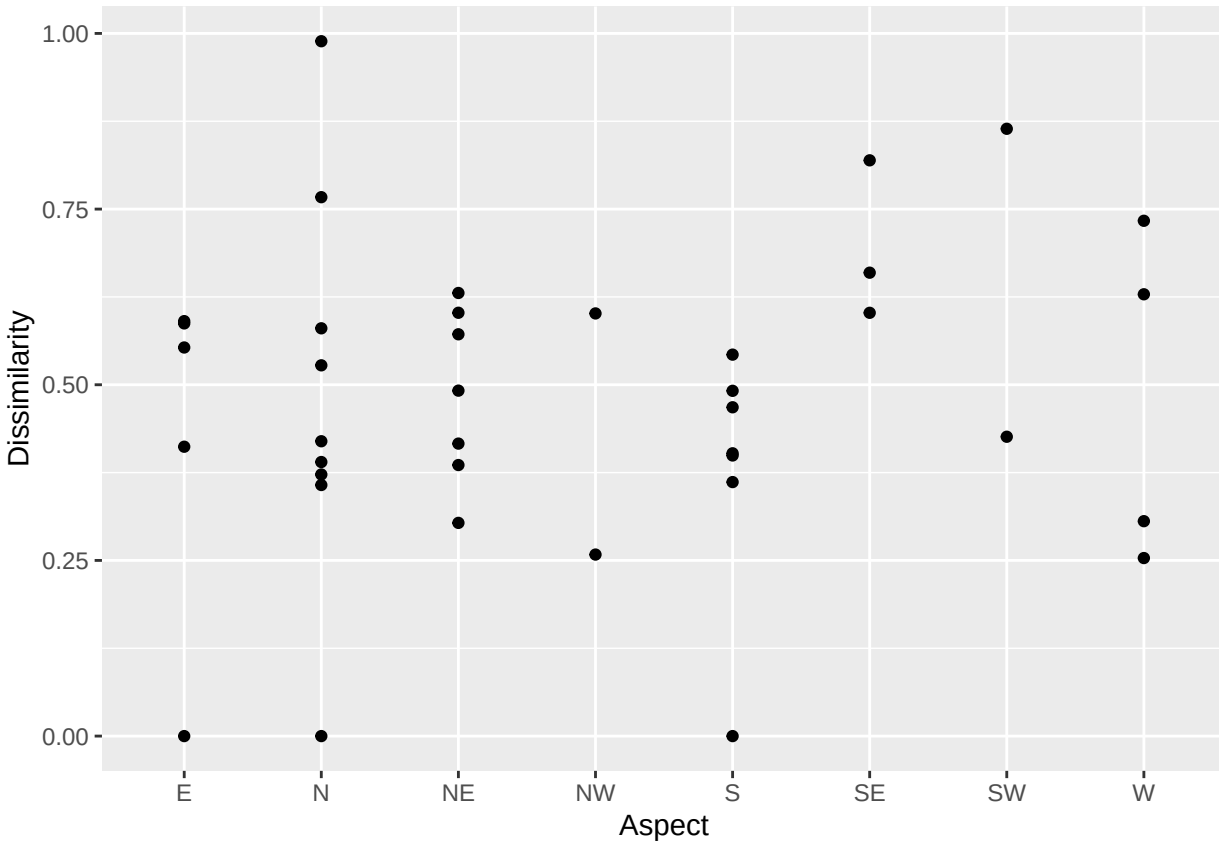



```r
summary(lm(Dissimilarity ~ Aspect, data = TeMi_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ Aspect, data = TeMi_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.48923 -0.10858  0.01857  0.13512  0.49966
```

```
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.428541   0.098478   4.352 0.000136 ***
## AspectN      0.060689   0.122824   0.494 0.624707
## AspectNE     0.057453   0.128939   0.446 0.658994
## AspectNW     0.001366   0.184236   0.007 0.994132
## AspectS     -0.047757   0.128939  -0.370 0.713611
## AspectSE     0.265227   0.160815   1.649 0.109191
## AspectSW     0.216613   0.184236   1.176 0.248652
## AspectW      0.051787   0.147718   0.351 0.728273
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2202 on 31 degrees of freedom
## Multiple R-squared:  0.1569, Adjusted R-squared:  -0.03346
## F-statistic: 0.8242 on 7 and 31 DF,  p-value: 0.5749
```

```r
ggplot(aes(x = Aspect, y = Dissimilarity), data = TeMi_Diss)+
  geom_point()
```

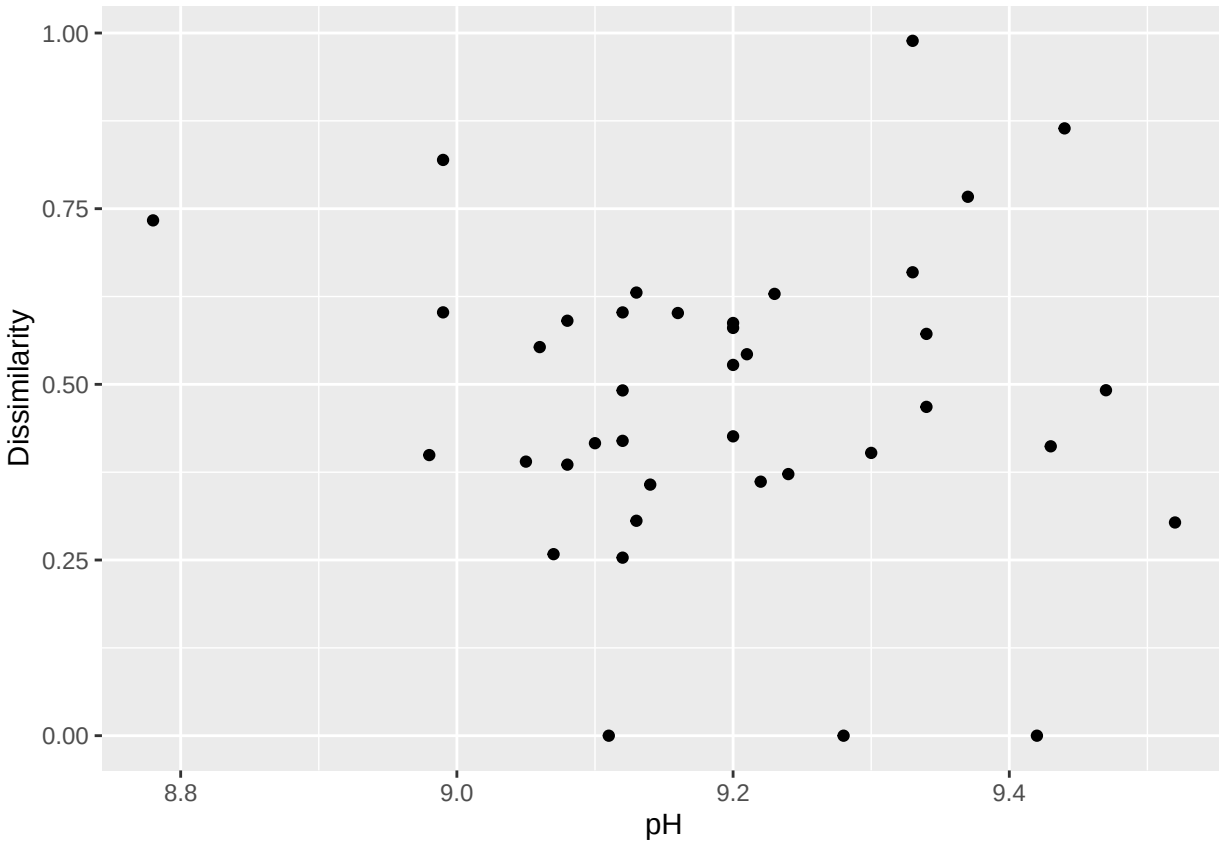



```r
######## INTERESTING! ######
summary(lm(Dissimilarity ~ pH, data = TeMi_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ pH, data = TeMi_Diss)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.48768 -0.10482  0.00446  0.11100  0.51806
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.18564    2.10841   0.562    0.577
## pH          -0.07661    0.22927  -0.334    0.740
##
## Residual standard error: 0.2192 on 37 degrees of freedom
## Multiple R-squared:  0.003009,   Adjusted R-squared:  -0.02394
## F-statistic: 0.1117 on 1 and 37 DF,  p-value: 0.7401
```

```
ggplot(aes(x = pH, y = Dissimilarity), data = TeMi_Diss)+
  geom_point()
```



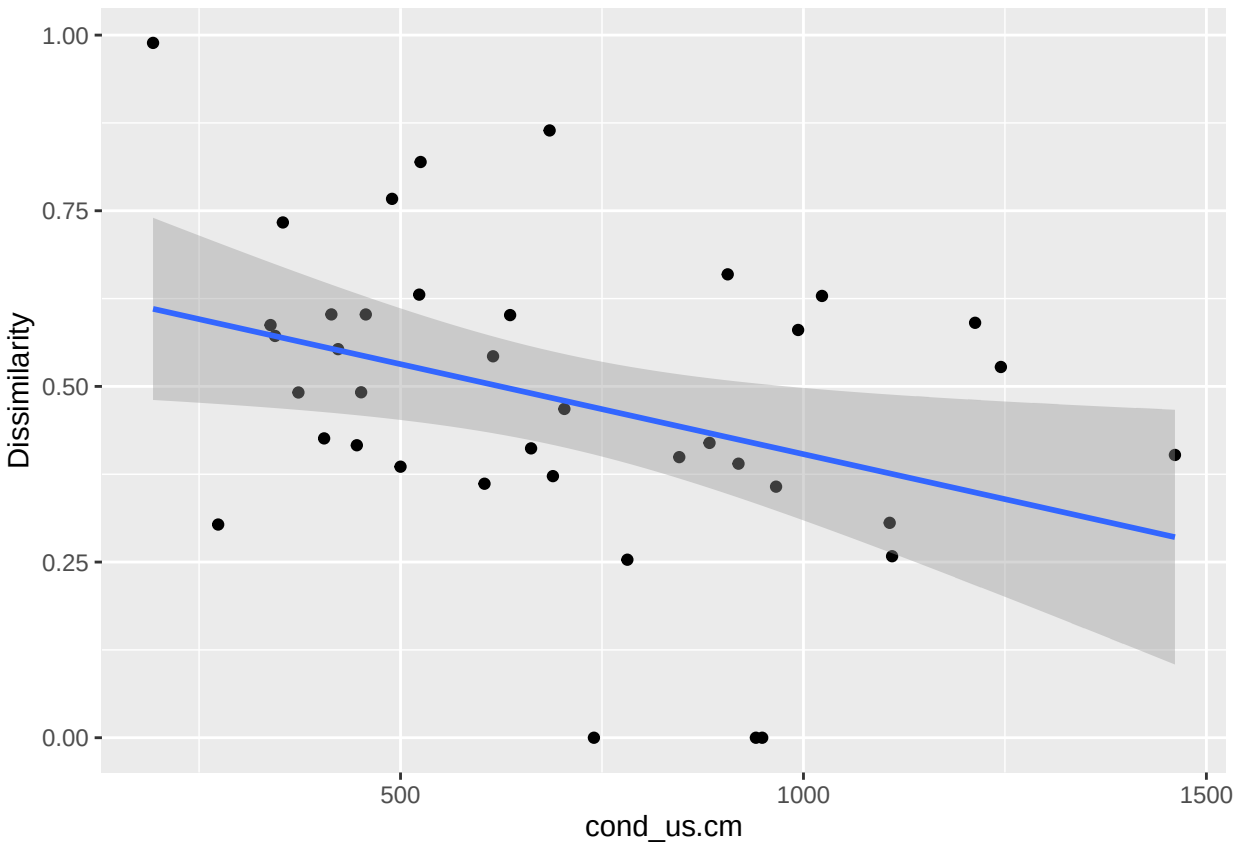

```
summary(lm(Dissimilarity ~ cond_us.cm, data = TeMi_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ cond_us.cm, data = TeMi_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.47020 -0.11401 -0.01166  0.14063  0.37999
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)  0.6597855  0.0827519   7.973 1.49e-09 ***
## cond_us.cm  -0.0002562  0.0001090  -2.351   0.0242 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2048 on 37 degrees of freedom
## Multiple R-squared:  0.1299, Adjusted R-squared:  0.1064
## F-statistic: 5.526 on 1 and 37 DF,  p-value: 0.02418
```

```r
ggplot(aes(x = cond_us.cm, y = Dissimilarity), data = TeMi_Diss)+
  geom_point()+
  geom_smooth(method = "lm")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



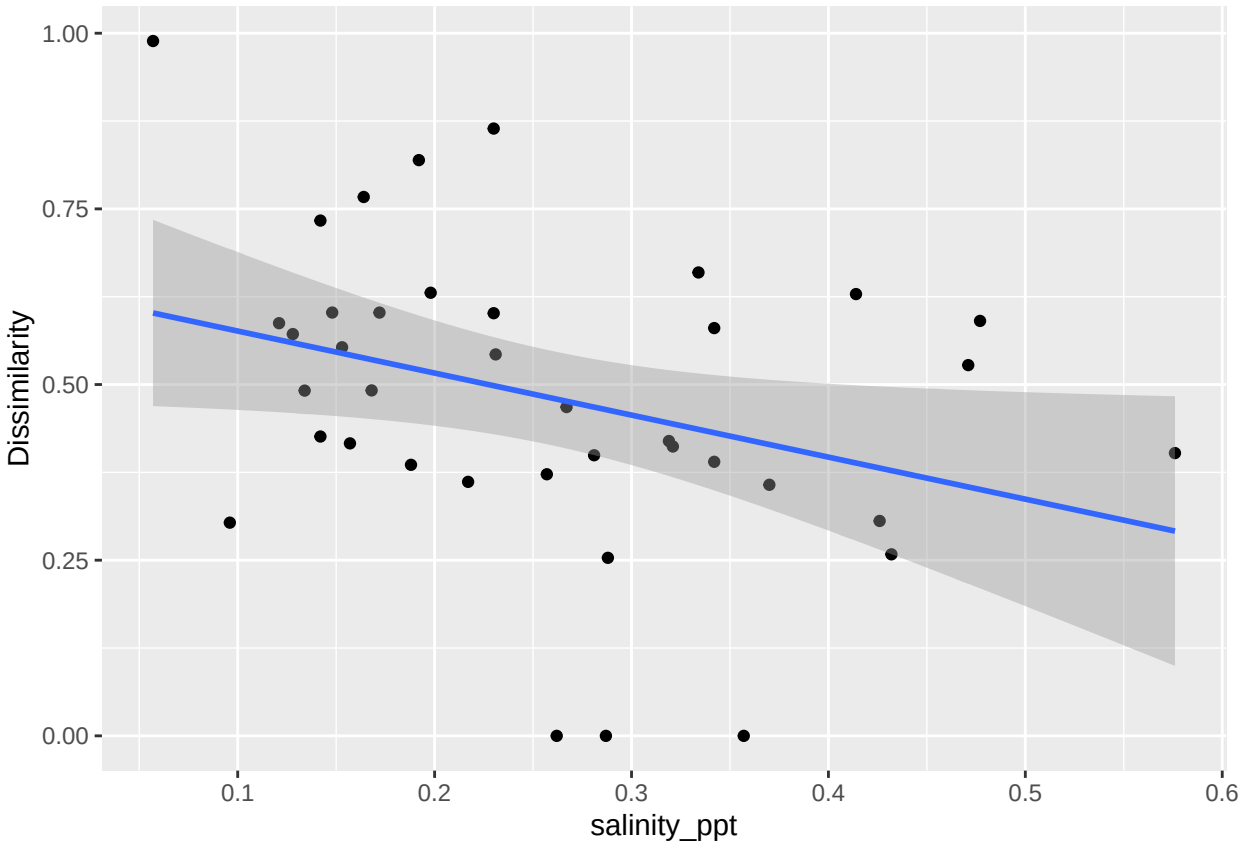

```r
summary(lm(Dissimilarity ~ salinity_ppt, data = TeMi_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ salinity_ppt, data = TeMi_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.47923 -0.11458 -0.00829  0.13102  0.38705
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.63592    0.07958   7.991 1.41e-09 ***
```

```
## salinity_ppt -0.59806    0.27958  -2.139   0.0391 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2071 on 37 degrees of freedom
## Multiple R-squared:  0.1101, Adjusted R-squared:  0.08601
## F-statistic: 4.576 on 1 and 37 DF,  p-value: 0.03909
```

```
ggplot(aes(x = salinity_ppt, y = Dissimilarity), data = TeMi_Diss)+
  geom_point()+
  geom_smooth(method = "lm")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



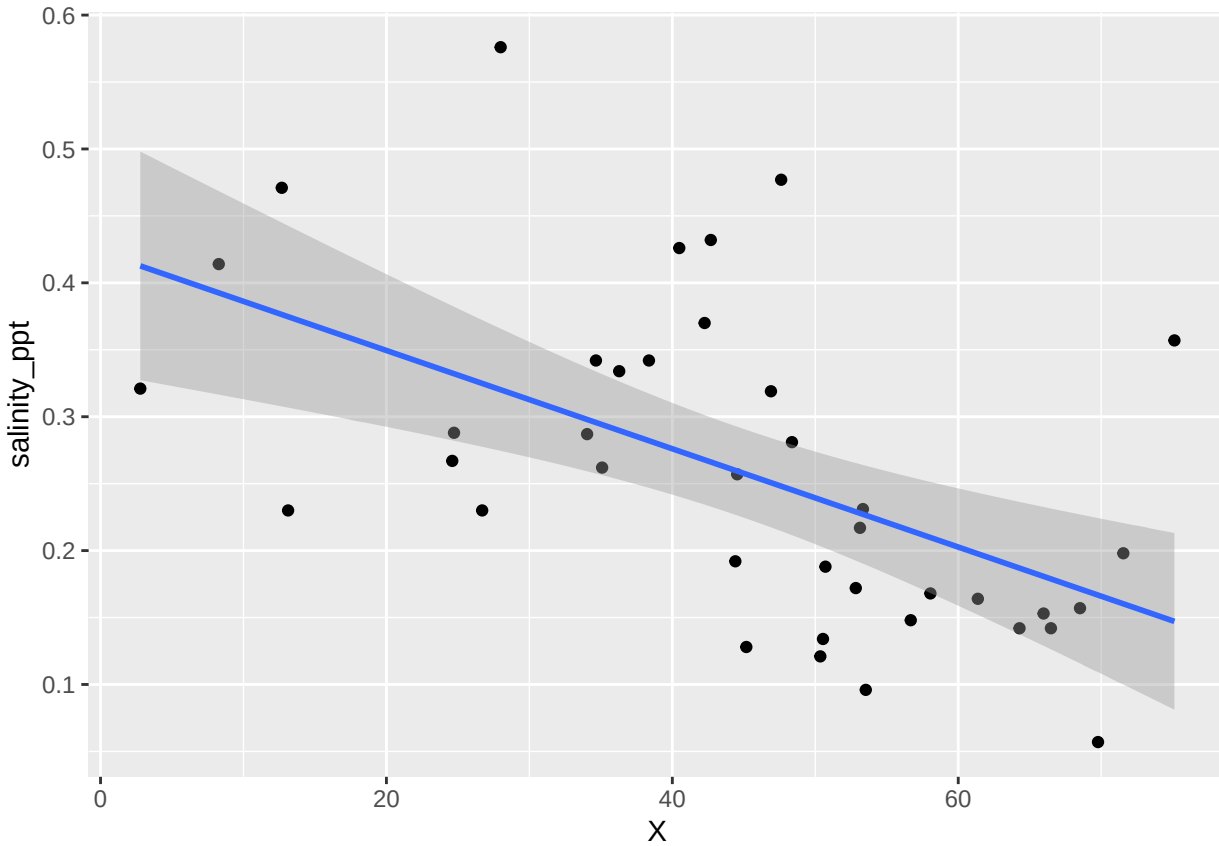
Combined

```
cor.test(x = TeMi_Diss$X, y = TeMi_Diss$salinity_ppt, method = "pearson")
```

```
##
##  Pearson's product-moment correlation
##
## data:  TeMi_Diss$X and TeMi_Diss$salinity_ppt
## t = -3.9617, df = 37, p-value = 0.0003263
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.7347511 -0.2781298
## sample estimates:
##        cor
## -0.5457544
```

```r
ggplot(aes(x = X, y = salinity_ppt), data = TeMi_Diss)+ #Weird that X isn't correlated with dissim
  geom_point()+
  geom_smooth(method = "lm")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

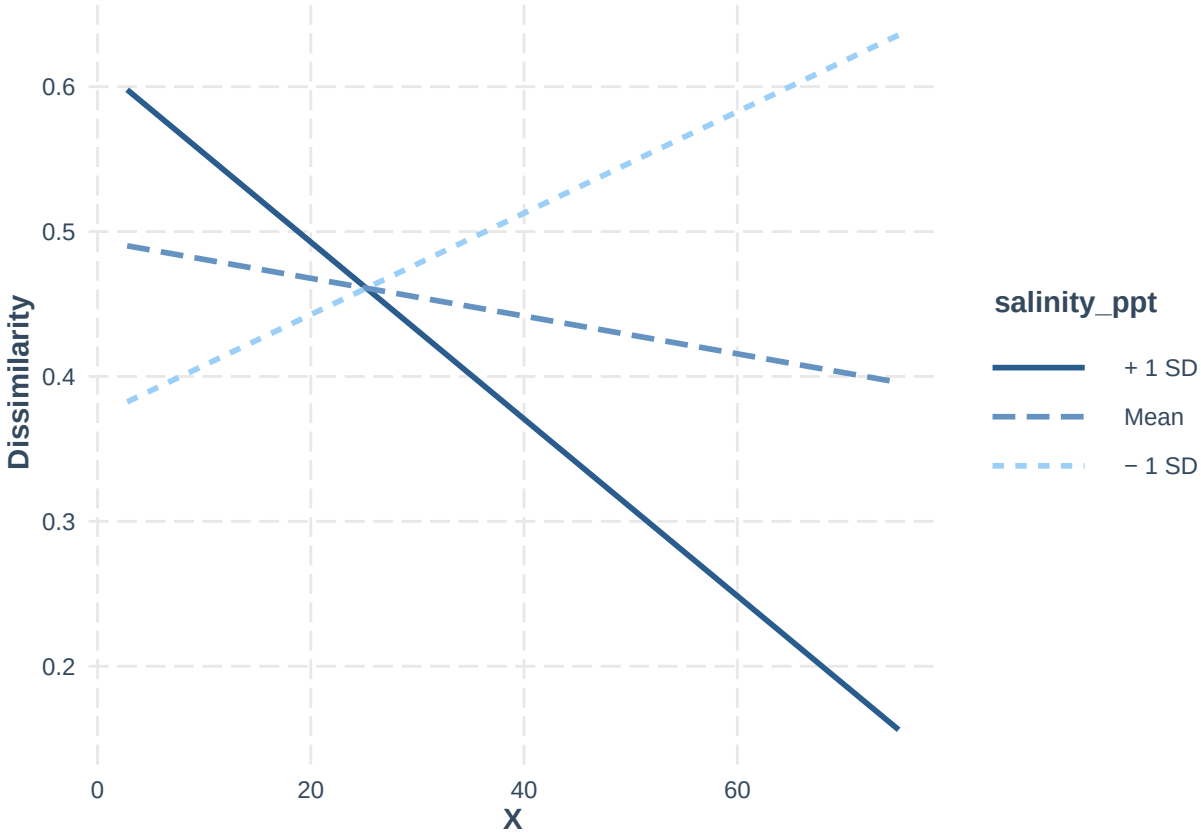



```r
mod.x.temi<-lm(Dissimilarity ~ salinity_ppt*X, data = TeMi_Diss)
summary(mod.x.temi) #WHAT!?!?!
```

```
##
## Call:
## lm(formula = Dissimilarity ~ salinity_ppt * X, data = TeMi_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.44680 -0.08725 -0.00999  0.10242  0.40148
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)     0.233019   0.279934   0.832   0.4108
## salinity_ppt    1.007856   0.822409   1.225   0.2286
## X               0.009041   0.005210   1.735   0.0915 .
## salinity_ppt:X -0.039978   0.017115  -2.336   0.0254 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1957 on 35 degrees of freedom
```

```
## Multiple R-squared:  0.2479, Adjusted R-squared:  0.1834
## F-statistic: 3.845 on 3 and 35 DF,  p-value: 0.01768
```

```
interact_plot(mod.x.temi, pred = X, modx = salinity_ppt)
```




```
TeMi.mod.1<- lm(Dissimilarity ~ SoilTemp + SoilMoist + Slope + Aspect + X + Z + pH, data = TeMi_Diss)
summary(TeMi.mod.1)
```

Negligible effects

```
##
## Call:
## lm(formula = Dissimilarity ~ SoilTemp + SoilMoist + Slope + Aspect +
##     X + Z + pH, data = TeMi_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.46820 -0.08120  0.04277  0.12199  0.36691
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.695694   3.060093   0.881   0.3868
## SoilTemp     0.088066   0.045154   1.950   0.0624 .
## SoilMoist    0.013885   0.049841   0.279   0.7829
## Slope        0.004875   0.005777   0.844   0.4068
## AspectN      0.049005   0.130136   0.377   0.7097
## AspectNE     0.089118   0.150427   0.592   0.5589
```

```
## AspectNW     0.046684   0.199363   0.234   0.8168
## AspectS     -0.052685   0.141462  -0.372   0.7127
## AspectSE     0.296035   0.169731   1.744   0.0934 .
## AspectSW     0.204018   0.201709   1.011   0.3215
## AspectW      0.052063   0.169360   0.307   0.7611
## X            0.001701   0.002833   0.600   0.5536
## Z           -0.005324   0.039920  -0.133   0.8950
## pH          -0.258973   0.323818  -0.800   0.4314
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2264 on 25 degrees of freedom
## Multiple R-squared:  0.2816, Adjusted R-squared:  -0.09202
## F-statistic: 0.7537 on 13 and 25 DF,  p-value: 0.6972
```

```
TeMi.Null.1 <- lm(Dissimilarity ~ 1, data = TeMi_Diss)
anova(TeMi.mod.1, TeMi.Null.1)
```

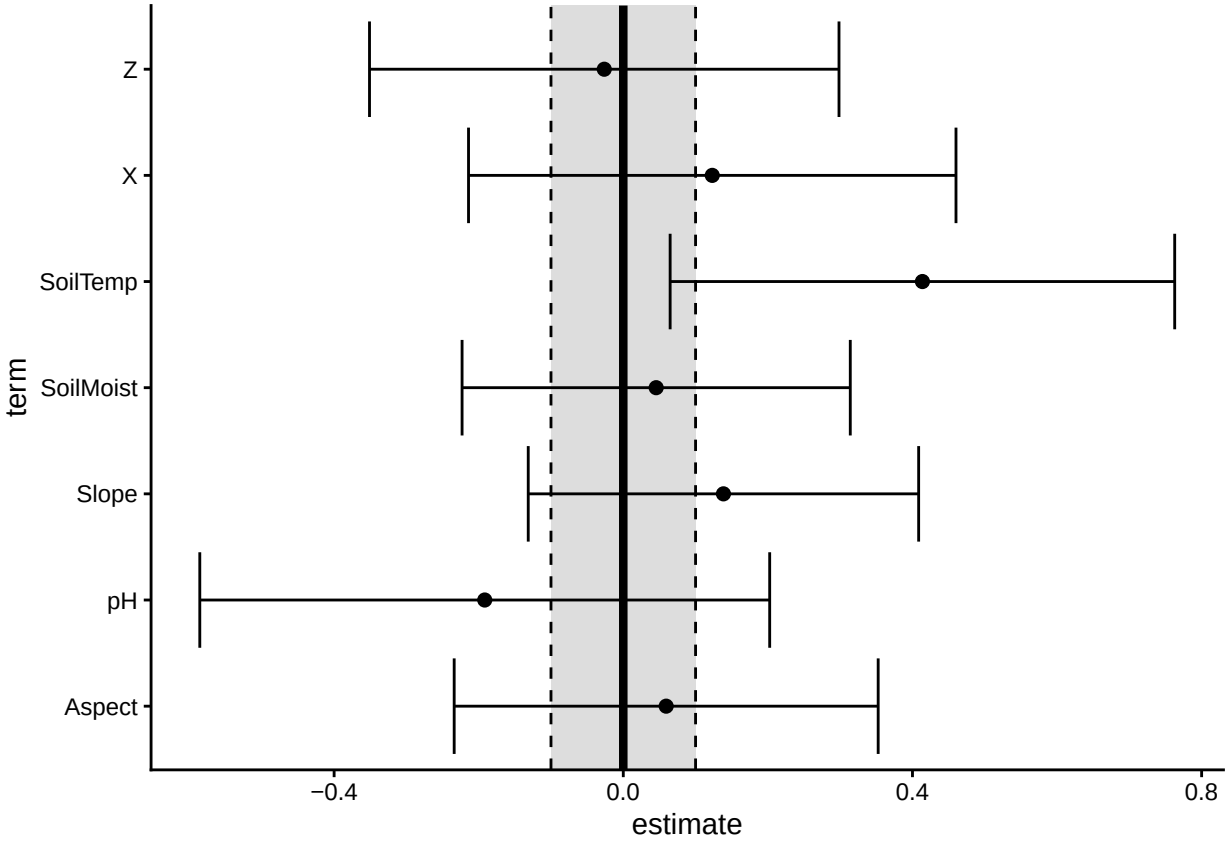
```
## Analysis of Variance Table
##
## Model 1: Dissimilarity ~ SoilTemp + SoilMoist + Slope + Aspect + X + Z +
##     pH
## Model 2: Dissimilarity ~ 1
##   Res.Df    RSS  Df Sum of Sq      F Pr(>F)
## 1     25 1.2809
## 2     38 1.7830 -13  -0.50202 0.7537 0.6972
```

```
TeMi.pre.1<- predictions(TeMi.mod.1)

TeMi.cmp<- avg_comparisons(TeMi.mod.1,  variables =  list(SoilTemp = "minmax", SoilMoist = "minmax", Sl


TeMi.cmp<-TeMi.cmp %>%
  group_by(term) %>%
  summarise_all(.funs = mean)
```

```
## Warning: There were 7 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 6 remaining warnings.
```

```
ggplot(TeMi.cmp, aes(x = estimate,
                 xmin = conf.low,
                 xmax = conf.high,
                 y = term)) +
     annotate("rect", xmin = -0.1, xmax = 0.1, ymin = -Inf, ymax = Inf, alpha = .2)+
  geom_vline(xintercept = c(-0.1, .1), linetype = "dashed") +
  geom_vline(xintercept = 0, linetype = "solid", linewidth = 1.5)+
  geom_errorbarh() +
  geom_point(size = 2) +
  theme_classic()
```

```
## Warning: Using the `size` aesthetic with geom_path was deprecated in ggplot2 3.4.0.
## i Please use the `linewidth` aesthetic instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Zmudowski

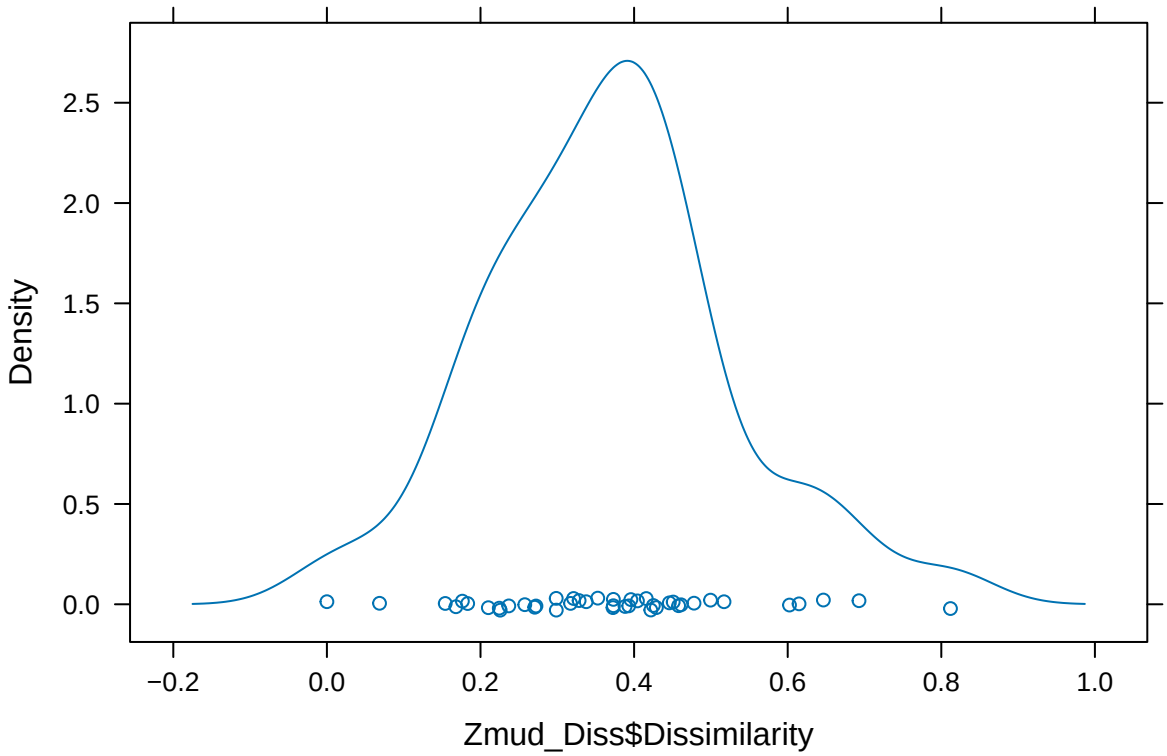
```
#I make unique identifiers for each plot
Zmud_Names<- colnames(Zmud_Cover)
Zmud_Names<-gsub('.{1}$', '', Zmud_Names)
Zmud_Names<-unique(gsub('Zmud.', '', Zmud_Names))
Zmud_Diss<-data.frame(Plot = rep("", length(Zmud_Names)), Dissimilarity= rep(0, length(Zmud_Names)))


for (i in 1:length(Zmud_Names)) {
  Temp_Cover<-Zmud_Cover %>% select(starts_with(paste0("Zmud.", Zmud_Names[i])))
  #Temp_Cover<- cbind(Temp_Cover[1], rowSums(Temp_Cover[2:5], na.rm = TRUE)/4)
  Temp_Diss<-vegdist(t(Temp_Cover), na.rm = TRUE)
  Temp_Av_Dis<-sum(as.matrix(Temp_Diss)[,1])/4
  Zmud_Diss$Plot[i]<- Zmud_Names[i]
  Zmud_Diss$Dissimilarity[i]<- ifelse(is.na(Temp_Av_Dis) == TRUE, 0, ifelse(Temp_Av_Dis == "NaN", 0, Ter

}
```

```
## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                    meaningless in method "bray"
```

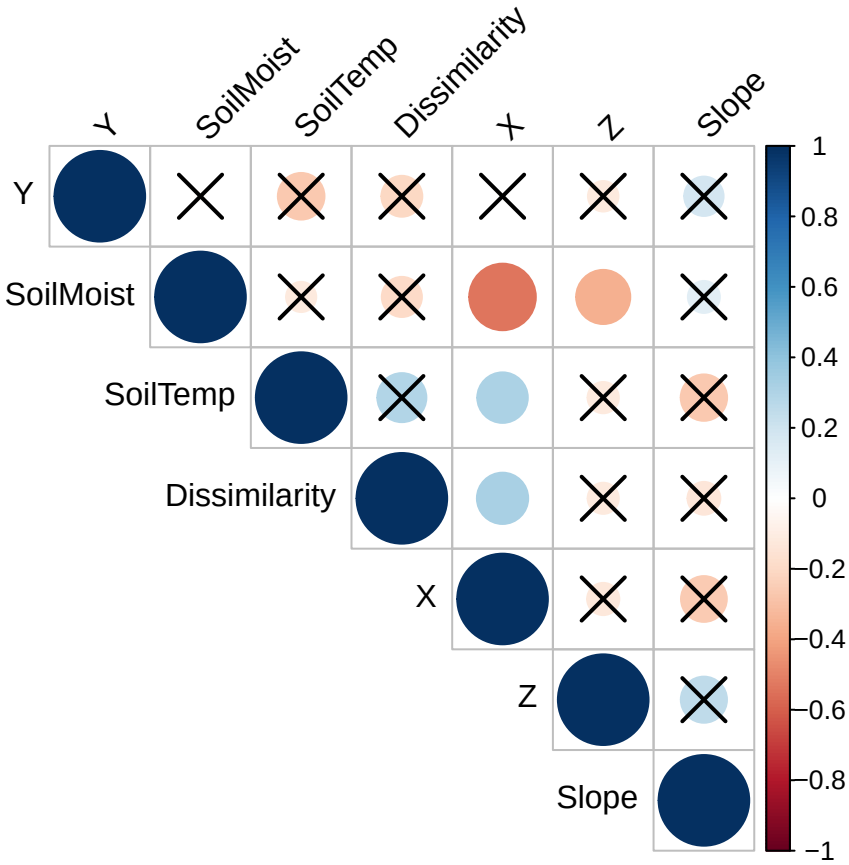
```
## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                    meaningless in method "bray"
```

```
## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): missing values in results
```

```r
densityplot(Zmud_Diss$Dissimilarity)
```

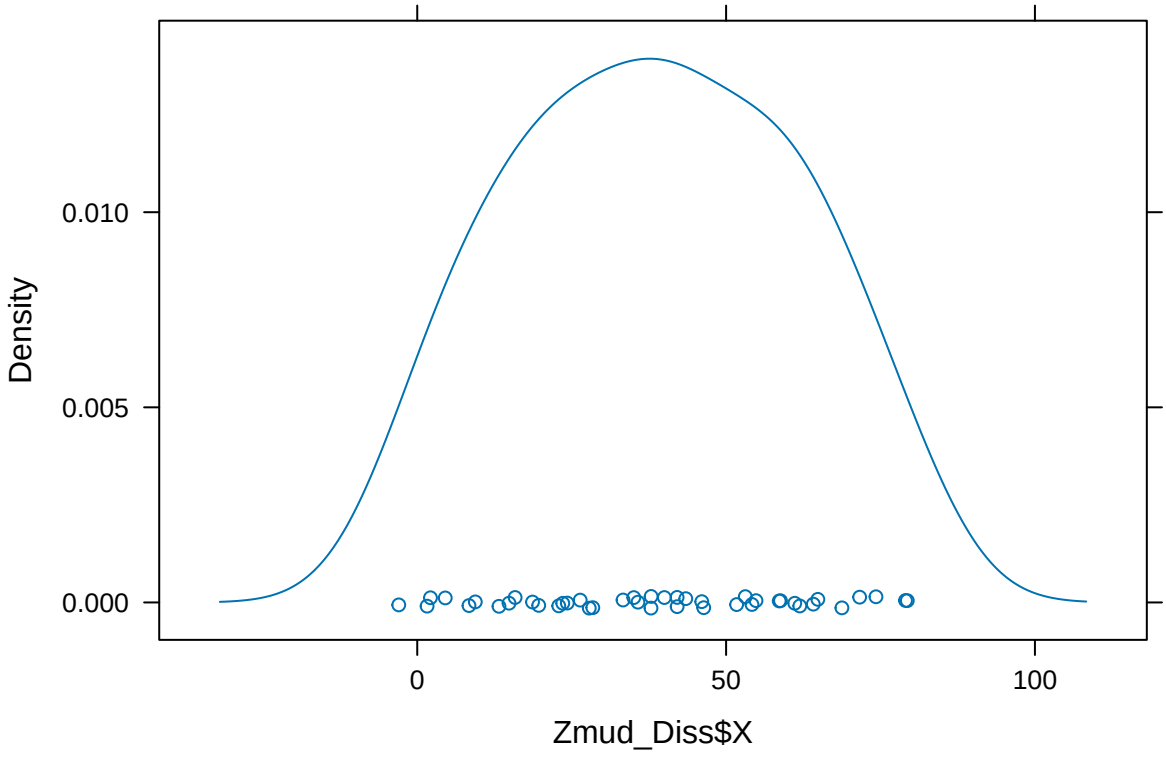



```r
Zmud_Diss$Plot<- as.numeric(Zmud_Diss$Plot)

Zmud_Diss<-merge(Zmud_Diss, Zmud_RelCords)


Zmud_Diss<-merge(Zmud_Diss, subset(DuneEnv_scaled, Site == "Zmud"))

Zmud_pcor<- Zmud_Diss %>%
  select(-c(Site, Plot, Aspect, Date, Collectors, dry_weight, Notes, pH, cond_us.cm, salinity_ppt))%>%
  as.matrix()%>%
  cor.mtest(use =  "complete")

Zmud_Diss %>%
  select(-c(Site, Plot, Aspect, Date, Collectors, dry_weight, Notes, pH, cond_us.cm, salinity_ppt))%>%
  as.matrix()%>%
  cor(use =  "complete") %>%
  corrplot(type = "upper", order = "hclust", tl.col = "black", tl.srt = 45, p.mat = Zmud_pcor$p)
```



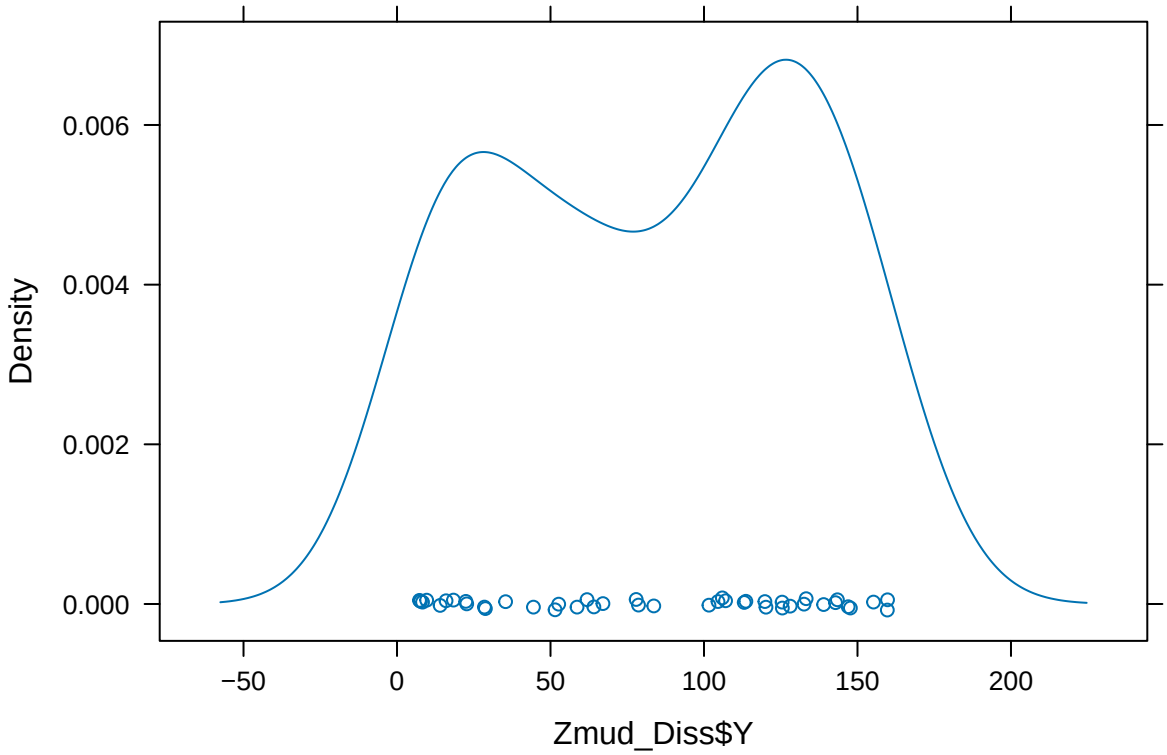
Graphs and models

```
densityplot(Zmud_Diss$X)
```

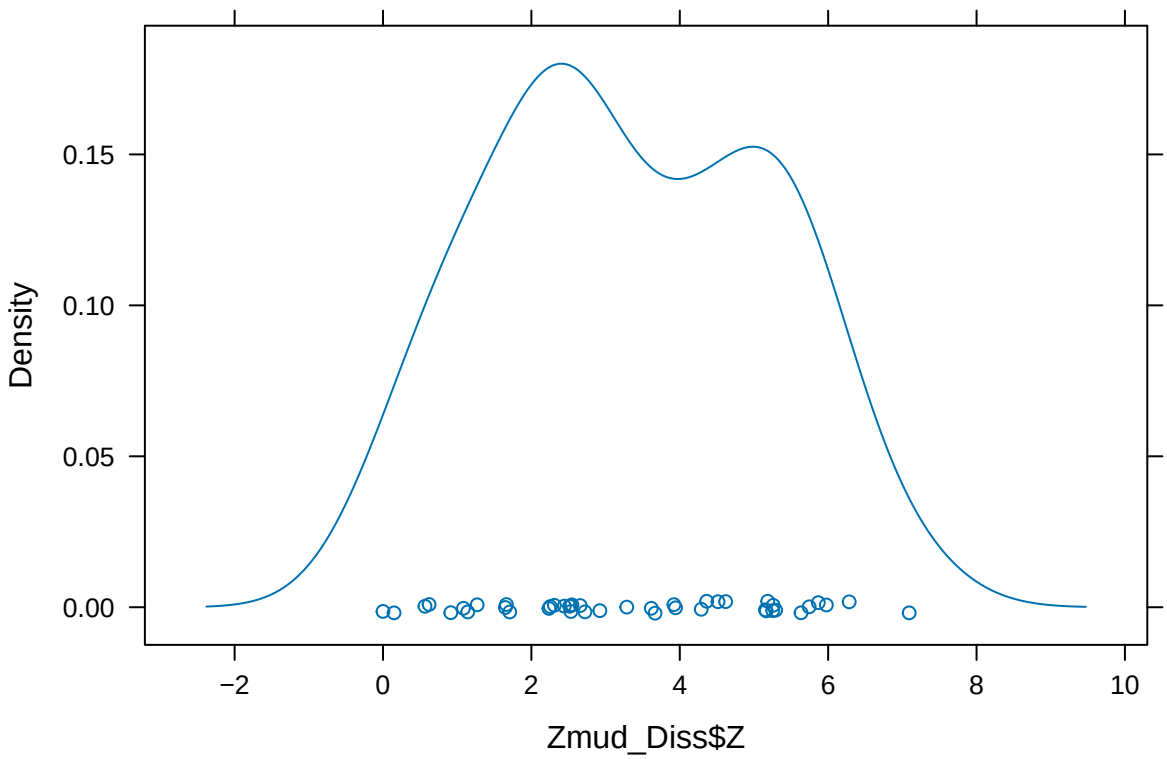


Distance

```
densityplot(Zmud_Diss$Y)
```




```
densityplot(Zmud_Diss$Z)
```

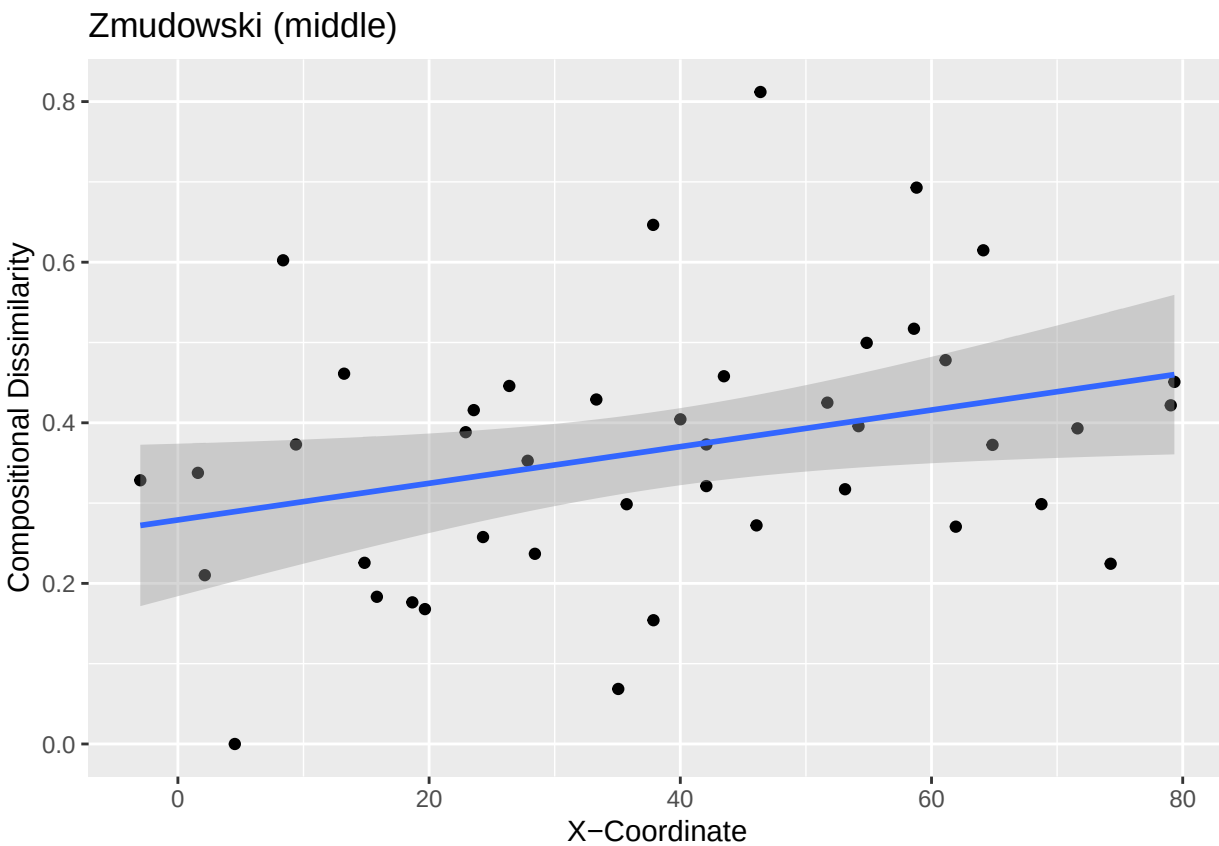



```r
summary(lm(Dissimilarity ~ X, data = Zmud_Diss)) # Distance from waterline SIG HERE!
```

```
##
## Call:
## lm(formula = Dissimilarity ~ X, data = Zmud_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.29045 -0.09714 -0.00685  0.07688  0.42712
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.279012   0.047011   5.935 5.35e-07 ***
## X           0.002281   0.001055   2.163   0.0364 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1555 on 41 degrees of freedom
## Multiple R-squared:  0.1024, Adjusted R-squared:  0.08055
## F-statistic:  4.68 on 1 and 41 DF,  p-value: 0.0364
```

```r
ggplot(aes(x = X, y = Dissimilarity), data = Zmud_Diss)+
  geom_point()+
  geom_smooth(method = "lm")+
  labs(x = "X-Coordinate", y = "Compositional Dissimilarity", title = "Zmudowski (middle)")
```

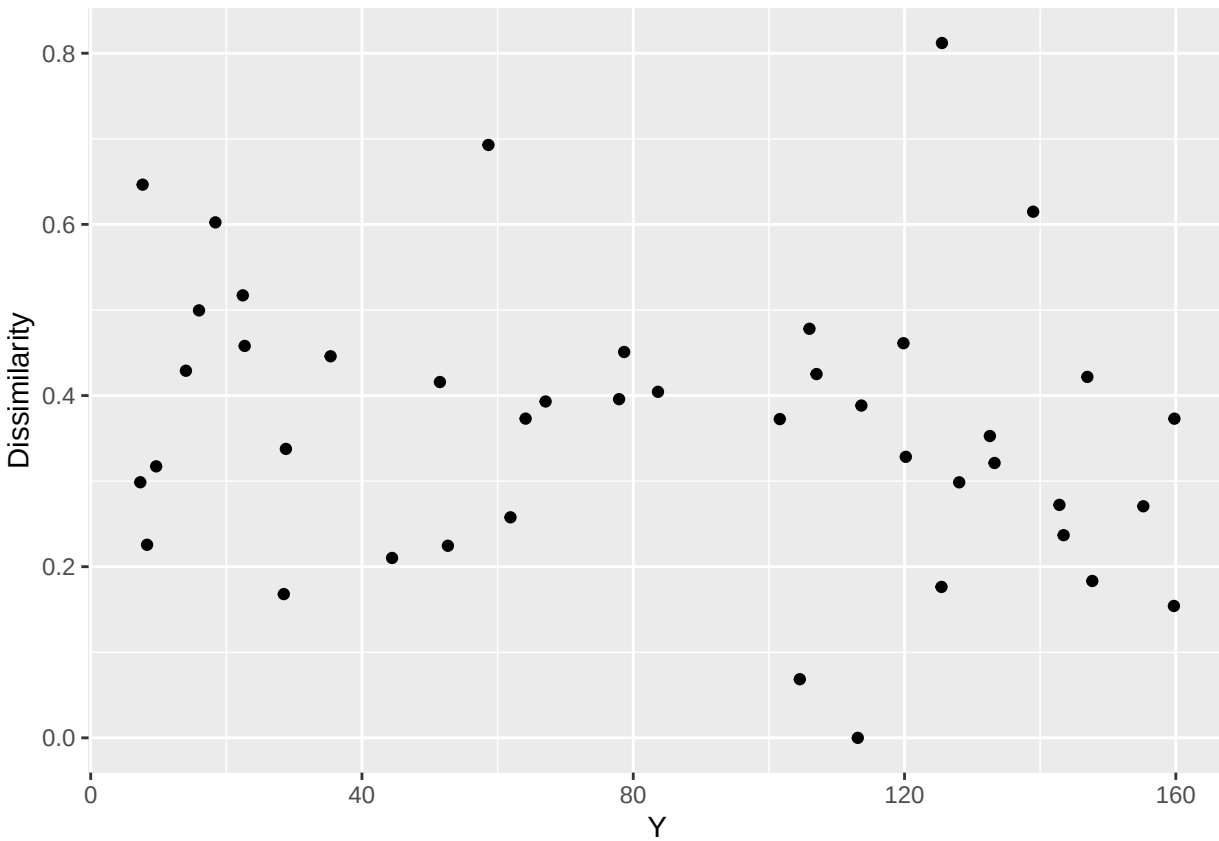
```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
summary(lm(Dissimilarity ~ Y, data = Zmud_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ Y, data = Zmud_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.34757 -0.10733  0.01730  0.07731  0.47241
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.4207881  0.0474042   8.877 4.29e-11 ***
## Y           -0.0006473  0.0004866  -1.330    0.191
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1608 on 41 degrees of freedom
## Multiple R-squared:  0.04137,    Adjusted R-squared:  0.01799
## F-statistic:  1.77 on 1 and 41 DF,  p-value: 0.1908
```

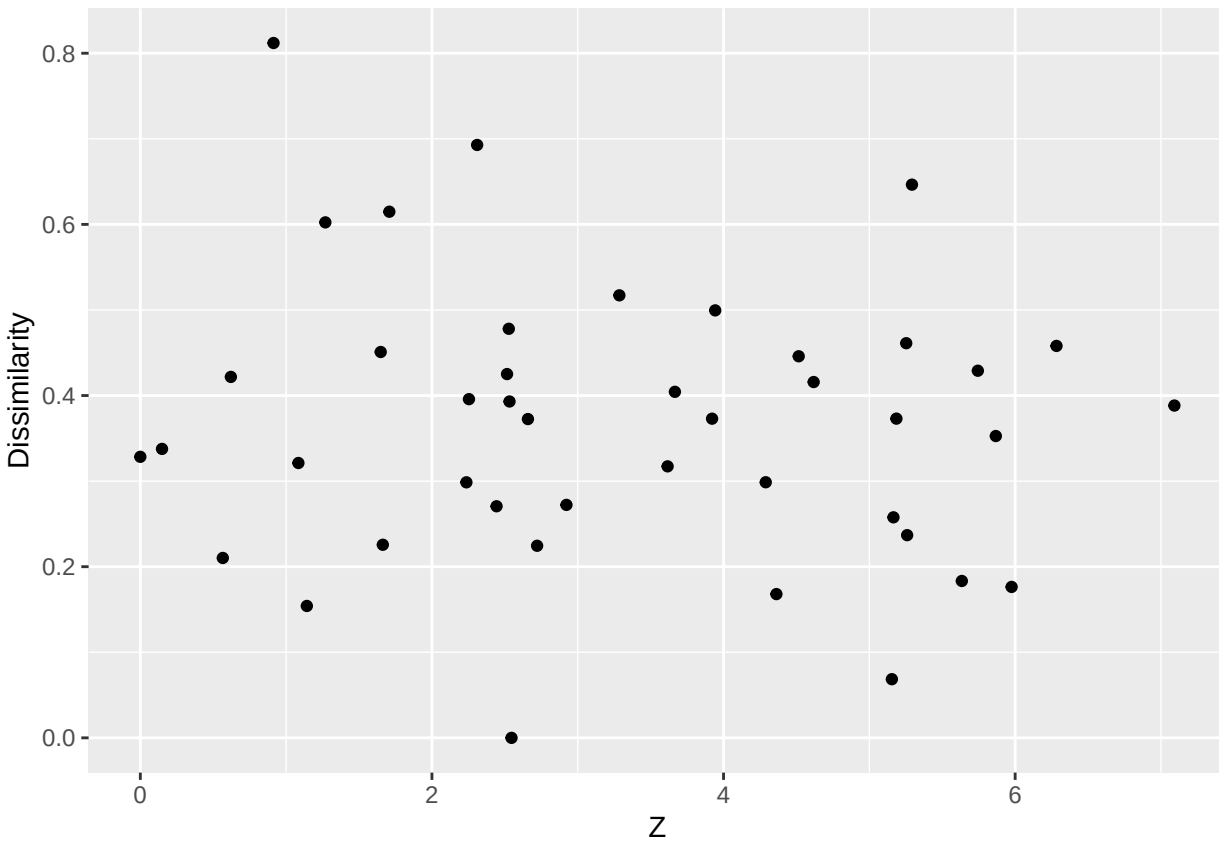
```r
ggplot(aes(x = Y, y = Dissimilarity), data = Zmud_Diss)+
  geom_point()
```




```r
summary(lm(Dissimilarity ~ Z, data = Zmud_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ Z, data = Zmud_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.37468 -0.10186  0.01237  0.08918  0.42056
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.40072    0.05106   7.848 1.07e-09 ***
## Z           -0.01023    0.01345  -0.760    0.451
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.163 on 41 degrees of freedom
## Multiple R-squared:  0.0139, Adjusted R-squared:  -0.01015
## F-statistic: 0.578 on 1 and 41 DF,  p-value: 0.4514
```

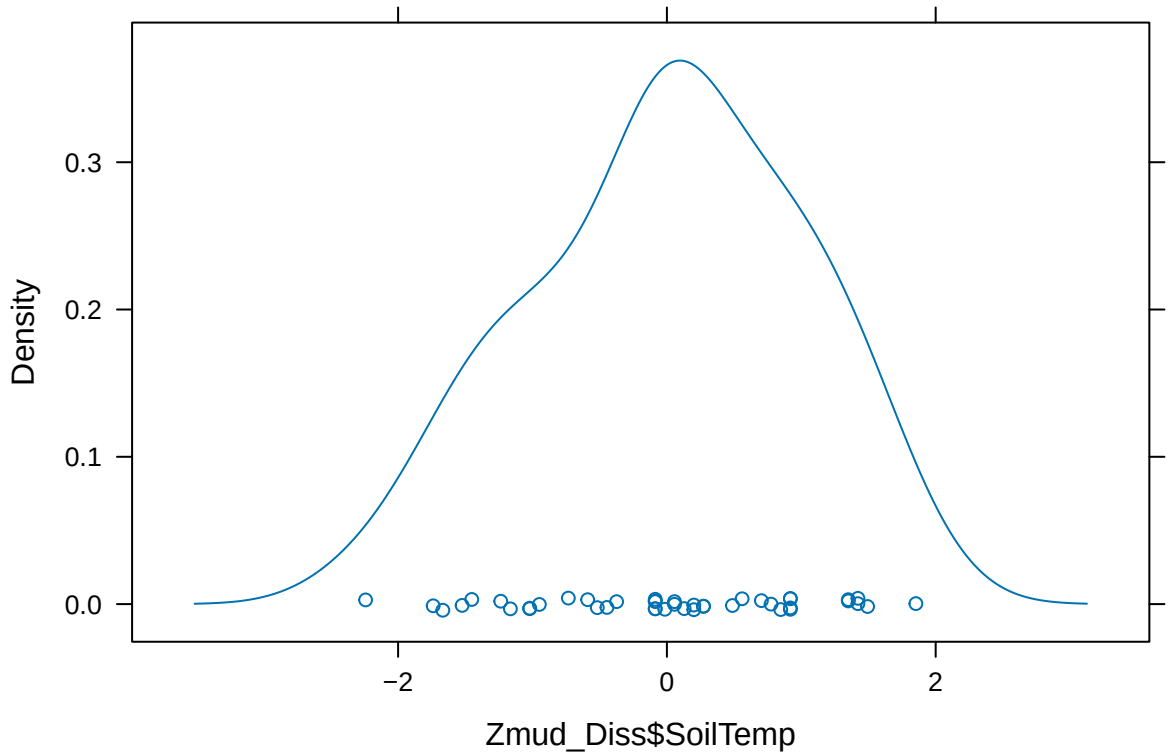
```r
ggplot(aes(x = Z, y = Dissimilarity), data = Zmud_Diss)+
  geom_point()
```




```r
summary(lm(Dissimilarity ~ X*Y*Z, data = Zmud_Diss))
```

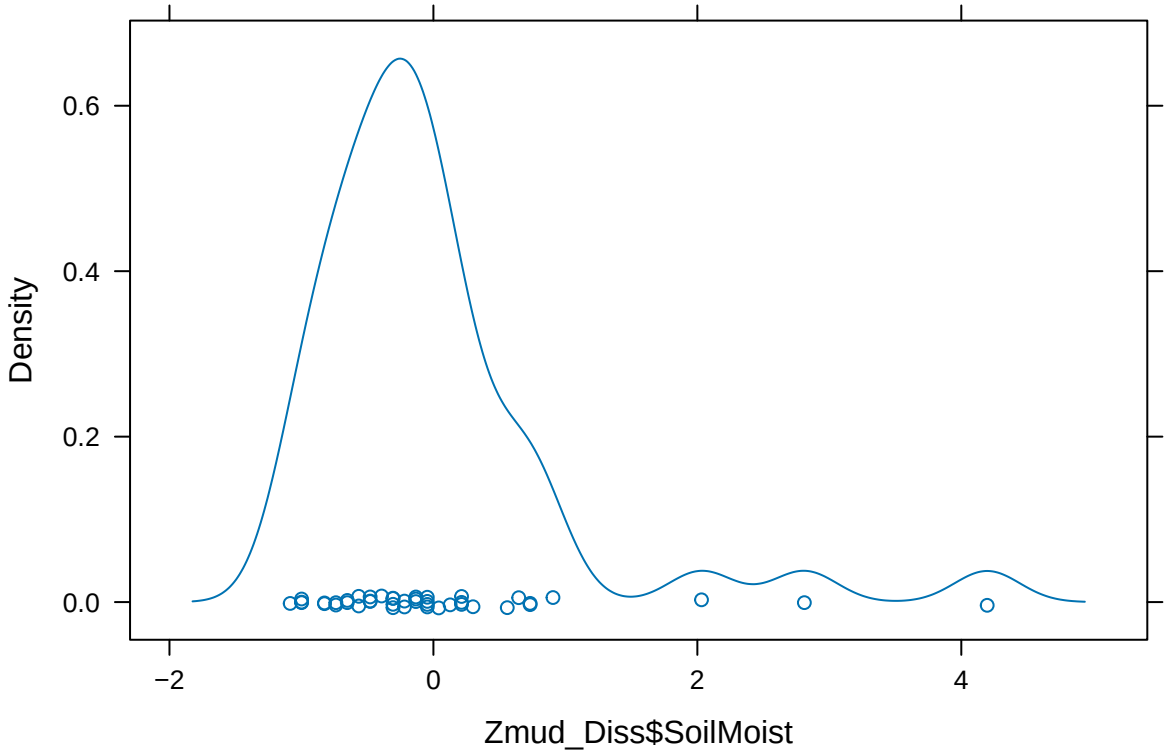
```
##
## Call:
## lm(formula = Dissimilarity ~ X * Y * Z, data = Zmud_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.29185 -0.10393 -0.00711  0.08695  0.40144
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.340e-01  1.513e-01   2.208   0.0339 *
## X            2.698e-03  5.264e-03   0.513   0.6115
## Y           -7.589e-04  1.721e-03  -0.441   0.6619
## Z            2.452e-02  5.702e-02   0.430   0.6698
## X:Y          1.397e-05  4.734e-05   0.295   0.7697
## X:Z         -6.088e-04  1.761e-03  -0.346   0.7316
## Y:Z         -9.026e-05  5.400e-04  -0.167   0.8682
## X:Y:Z       -3.951e-06  1.594e-05  -0.248   0.8057
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1598 on 35 degrees of freedom
## Multiple R-squared:  0.1912, Adjusted R-squared:  0.02944
## F-statistic: 1.182 on 7 and 35 DF,  p-value: 0.3381
```

```r
densityplot(Zmud_Diss$SoilTemp)
```

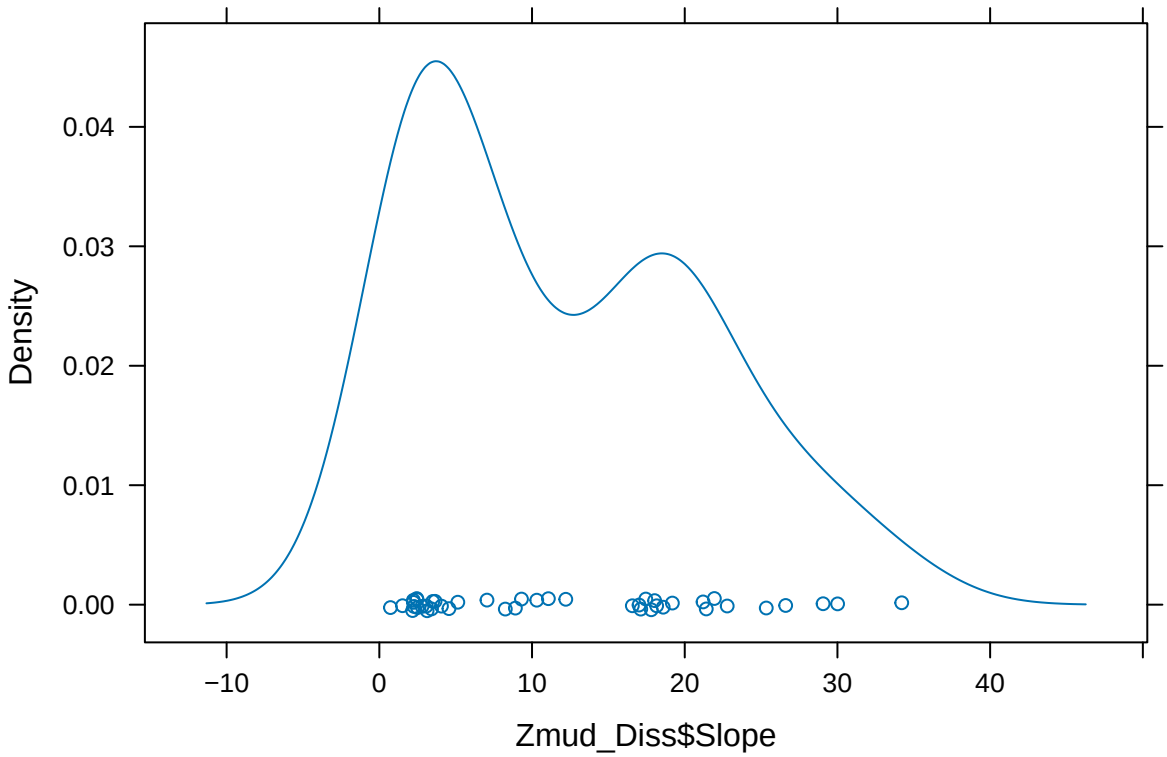


Environmental

```r
densityplot(Zmud_Diss$SoilMoist)
```



```r
densityplot(Zmud_Diss$Slope)
```




```r
summary(lm(Dissimilarity ~ SoilTemp+SoilMoist+Slope+Aspect, data = Zmud_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ SoilTemp + SoilMoist + Slope + Aspect,
##     data = Zmud_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.32694 -0.09430 -0.01110  0.05693  0.35093
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.302844   0.109728   2.760  0.00936 **
## SoilTemp     0.049177   0.027979   1.758  0.08808 .
## SoilMoist   -0.023532   0.027859  -0.845  0.40438
## Slope        0.000695   0.003040   0.229  0.82055
## AspectN      0.151999   0.110299   1.378  0.17746
## AspectNE    -0.003859   0.118886  -0.032  0.97430
## AspectS      0.075705   0.123780   0.612  0.54499
## AspectSE    -0.050228   0.148740  -0.338  0.73773
## AspectSW     0.053095   0.116440   0.456  0.65138
## AspectW      0.034952   0.109105   0.320  0.75072
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1602 on 33 degrees of freedom
## Multiple R-squared:  0.2334, Adjusted R-squared:  0.02434
```

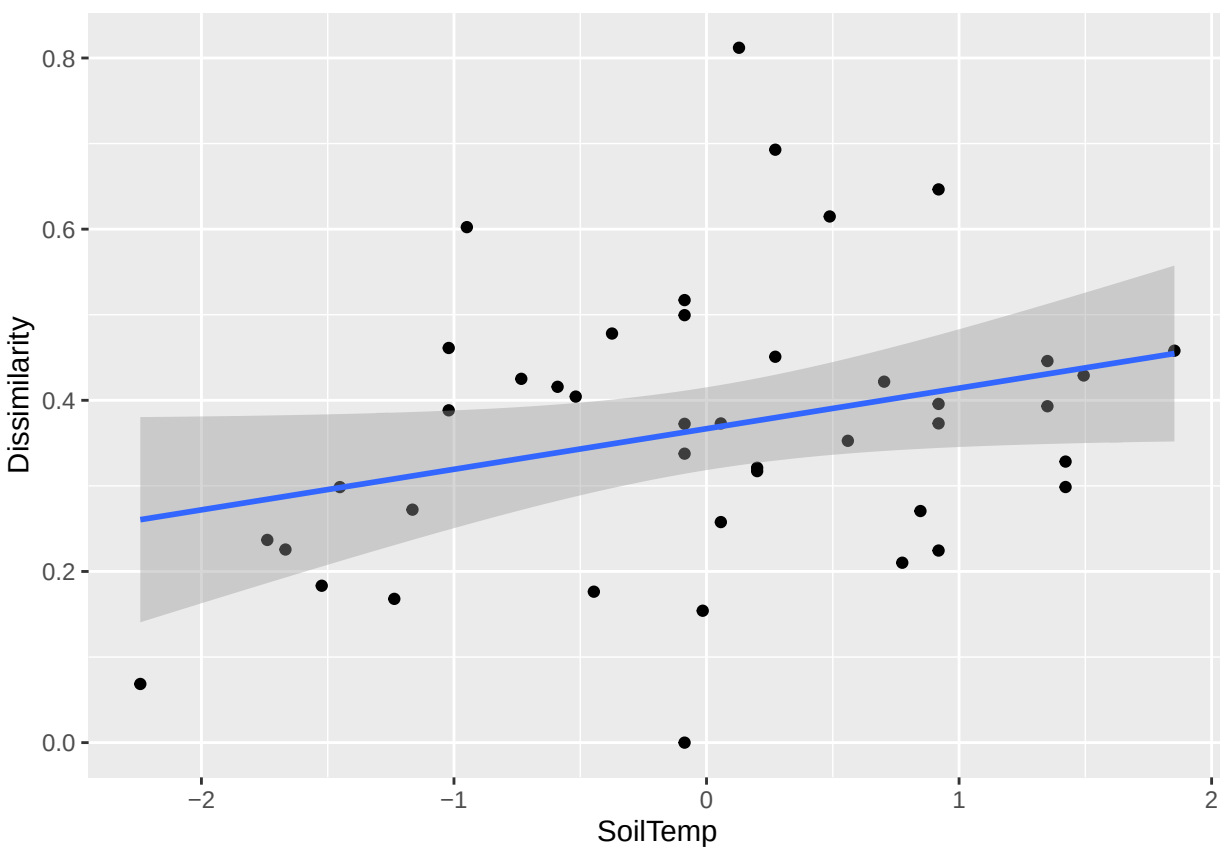
```
## F-statistic: 1.116 on 9 and 33 DF,  p-value: 0.3789
```

```r
summary(lm(Dissimilarity ~ SoilTemp, data = Zmud_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ SoilTemp, data = Zmud_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.36270 -0.10855 -0.01459  0.07406  0.43903
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.36682    0.02394  15.320   <2e-16 ***
## SoilTemp     0.04742    0.02423   1.957   0.0572 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.157 on 41 degrees of freedom
## Multiple R-squared:  0.08544,    Adjusted R-squared:  0.06314
## F-statistic:  3.83 on 1 and 41 DF,  p-value: 0.05716
```

```r
ggplot(aes(x = SoilTemp, y = Dissimilarity), data = Zmud_Diss)+
  geom_point()+
  geom_smooth(method = "lm")
```

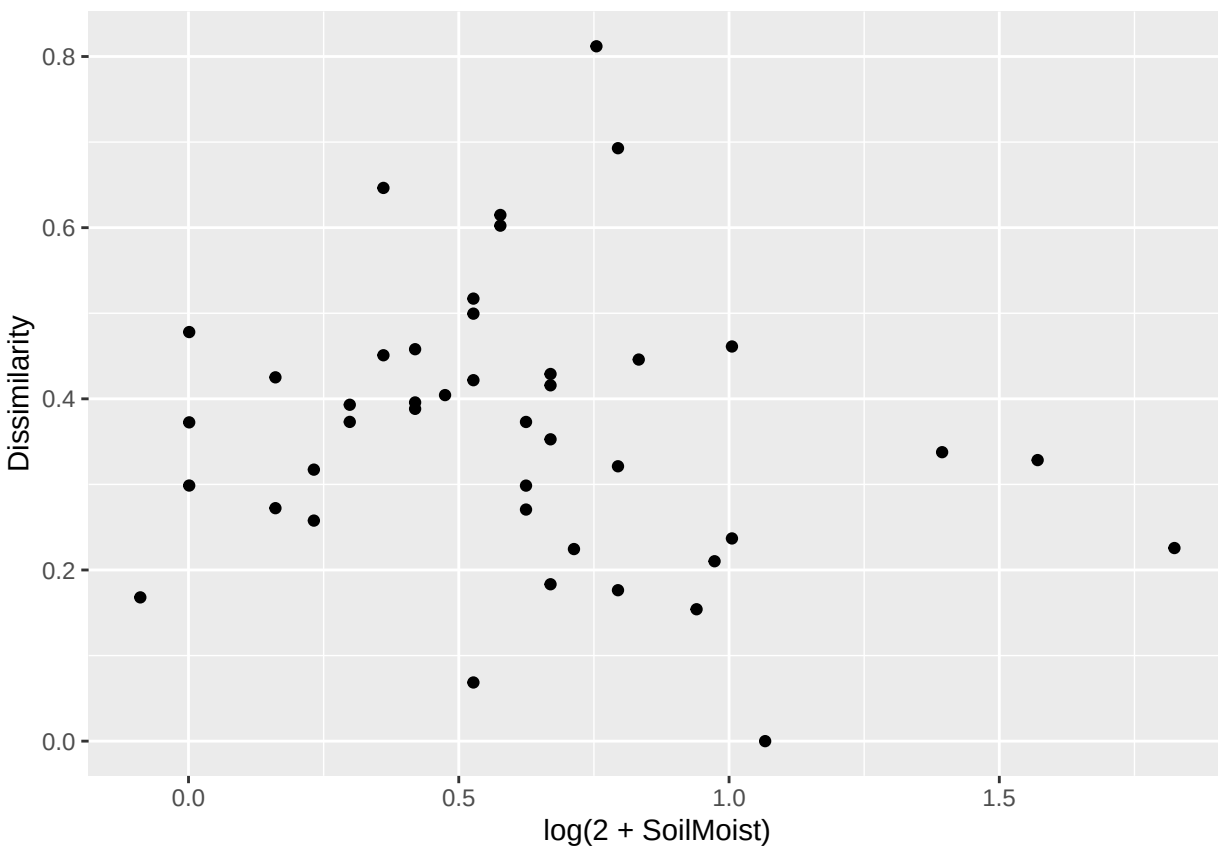
```
## `geom_smooth()` using formula = 'y ~ x'
```



```r
summary(lm(Dissimilarity ~ log(2+SoilMoist), data = Zmud_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ log(2 + SoilMoist), data = Zmud_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.33736 -0.10556  0.00738  0.07057  0.45466
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)          0.40544    0.04518   8.975 3.18e-11 ***
## log(2 + SoilMoist) -0.06382    0.06247  -1.022    0.313
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1621 on 41 degrees of freedom
## Multiple R-squared:  0.02483,    Adjusted R-squared:  0.001041
## F-statistic: 1.044 on 1 and 41 DF,  p-value: 0.3129
```

```r
ggplot(aes(x = log(2+SoilMoist), y = Dissimilarity), data = Zmud_Diss)+
  geom_point()
```




```r
summary(lm(Dissimilarity ~ SoilTemp*SoilMoist, data = Zmud_Diss))
```

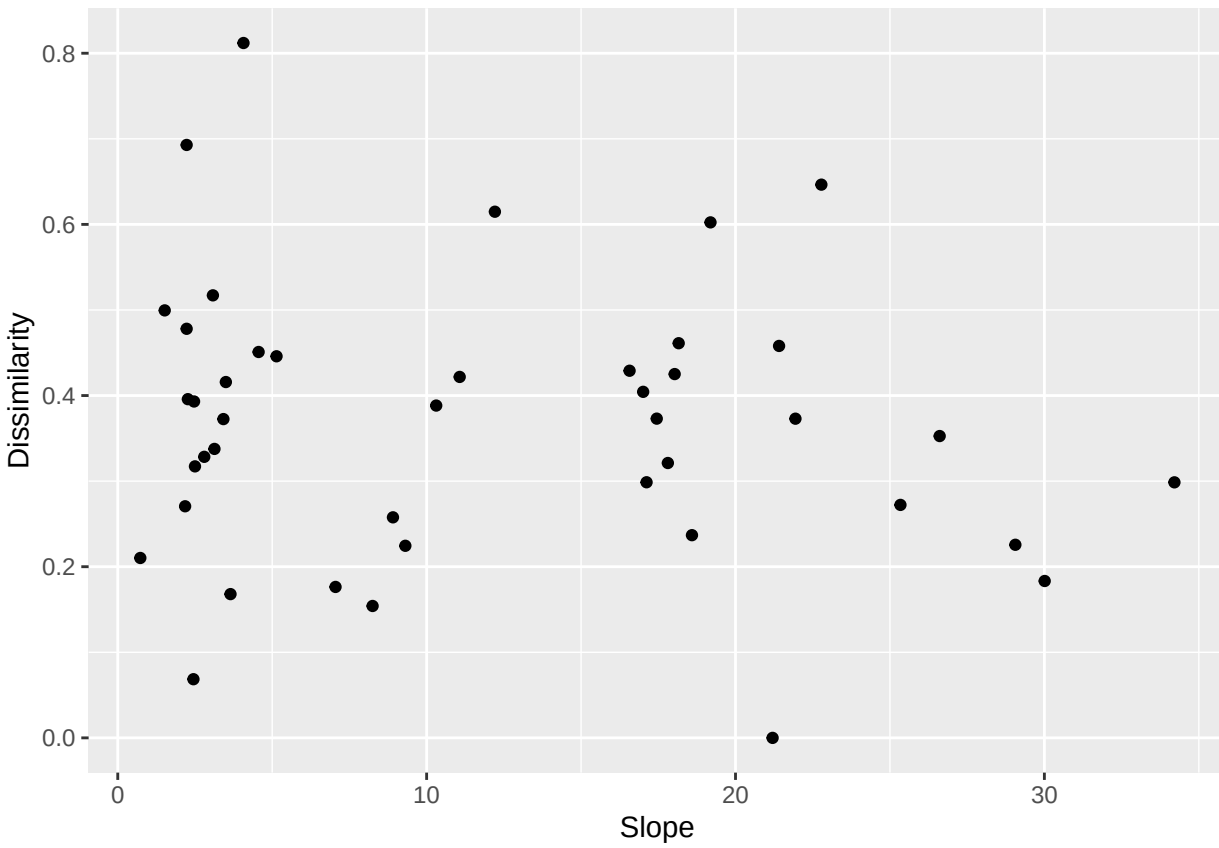
```
##
## Call:
```

```
## lm(formula = Dissimilarity ~ SoilTemp * SoilMoist, data = Zmud_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.33457 -0.09773 -0.00732  0.06579  0.44427
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)         0.365842   0.024213  15.110   <2e-16 ***
## SoilTemp            0.045814   0.024736   1.852   0.0716 .
## SoilMoist          -0.030904   0.026111  -1.184   0.2437
## SoilTemp:SoilMoist -0.008999   0.019557  -0.460   0.6480
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1582 on 39 degrees of freedom
## Multiple R-squared:  0.1172, Adjusted R-squared:  0.04933
## F-statistic: 1.726 on 3 and 39 DF,  p-value: 0.1774
```

```r
summary(lm(Dissimilarity ~ Slope, data = Zmud_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ Slope, data = Zmud_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.34595 -0.10882  0.00739  0.07235  0.42757
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.393495   0.040143   9.802 2.63e-12 ***
## Slope       -0.002243   0.002652  -0.846    0.403
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1628 on 41 degrees of freedom
## Multiple R-squared:  0.01714,    Adjusted R-squared:  -0.006829
## F-statistic: 0.7151 on 1 and 41 DF,  p-value: 0.4027
```

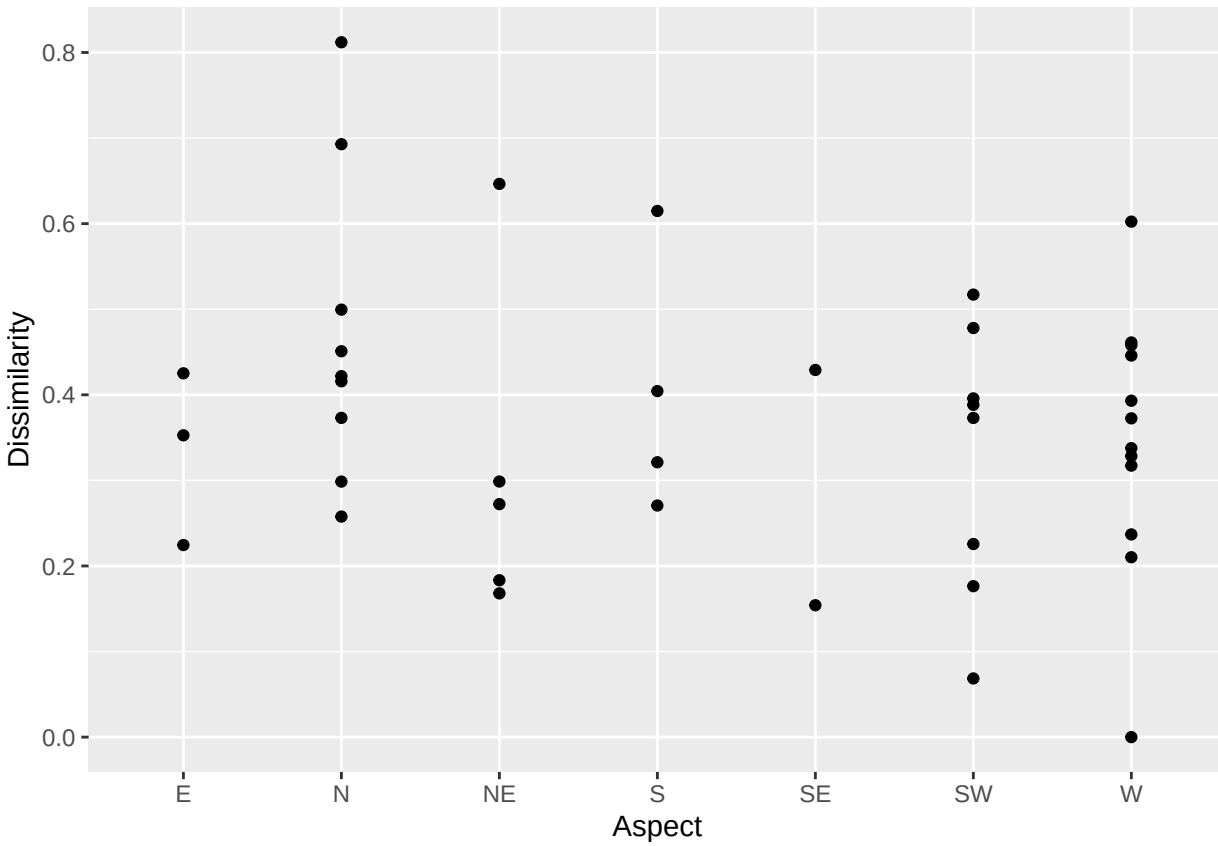
```r
ggplot(aes(x = Slope, y = Dissimilarity), data = Zmud_Diss)+
  geom_point()
```




```r
summary(lm(Dissimilarity ~ Aspect, data = Zmud_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ Aspect, data = Zmud_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.34694 -0.10986 -0.01505  0.09498  0.34282
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.334079   0.094297   3.543  0.00112 **
## AspectN      0.135042   0.108884   1.240  0.22291
## AspectNE    -0.020349   0.119277  -0.171  0.86549
## AspectS      0.068655   0.124743   0.550  0.58546
## AspectSE    -0.042526   0.149096  -0.285  0.77711
## AspectSW    -0.006236   0.110573  -0.056  0.95534
## AspectW      0.012862   0.105427   0.122  0.90358
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1633 on 36 degrees of freedom
## Multiple R-squared:  0.1311, Adjusted R-squared:  -0.01373
## F-statistic: 0.9052 on 6 and 36 DF,  p-value: 0.5021
```

```r
ggplot(aes(x = Aspect, y = Dissimilarity), data = Zmud_Diss)+
  geom_point()
```



Combined

```r
summary(lm(Dissimilarity ~ SoilTemp*X, data = Zmud_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ SoilTemp * X, data = Zmud_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.29434 -0.09002 -0.02199  0.05640  0.41752
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.290133   0.049258   5.890 7.36e-07 ***
## SoilTemp     0.071412   0.051326   1.391   0.1720
## X            0.002198   0.001199   1.833   0.0745 .
## SoilTemp:X  -0.001146   0.001396  -0.821   0.4165
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1545 on 39 degrees of freedom
## Multiple R-squared:  0.1582, Adjusted R-squared:  0.09344
## F-statistic: 2.443 on 3 and 39 DF,  p-value: 0.07855
```

```r
summary(lm(Dissimilarity ~ SoilMoist*X, data = Zmud_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ SoilMoist * X, data = Zmud_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.29630 -0.09496 -0.00137  0.07938  0.41503
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.2853818  0.0549651   5.192 6.82e-06 ***
## SoilMoist   -0.0167655  0.0370466  -0.453   0.6534
## X            0.0023537  0.0013385   1.758   0.0865 .
## SoilMoist:X  0.0007635  0.0015228   0.501   0.6189
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1589 on 39 degrees of freedom
## Multiple R-squared:  0.1089, Adjusted R-squared:  0.0404
## F-statistic: 1.589 on 3 and 39 DF,  p-value: 0.2074
```

```r
summary(lm(Dissimilarity ~ SoilTemp*Y, data = Zmud_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ SoilTemp * Y, data = Zmud_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.35295 -0.11106 -0.00468  0.07524  0.45298
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.4065707  0.0486121   8.364 3.12e-10 ***
## SoilTemp     0.0209491  0.0454729   0.461    0.648
## Y           -0.0004349  0.0004987  -0.872    0.389
## SoilTemp:Y   0.0002654  0.0004846   0.548    0.587
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1589 on 39 degrees of freedom
## Multiple R-squared:  0.1094, Adjusted R-squared:  0.04089
## F-statistic: 1.597 on 3 and 39 DF,  p-value: 0.2057
```

```r
summary(lm(Dissimilarity ~ SoilMoist*Y, data = Zmud_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ SoilMoist * Y, data = Zmud_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.31423 -0.10345  0.00913  0.07086  0.47729
```

```
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.209e-01  4.783e-02   8.799 8.41e-11 ***
## SoilMoist   -2.844e-02  3.577e-02  -0.795    0.431
## Y           -6.488e-04  4.914e-04  -1.320    0.194
## SoilMoist:Y -7.299e-05  4.933e-04  -0.148    0.883
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1613 on 39 degrees of freedom
## Multiple R-squared:  0.08137,    Adjusted R-squared:  0.01071
## F-statistic: 1.152 on 3 and 39 DF,  p-value: 0.3405
```

```r
summary(lm(Dissimilarity ~ SoilTemp*Z, data = Zmud_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ SoilTemp * Z, data = Zmud_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.37643 -0.08509 -0.03045  0.07686  0.41895
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.403076   0.050210   8.028 8.69e-10 ***
## SoilTemp    -0.024725   0.058350  -0.424    0.674
## Z           -0.009794   0.013167  -0.744    0.461
## SoilTemp:Z   0.017433   0.013107   1.330    0.191
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1569 on 39 degrees of freedom
## Multiple R-squared:  0.1318, Adjusted R-squared:  0.06505
## F-statistic: 1.974 on 3 and 39 DF,  p-value: 0.1338
```

```r
summary(lm(Dissimilarity ~ SoilMoist*Z, data = Zmud_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ SoilMoist * Z, data = Zmud_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.35068 -0.09625  0.00926  0.09395  0.40148
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.434130   0.055128   7.875 1.39e-09 ***
## SoilMoist   -0.056429   0.041486  -1.360    0.182
## Z           -0.018950   0.014264  -1.328    0.192
## SoilMoist:Z  0.006903   0.017894   0.386    0.702
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 0.1613 on 39 degrees of freedom
## Multiple R-squared:  0.08211,    Adjusted R-squared:  0.0115
## F-statistic: 1.163 on 3 and 39 DF,  p-value: 0.3362
```

```
Zmud.mod.1<- lm(Dissimilarity ~ SoilTemp + SoilMoist + Slope + Aspect + X + Z, data = Zmud_Diss)
summary(Zmud.mod.1)
```

Negligible effects

```
##
## Call:
## lm(formula = Dissimilarity ~ SoilTemp + SoilMoist + Slope + Aspect +
##     X + Z, data = Zmud_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.32321 -0.11702 -0.01094  0.07883  0.34179
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.2753405  0.1563741   1.761   0.0881 .
## SoilTemp     0.0444959  0.0298302   1.492   0.1459
## SoilMoist   -0.0223662  0.0373333  -0.599   0.5535
## Slope        0.0015574  0.0032906   0.473   0.6393
## AspectN      0.1537082  0.1128200   1.362   0.1829
## AspectNE     0.0104498  0.1231810   0.085   0.9329
## AspectS      0.0678871  0.1274385   0.533   0.5980
## AspectSE    -0.0325557  0.1536384  -0.212   0.8336
## AspectSW     0.0721878  0.1209467   0.597   0.5549
## AspectW      0.0557318  0.1164726   0.478   0.6357
## X            0.0008466  0.0016386   0.517   0.6091
## Z           -0.0080852  0.0173571  -0.466   0.6446
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1634 on 31 degrees of freedom
## Multiple R-squared:  0.2507, Adjusted R-squared:  -0.01522
## F-statistic: 0.9428 on 11 and 31 DF,  p-value: 0.5148
```

```
Zmud.Null.1 <- lm(Dissimilarity ~ 1, data = Zmud_Diss)
anova(Zmud.mod.1, Zmud.Null.1)
```

```
## Analysis of Variance Table
##
## Model 1: Dissimilarity ~ SoilTemp + SoilMoist + Slope + Aspect + X + Z
## Model 2: Dissimilarity ~ 1
##   Res.Df     RSS  Df Sum of Sq      F Pr(>F)
## 1     31 0.82816
## 2     42 1.10521 -11  -0.27705 0.9428 0.5148
```

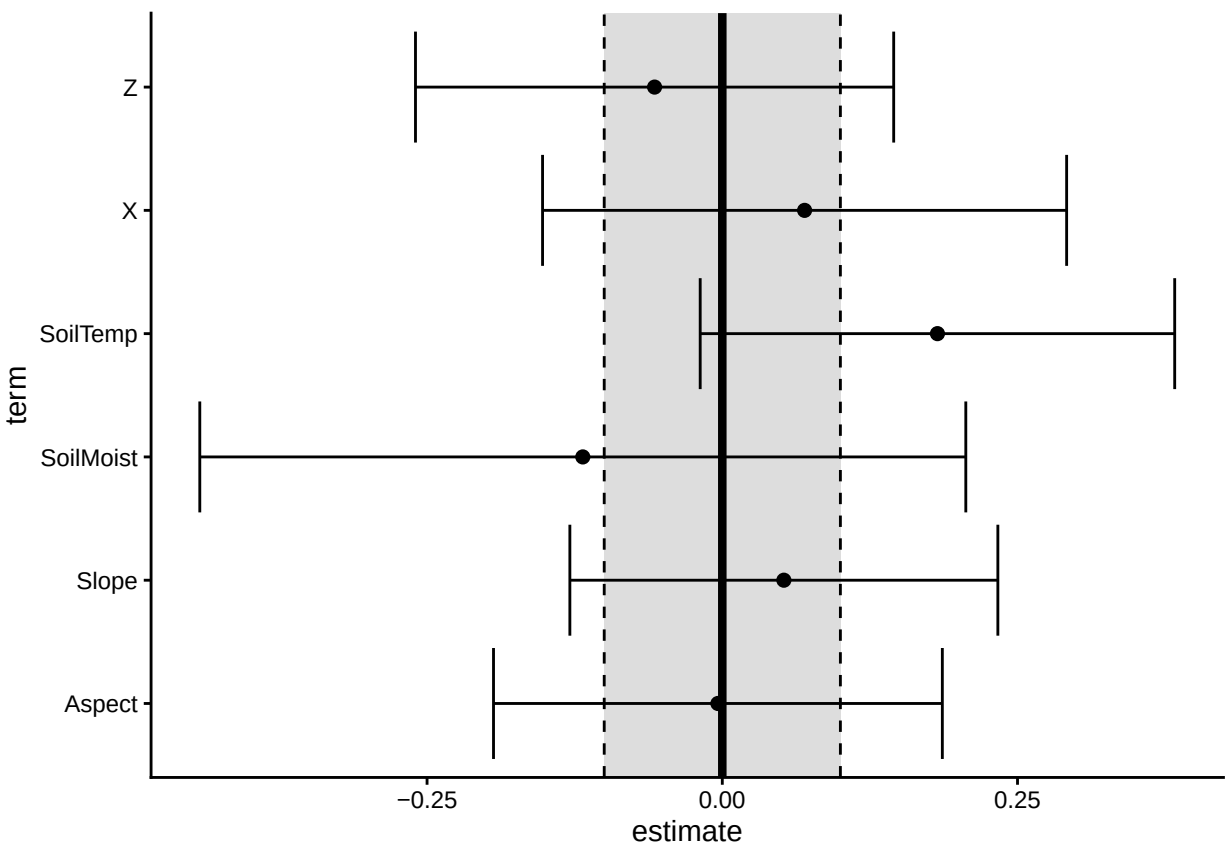
```
Zmud.cmp<- avg_comparisons(Zmud.mod.1,  variables =  list(SoilTemp = "minmax", SoilMoist = "minmax", Sl
```

```
Zmud.cmp<-Zmud.cmp %>%
```

```
  group_by(term) %>%
  summarise_all(.funs = mean)
```

```
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.
```

```
ggplot(Zmud.cmp, aes(x = estimate,
                 xmin = conf.low,
                 xmax = conf.high,
                 y = term)) +
     annotate("rect", xmin = -0.1, xmax = 0.1, ymin = -Inf, ymax = Inf, alpha = .2)+
  geom_vline(xintercept = c(-0.1, .1), linetype = "dashed") +
  geom_vline(xintercept = 0, linetype = "solid", linewidth = 1.5)+
  geom_errorbarh() +
  geom_point(size = 2) +
  theme_classic()
```



Guadalupe-Nipomo

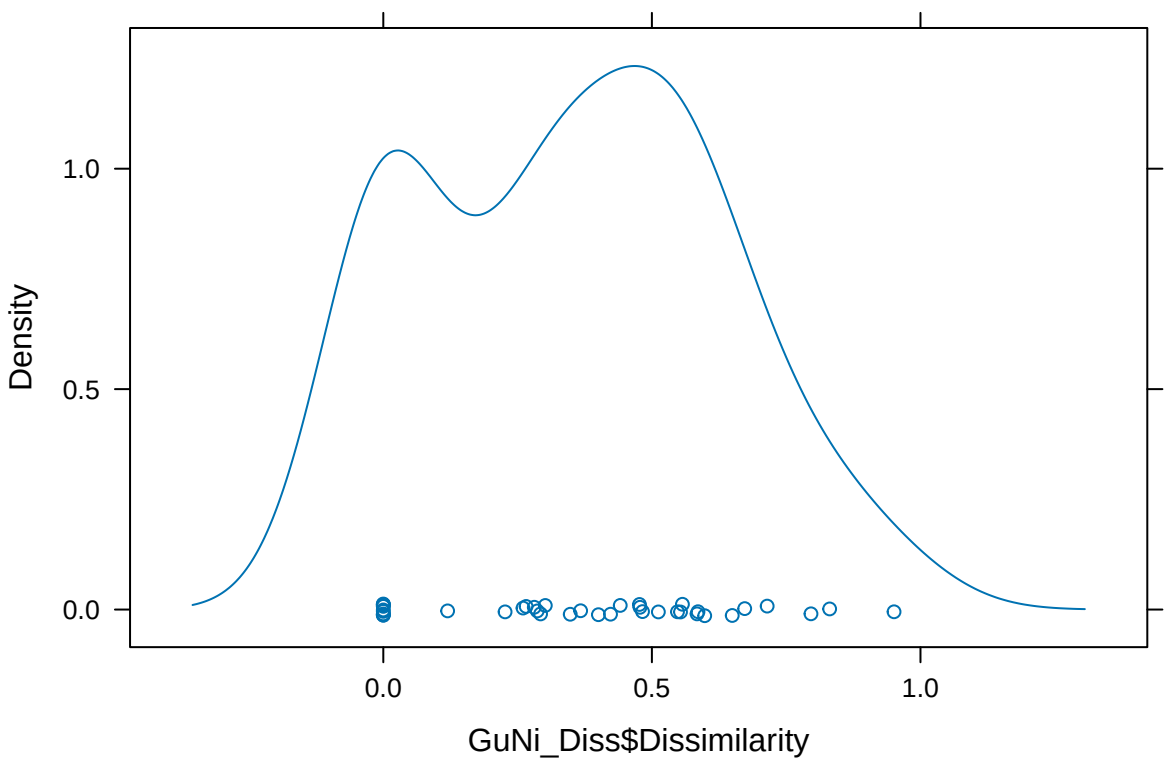
```
#I make unique identifiers for each plot
GuNi_Names<- colnames(GuNi_Cover)
GuNi_Names<-gsub('.{1}$', '', GuNi_Names)
GuNi_Names<-unique(gsub('GuNi.', '', GuNi_Names))
```

```r
GuNi_Diss<-data.frame(Plot = rep("", length(GuNi_Names)), Dissimilarity= rep(0, length(GuNi_Names)))


for (i in 1:length(GuNi_Names)) {
  Temp_Cover<-GuNi_Cover %>% select(starts_with(paste0("GuNi.", GuNi_Names[i])))
  #Temp_Cover<- cbind(Temp_Cover[1], rowSums(Temp_Cover[2:5], na.rm = TRUE)/4)
  Temp_Diss<-vegdist(t(Temp_Cover), na.rm = TRUE)
  Temp_Av_Dis<-sum(as.matrix(Temp_Diss)[,1])/4
  GuNi_Diss$Plot[i]<- GuNi_Names[i]
  GuNi_Diss$Dissimilarity[i]<- ifelse(is.na(Temp_Av_Dis) == TRUE, 0, ifelse(Temp_Av_Dis == "NaN", 0, Ter

}
```

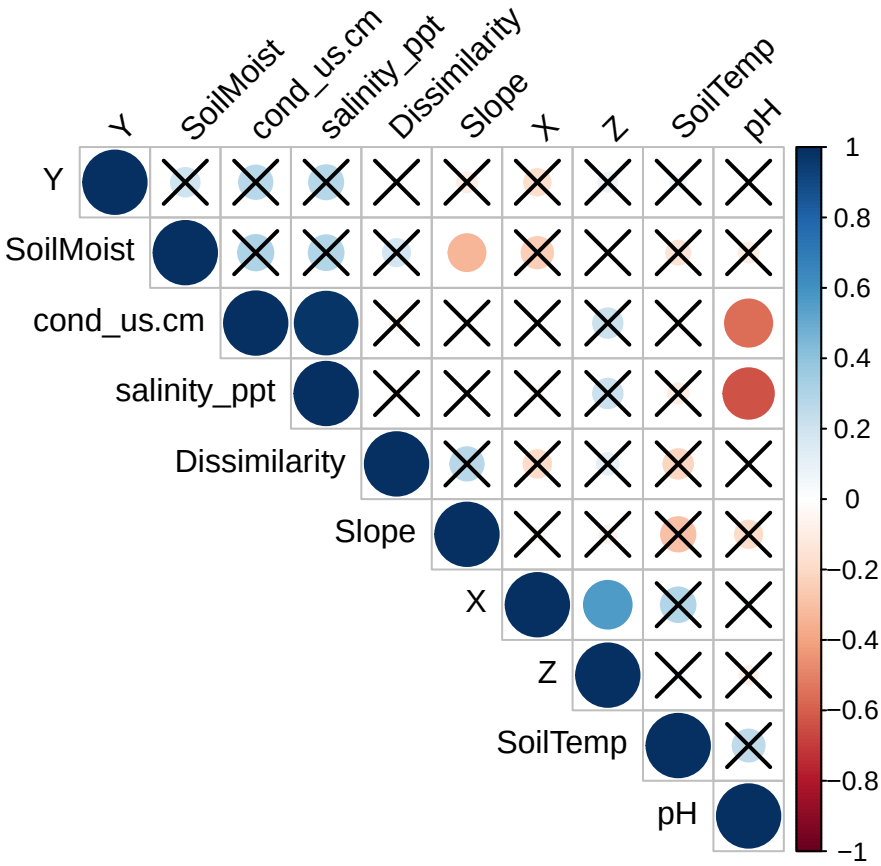
```
## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): missing values in results

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"
## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"
## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): missing values in results

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): missing values in results

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"
## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"
## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): missing values in results

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): missing values in results

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): missing values in results

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): missing values in results

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"
## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"
```

```
## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): missing values in results

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"
## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): missing values in results

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"
## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): missing values in results

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"
## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): missing values in results

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): missing values in results

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): you have empty rows: their dissimilarities may be
##                  meaningless in method "bray"

## Warning in vegdist(t(Temp_Cover), na.rm = TRUE): missing values in results
```

```r
densityplot(GuNi_Diss$Dissimilarity)
```

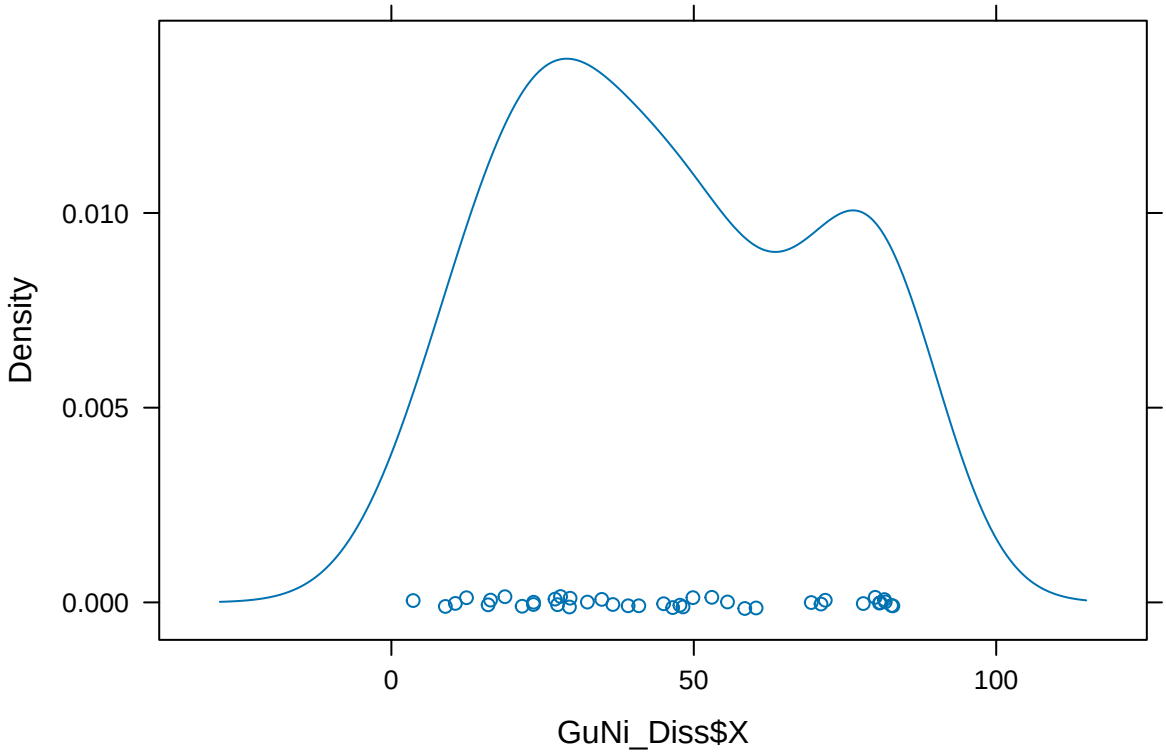


```
GuNi_Diss$Plot<- as.numeric(GuNi_Diss$Plot)

GuNi_Diss<-merge(GuNi_Diss, GuNi_RelCords)

GuNi_Diss<-merge(GuNi_Diss, subset(DuneEnv_scaled, Site == "GuNi"))


GuNi_pcor<- GuNi_Diss %>%
  select(-c(Site, Plot, Aspect, Date, Collectors, dry_weight, Notes))%>%
  as.matrix()%>%
  cor.mtest(use =  "complete")


GuNi_Diss %>%
  select(-c(Site, Plot, Aspect, Date, Collectors, dry_weight, Notes))%>%
  as.matrix()%>%
  cor(use =  "complete")%>%
  corrplot( type = "upper", order = "hclust",  tl.col = "black", tl.srt = 45, p.mat = GuNi_pcor$p)
```



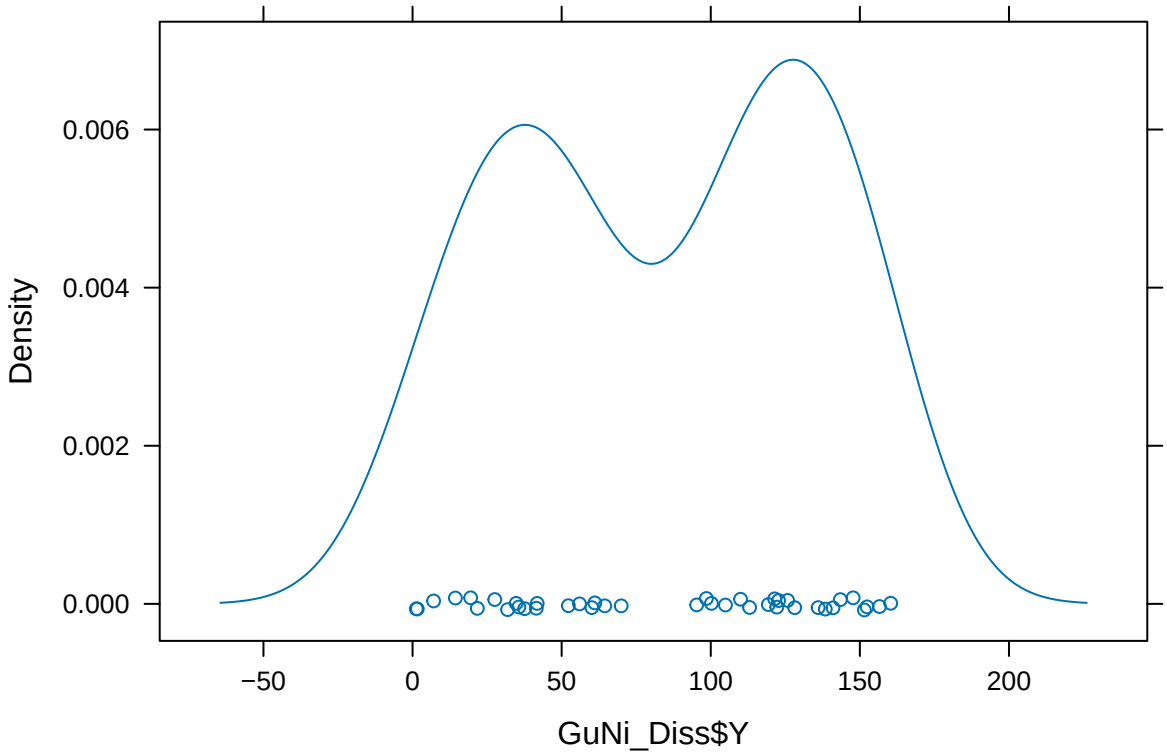
Graphs and models

```
densityplot(GuNi_Diss$X)
```

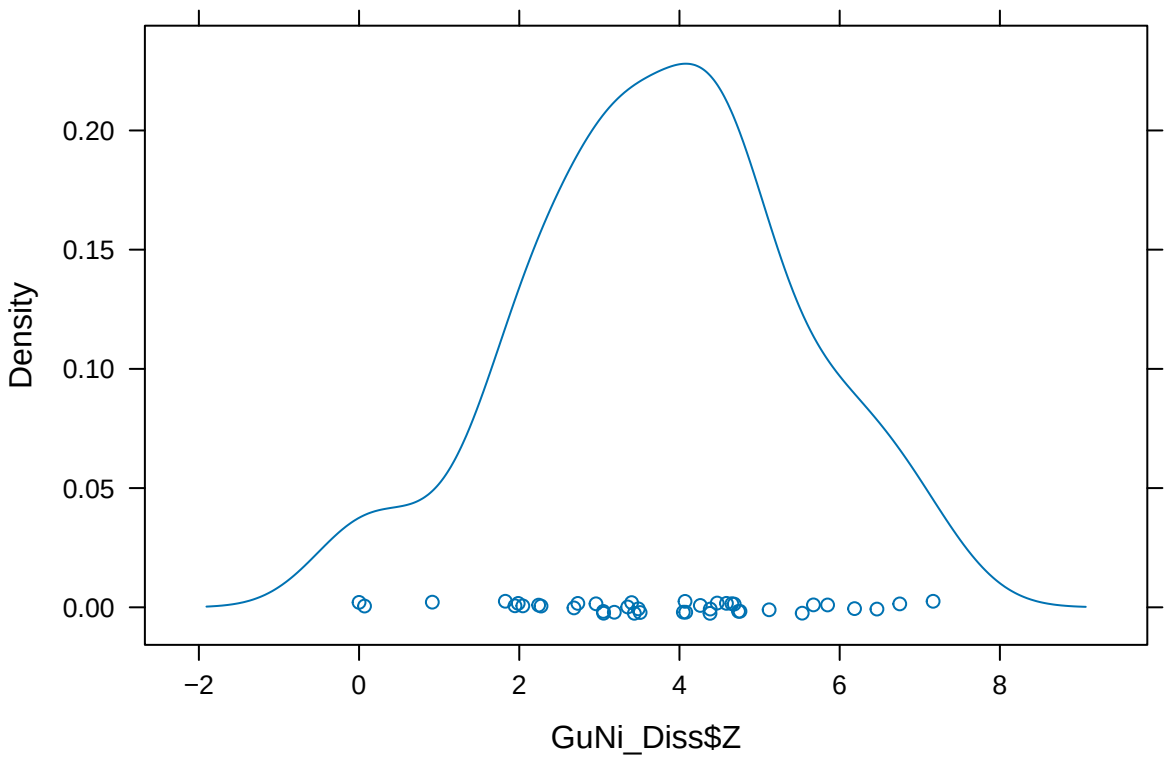


Distance

```
densityplot(GuNi_Diss$Y)
```



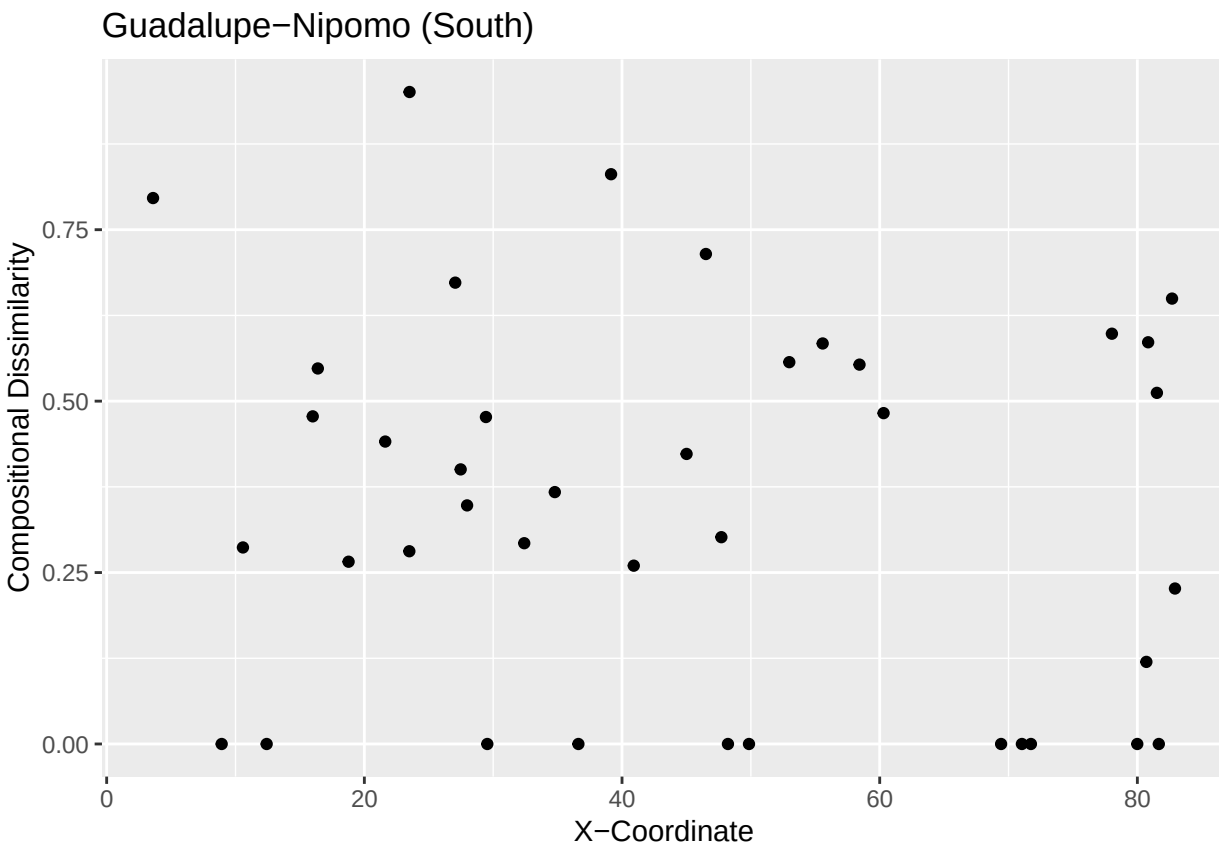

```
densityplot(GuNi_Diss$Z)
```




```r
summary(lm(Dissimilarity ~ X, data = GuNi_Diss)) # Distance from waterline not sig
```

```
##
## Call:
## lm(formula = Dissimilarity ~ X, data = GuNi_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.41044 -0.28997 -0.01538  0.22324  0.56467
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.425305   0.091505   4.648 3.96e-05 ***
## X           -0.001666   0.001782  -0.935    0.356
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2755 on 38 degrees of freedom
## Multiple R-squared:  0.02247,    Adjusted R-squared:  -0.003259
## F-statistic: 0.8733 on 1 and 38 DF,  p-value: 0.3559
```

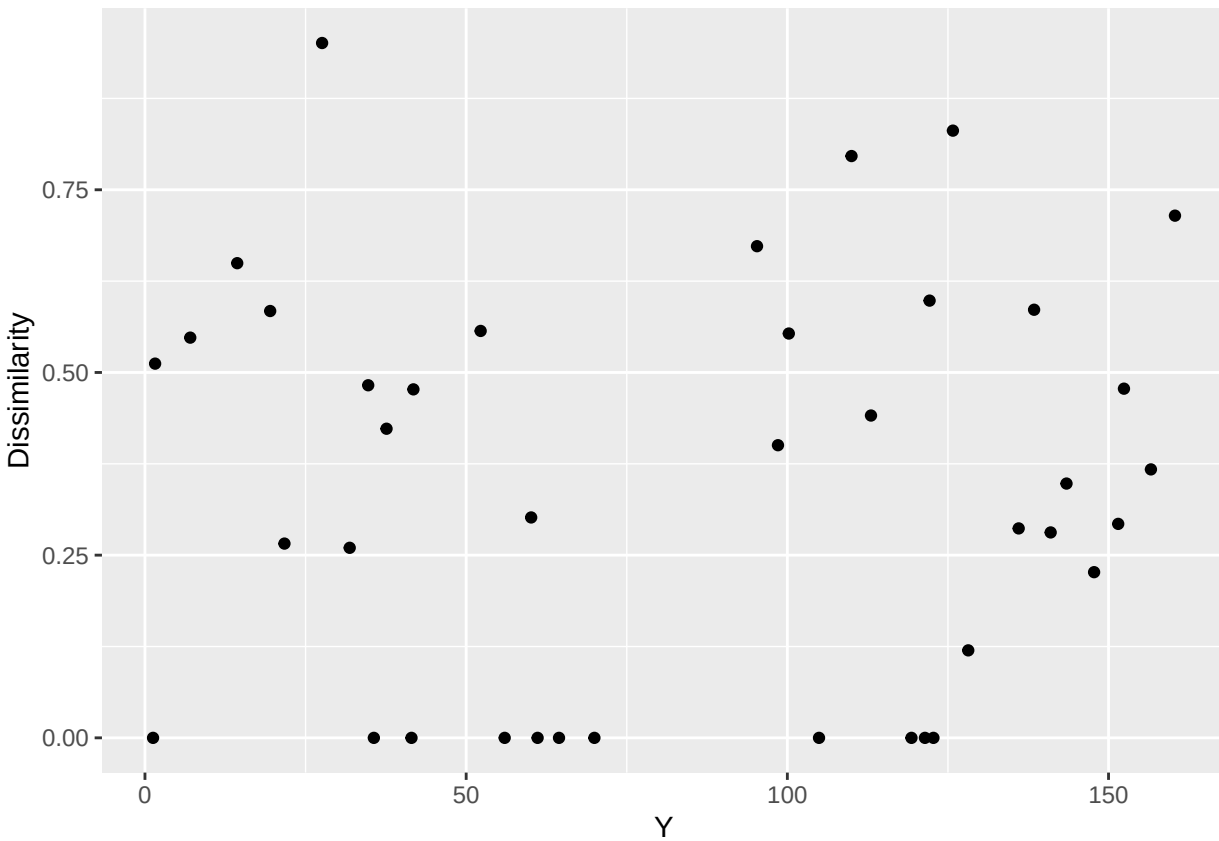
```r
ggplot(aes(x = X, y = Dissimilarity), data = GuNi_Diss)+
  geom_point()+
  labs(x = "X-Coordinate", y = "Compositional Dissimilarity", title = "Guadalupe-Nipomo (South)")
```




```r
summary(lm(Dissimilarity ~ Y, data = GuNi_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ Y, data = GuNi_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.35305 -0.34443  0.00253  0.20486  0.60504
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 3.437e-01  8.599e-02   3.997 0.000285 ***
## Y           7.634e-05  8.767e-04   0.087 0.931072
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2787 on 38 degrees of freedom
## Multiple R-squared:  0.0001995,  Adjusted R-squared:  -0.02611
## F-statistic: 0.007581 on 1 and 38 DF,  p-value: 0.9311
```

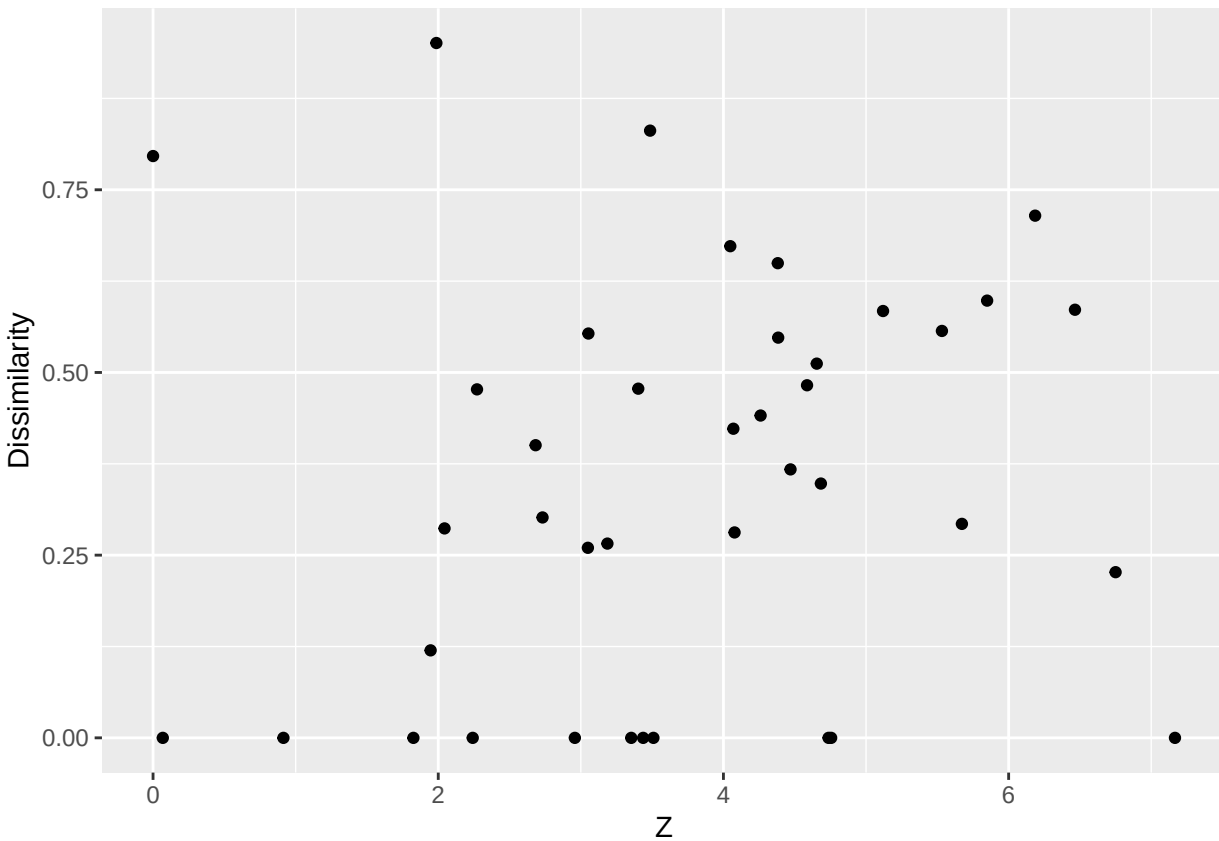
```r
ggplot(aes(x = Y, y = Dissimilarity), data = GuNi_Diss)+
  geom_point()
```




```r
summary(lm(Dissimilarity ~ Z, data = GuNi_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ Z, data = GuNi_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.42947 -0.26950 -0.01166  0.17519  0.64167
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.26300    0.10740   2.449   0.0191 *
## Z            0.02323    0.02617   0.888   0.3804
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2758 on 38 degrees of freedom
## Multiple R-squared:  0.02031,    Adjusted R-squared:  -0.005474
## F-statistic: 0.7877 on 1 and 38 DF,  p-value: 0.3804
```

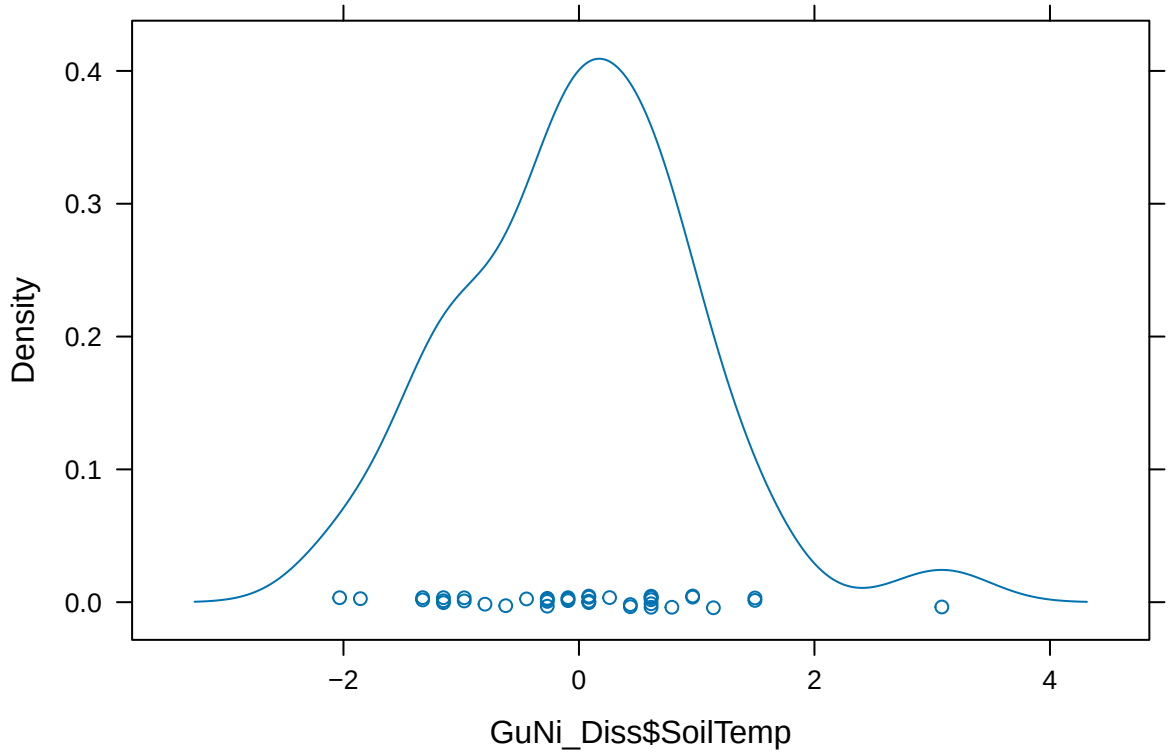
```r
ggplot(aes(x = Z, y = Dissimilarity), data = GuNi_Diss)+
  geom_point()
```




```r
summary(lm(Dissimilarity ~ X+Y+Z,  data = GuNi_Diss))
```

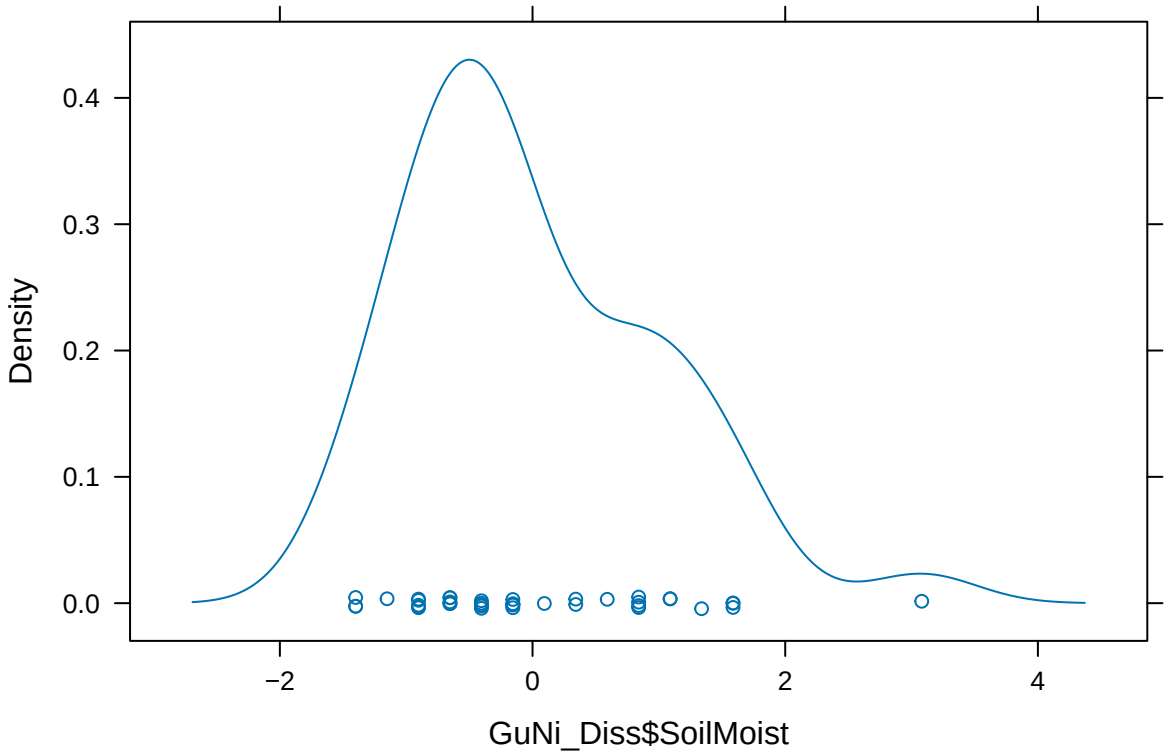
```
##
## Call:
## lm(formula = Dissimilarity ~ X + Y + Z, data = GuNi_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.41244 -0.23393 -0.01105  0.18109  0.59215
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.3489149  0.1299594   2.685   0.0109 *
## X           -0.0042107  0.0022277  -1.890   0.0668 .
## Y           -0.0004334  0.0008847  -0.490   0.6272
## Z            0.0607428  0.0325442   1.866   0.0701 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2703 on 36 degrees of freedom
## Multiple R-squared:  0.1088, Adjusted R-squared:  0.03449
## F-statistic: 1.464 on 3 and 36 DF,  p-value: 0.2405
```

```r
densityplot(GuNi_Diss$SoilTemp)
```

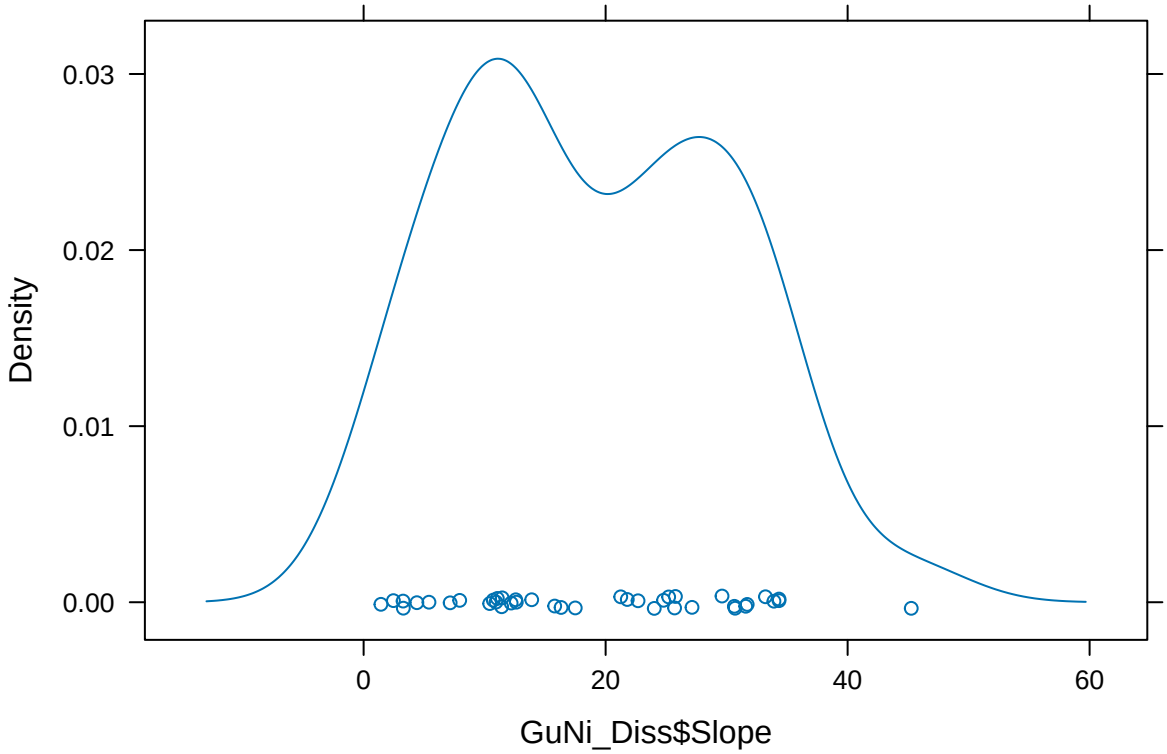


Environmental

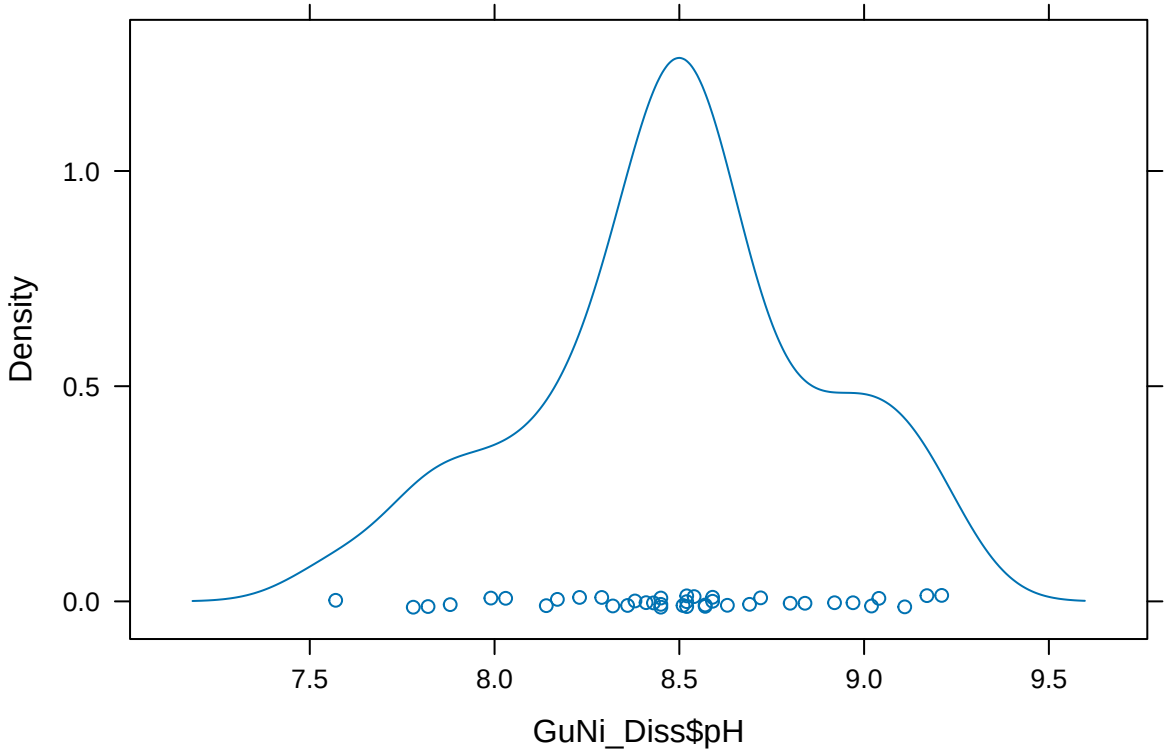
```
densityplot(GuNi_Diss$SoilMoist)
```



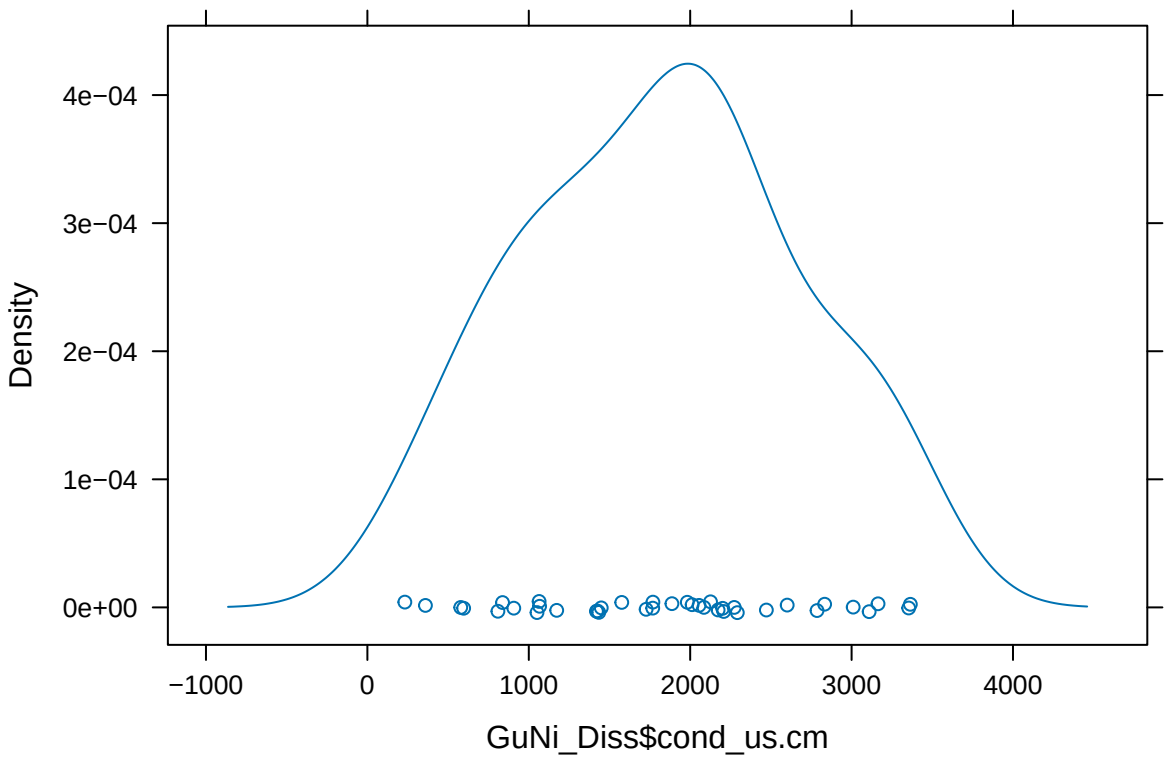

```
densityplot(GuNi_Diss$Slope)
```



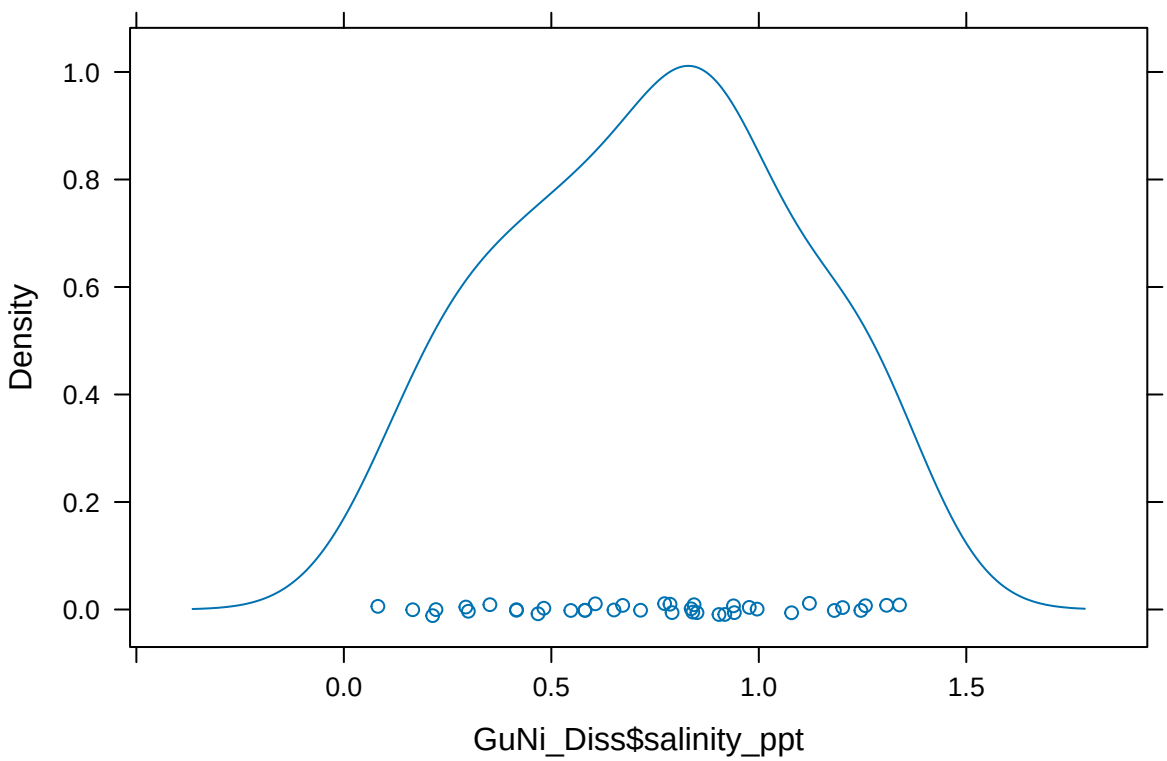

```
densityplot(GuNi_Diss$pH)
```




```
densityplot(GuNi_Diss$cond_us.cm)
```




```
densityplot(GuNi_Diss$salinity_ppt)
```



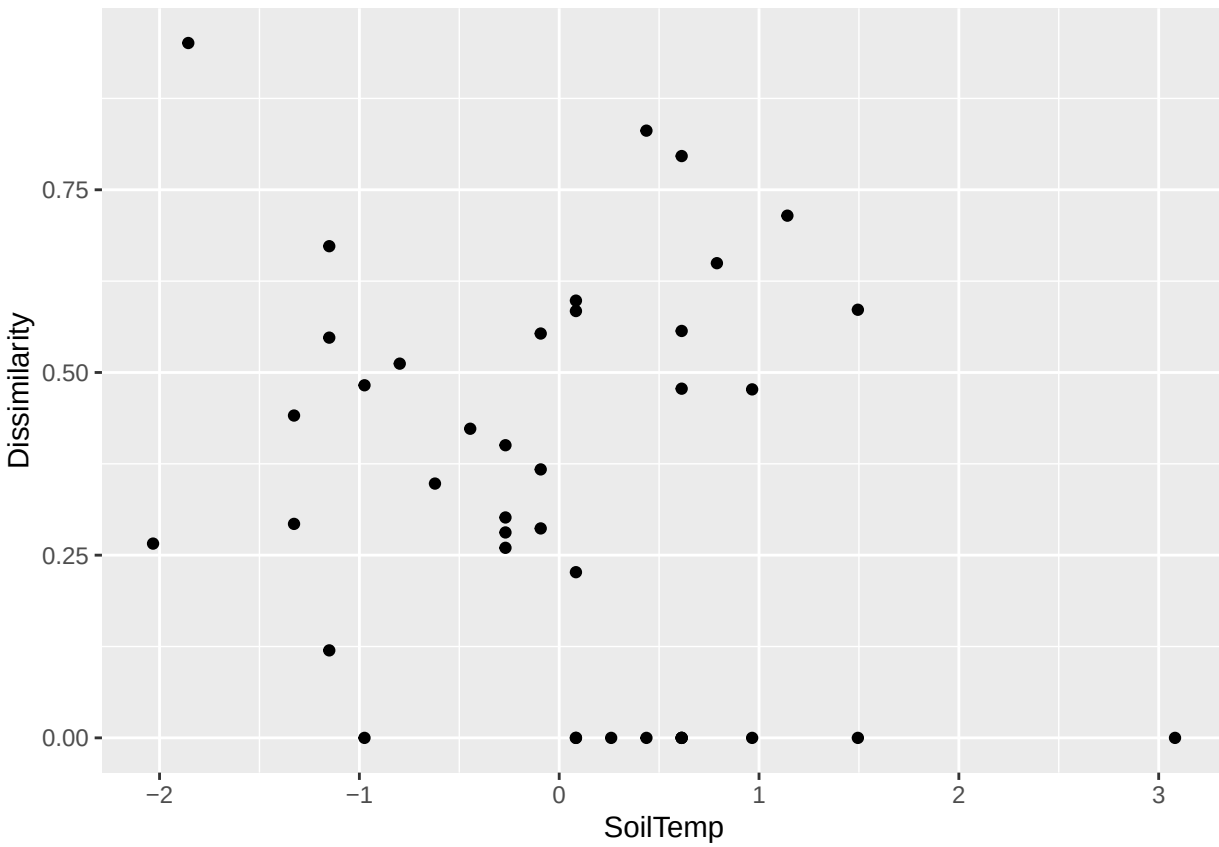

```
summary(lm(Dissimilarity ~ SoilTemp+SoilMoist+Slope+Aspect+pH+salinity_ppt, data = GuNi_Diss))

##
## Call:
```

```
## lm(formula = Dissimilarity ~ SoilTemp + SoilMoist + Slope + Aspect +
##     pH + salinity_ppt, data = GuNi_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.40481 -0.13876 -0.00477  0.14910  0.39790
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.402021   1.737859   0.231   0.8188
## SoilTemp     -0.051178   0.054305  -0.942   0.3543
## SoilMoist     0.110414   0.051923   2.126   0.0428 *
## Slope         0.008185   0.004813   1.701   0.1005
## AspectNE      0.073767   0.232469   0.317   0.7534
## AspectNW      0.098057   0.208172   0.471   0.6414
## AspectS       0.112880   0.140333   0.804   0.4282
## AspectSE      0.327631   0.195168   1.679   0.1047
## AspectSW      0.054148   0.148836   0.364   0.7188
## AspectW      -0.107410   0.153068  -0.702   0.4889
## pH           -0.014779   0.196270  -0.075   0.9405
## salinity_ppt -0.193737   0.190547  -1.017   0.3183
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2637 on 27 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:  0.3498, Adjusted R-squared:  0.08497
## F-statistic: 1.321 on 11 and 27 DF,  p-value: 0.2664
```

```r
summary(lm(Dissimilarity ~ SoilTemp, data = GuNi_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ SoilTemp, data = GuNi_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.40716 -0.27034 -0.01336  0.20803  0.50637
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.35011    0.04306   8.132 7.66e-10 ***
## SoilTemp    -0.05855    0.04360  -1.343    0.187
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2723 on 38 degrees of freedom
## Multiple R-squared:  0.0453, Adjusted R-squared:  0.02017
## F-statistic: 1.803 on 1 and 38 DF,  p-value: 0.1873
```

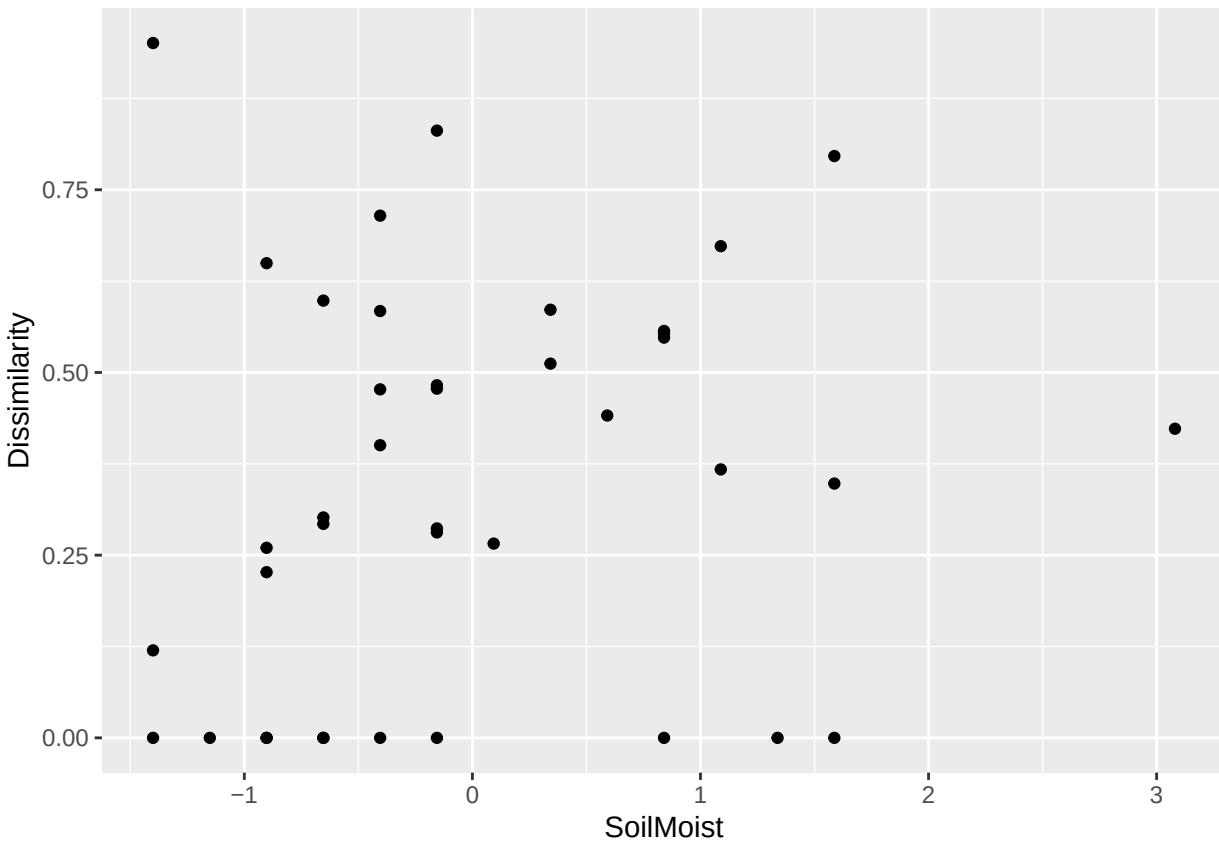
```r
ggplot(aes(x = SoilTemp, y = Dissimilarity), data = GuNi_Diss)+
  geom_point()
```




```r
summary(lm(Dissimilarity ~ SoilMoist, data = GuNi_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ SoilMoist, data = GuNi_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.42150 -0.28991 -0.02983  0.16625  0.66371
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.35011    0.04347   8.054 9.69e-10 ***
## SoilMoist    0.04499    0.04403   1.022    0.313
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2749 on 38 degrees of freedom
## Multiple R-squared:  0.02675,    Adjusted R-squared:  0.001134
## F-statistic: 1.044 on 1 and 38 DF,  p-value: 0.3133
```

```r
ggplot(aes(x = SoilMoist, y = Dissimilarity), data = GuNi_Diss)+
  geom_point()
```




```r
summary(lm(Dissimilarity ~ SoilTemp*SoilMoist, data = GuNi_Diss))
```

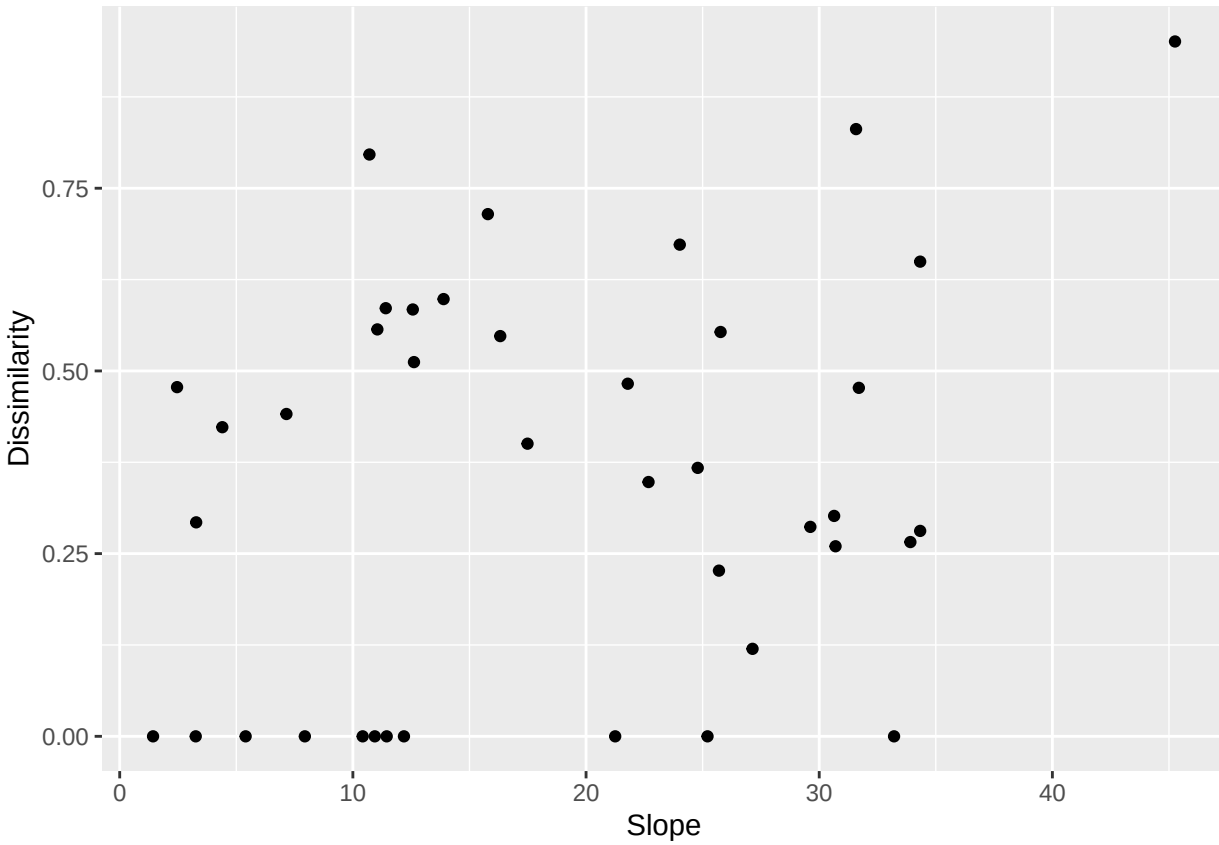
```
##
## Call:
## lm(formula = Dissimilarity ~ SoilTemp * SoilMoist, data = GuNi_Diss)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.5054 -0.1848 -0.0259  0.1926  0.4941
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)         0.36212    0.04305   8.411  5.1e-10 ***
## SoilTemp           -0.02354    0.04726  -0.498    0.621
## SoilMoist           0.05922    0.04552   1.301    0.202
## SoilTemp:SoilMoist  0.08607    0.05456   1.577    0.123
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.268 on 36 degrees of freedom
## Multiple R-squared:  0.1239, Adjusted R-squared:  0.05093
## F-statistic: 1.698 on 3 and 36 DF,  p-value: 0.1848
```

```r
summary(lm(Dissimilarity ~ Slope, data = GuNi_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ Slope, data = GuNi_Diss)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.44223 -0.25293  0.01124  0.21895  0.49872
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.228504   0.084370   2.708   0.0101 *
## Slope       0.006436   0.003856   1.669   0.1034
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.269 on 38 degrees of freedom
## Multiple R-squared:  0.06829,    Adjusted R-squared:  0.04377
## F-statistic: 2.785 on 1 and 38 DF,  p-value: 0.1034
```

```r
ggplot(aes(x = Slope, y = Dissimilarity), data = GuNi_Diss)+
  geom_point()
```

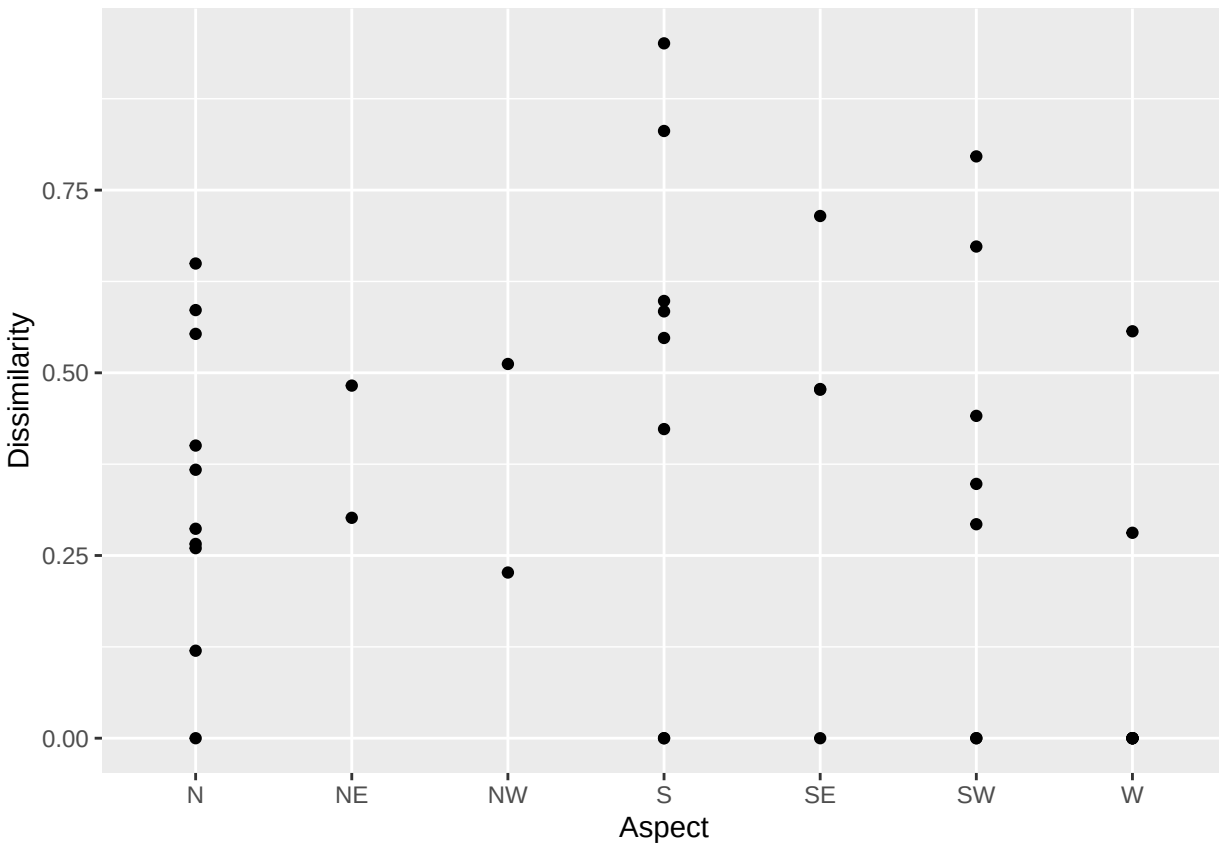



```r
summary(lm(Dissimilarity ~ Aspect, data = GuNi_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ Aspect, data = GuNi_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.49185 -0.11970  0.00103  0.14736  0.45897
```

```
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.34887    0.08520   4.095 0.000257 ***
## AspectNE     0.04317    0.20869   0.207 0.837375
## AspectNW     0.02053    0.20869   0.098 0.922243
## AspectS      0.14298    0.12780   1.119 0.271285
## AspectSE     0.06844    0.15939   0.429 0.670413
## AspectSW     0.01553    0.13277   0.117 0.907605
## AspectW     -0.22917    0.13277  -1.726 0.093691 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2694 on 33 degrees of freedom
## Multiple R-squared:  0.1884, Adjusted R-squared:  0.04086
## F-statistic: 1.277 on 6 and 33 DF,  p-value: 0.2947
```

```r
ggplot(aes(x = Aspect, y = Dissimilarity), data = GuNi_Diss)+
  geom_point()
```



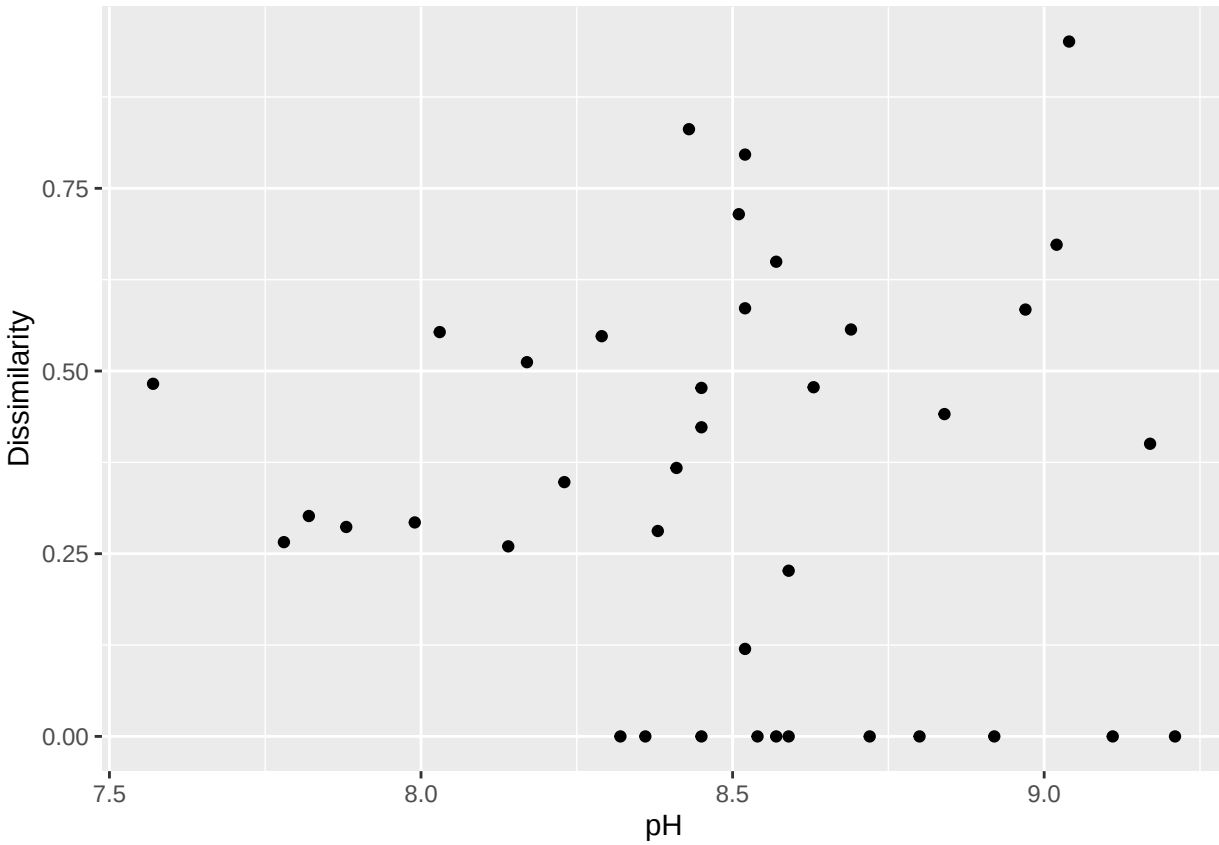

```r
summary(lm(Dissimilarity ~ pH, data = GuNi_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ pH, data = GuNi_Diss)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
```

```
## -0.3480 -0.3271 -0.0023  0.1984  0.6207
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.55505    0.98803   0.562    0.578
## pH          -0.02488    0.11623  -0.214    0.832
##
## Residual standard error: 0.2792 on 37 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:  0.001237,   Adjusted R-squared:  -0.02576
## F-statistic: 0.04583 on 1 and 37 DF,  p-value: 0.8317
```

```
ggplot(aes(x = pH, y = Dissimilarity), data = GuNi_Diss)+
  geom_point()
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



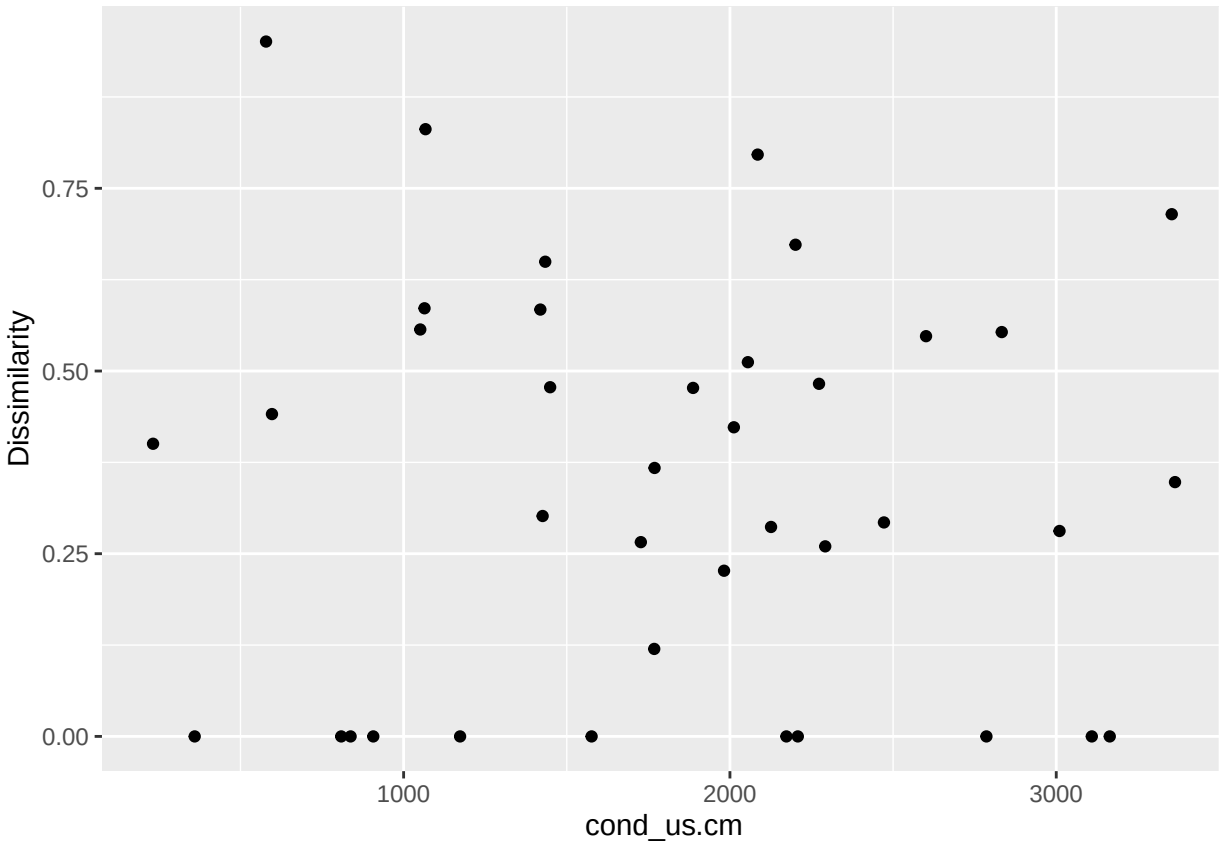

```
summary(lm(Dissimilarity ~ cond_us.cm, data = GuNi_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ cond_us.cm, data = GuNi_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.36843 -0.32169  0.02264  0.20848  0.58607
##
```

```
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.745e-01  1.076e-01   3.479   0.0013 **
## cond_us.cm  -1.683e-05  5.361e-05  -0.314   0.7553
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.279 on 37 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:  0.002656,   Adjusted R-squared:  -0.0243
## F-statistic: 0.09855 on 1 and 37 DF,  p-value: 0.7553
```

```r
ggplot(aes(x = cond_us.cm, y = Dissimilarity), data = GuNi_Diss)+
  geom_point()
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



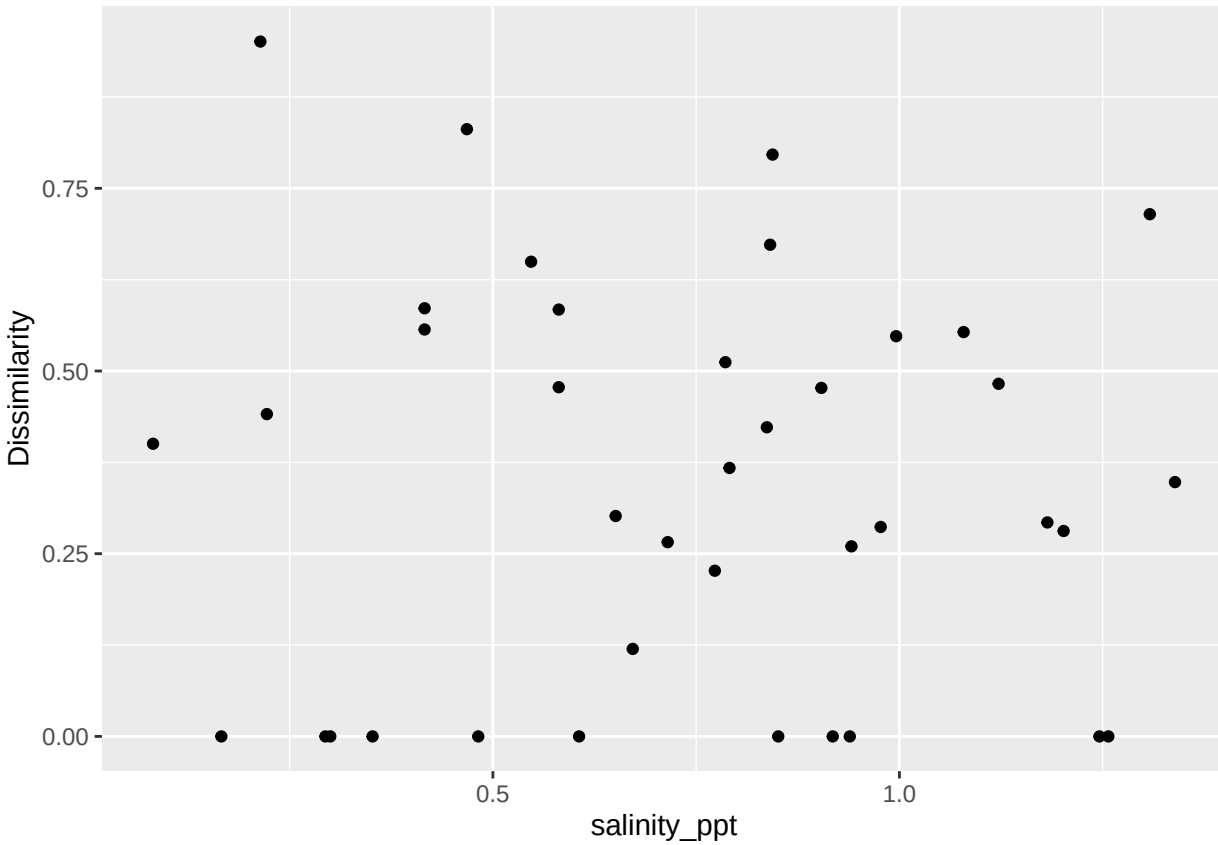

```r
summary(lm(Dissimilarity ~ salinity_ppt, data = GuNi_Diss))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ salinity_ppt, data = GuNi_Diss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.35799 -0.33109  0.01904  0.20762  0.59402
##
```

```
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.36211    0.10728   3.375  0.00174 **
## salinity_ppt -0.02478    0.13160  -0.188  0.85165
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2793 on 37 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:  0.0009576,  Adjusted R-squared:  -0.02604
## F-statistic: 0.03547 on 1 and 37 DF,  p-value: 0.8517
```

```
ggplot(aes(x = salinity_ppt, y = Dissimilarity), data = GuNi_Diss)+
  geom_point()
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```




```
GuNi.mod.1<- lm(Dissimilarity ~ SoilTemp + SoilMoist + Slope + Aspect + X + Z + pH, data = subset(GuNi_
summary(GuNi.mod.1)
```

Negligible effects

```
##
## Call:
## lm(formula = Dissimilarity ~ SoilTemp + SoilMoist + Slope + Aspect +
##     X + Z + pH, data = subset(GuNi_Diss, is.na(pH) == FALSE))
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.37592 -0.14293 -0.02795  0.14086  0.42223
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.085642   1.285948  -0.844    0.406
## SoilTemp    -0.014259   0.061869  -0.230    0.820
## SoilMoist    0.074680   0.051798   1.442    0.161
## Slope        0.006506   0.004948   1.315    0.200
## AspectNE     0.172607   0.230600   0.749    0.461
## AspectNW     0.119512   0.220173   0.543    0.592
## AspectS      0.038897   0.146521   0.265    0.793
## AspectSE     0.121380   0.212616   0.571    0.573
## AspectSW    -0.079264   0.162174  -0.489    0.629
## AspectW     -0.222775   0.166855  -1.335    0.193
## X           -0.003959   0.003061  -1.293    0.207
## Z            0.035268   0.035845   0.984    0.334
## pH           0.161223   0.152044   1.060    0.299
##
## Residual standard error: 0.2651 on 26 degrees of freedom
## Multiple R-squared:  0.3673, Adjusted R-squared:  0.07521
## F-statistic: 1.258 on 12 and 26 DF,  p-value: 0.2998
```

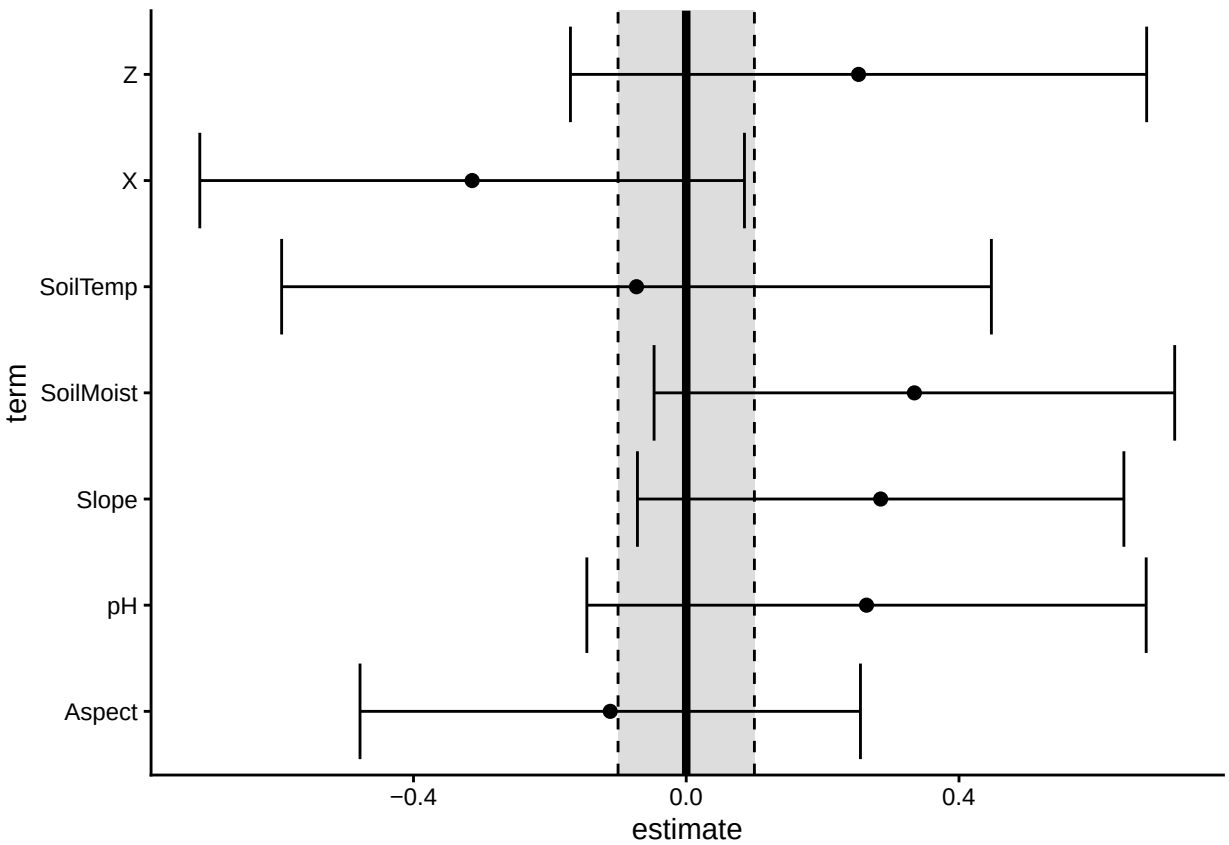
```
GuNi.Null.1 <- lm(Dissimilarity ~ 1, data = subset(GuNi_Diss, is.na(pH) == FALSE))
anova(GuNi.mod.1, GuNi.Null.1)
```

```
## Analysis of Variance Table
##
## Model 1: Dissimilarity ~ SoilTemp + SoilMoist + Slope + Aspect + X + Z +
##     pH
## Model 2: Dissimilarity ~ 1
##   Res.Df    RSS  Df Sum of Sq      F Pr(>F)
## 1     26 1.8275
## 2     38 2.8882 -12   -1.0607 1.2575 0.2998
```

```
GuNi.pre.1<- predictions(GuNi.mod.1)

GuNi.cmp<- avg_comparisons(GuNi.mod.1,  variables =  list(SoilTemp = "minmax", SoilMoist = "minmax", Sl


GuNi.cmp<-GuNi.cmp %>%
  group_by(term) %>%
  summarise_all(.funs = mean)
```

```
## Warning: There were 7 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 6 remaining warnings.
```

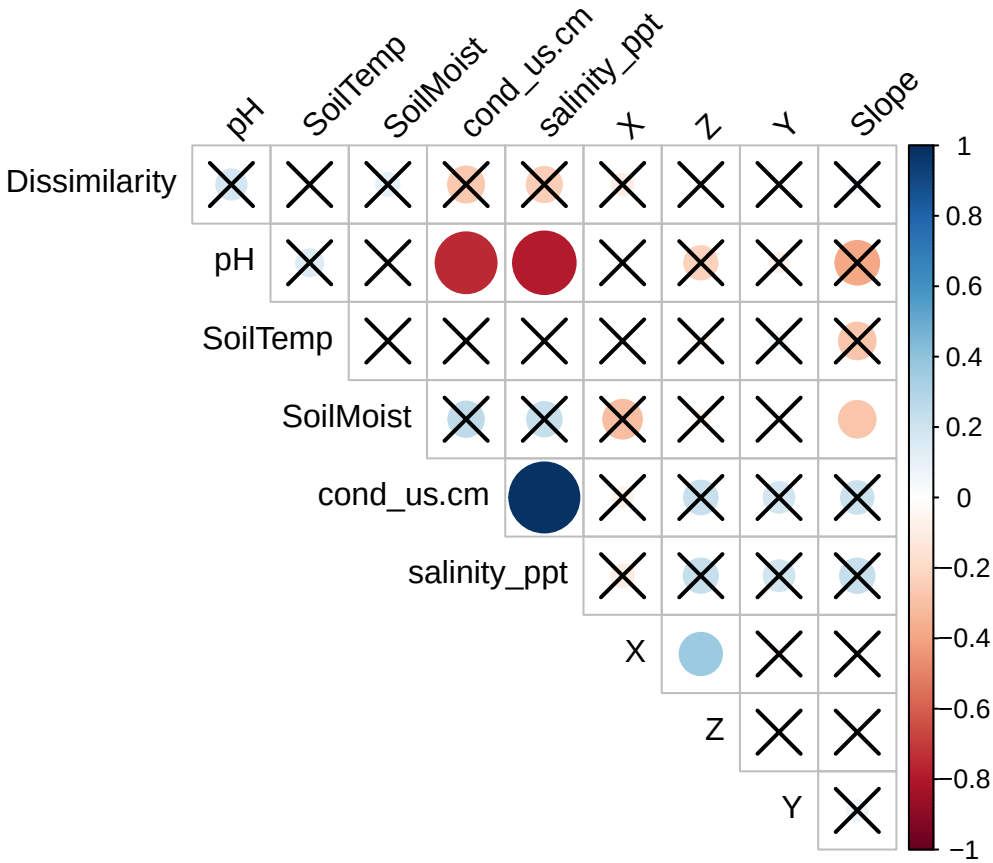
```
ggplot(GuNi.cmp, aes(x = estimate,
                     xmin = conf.low,
                     xmax = conf.high,
                     y = term)) +
     annotate("rect", xmin = -0.1, xmax = 0.1, ymin = -Inf, ymax = Inf, alpha = .2)+
  geom_vline(xintercept = c(-0.1, .1), linetype = "dashed") +
  geom_vline(xintercept = 0, linetype = "solid", linewidth = 1.5)+
  geom_errorbarh() +
  geom_point(size = 2) +
  theme_classic()
```



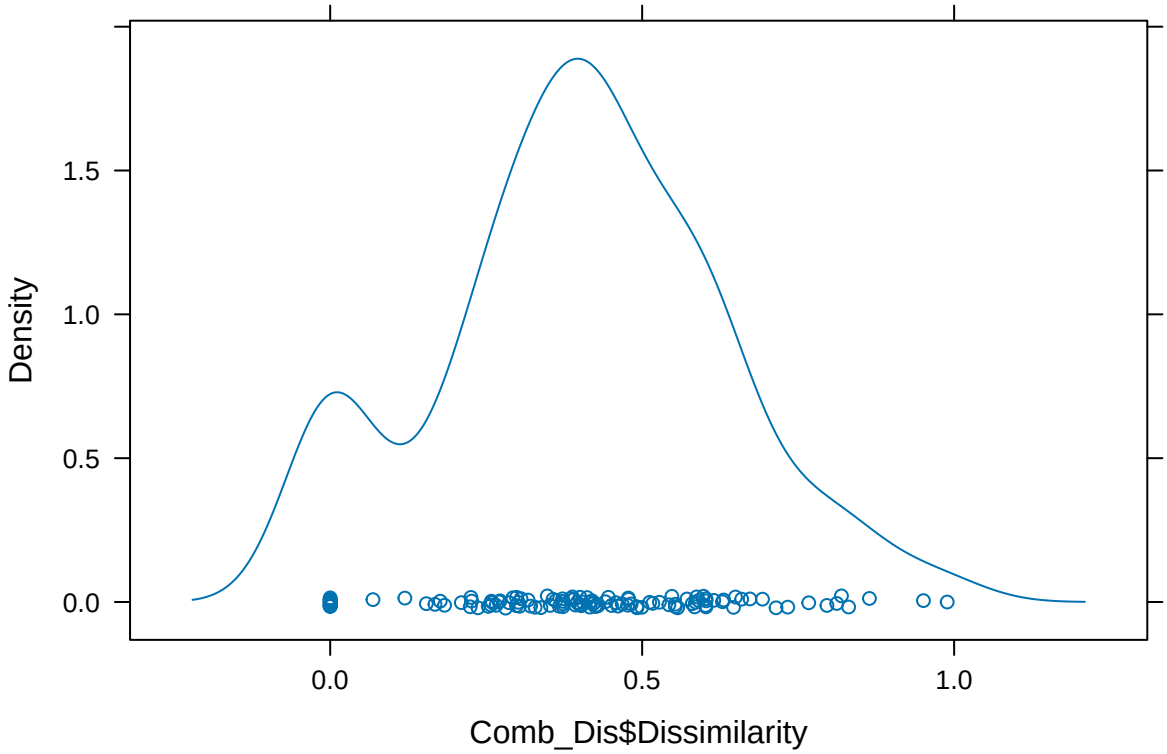
Mixed

```
Comb_Dis<-rbind(TeMi_Diss, Zmud_Diss, GuNi_Diss)

Comb_pcor<- Comb_Dis %>%
  select(-c(Site, Plot, Aspect, Date, Collectors, dry_weight, Notes))%>%
  as.matrix()%>%
  cor.mtest(use =  "complete")


Comb_Dis %>%
  select(-c(Site, Plot, Aspect, Date, Collectors, dry_weight, Notes))%>%
  as.matrix()%>%
  cor(use =  "complete")%>%
```

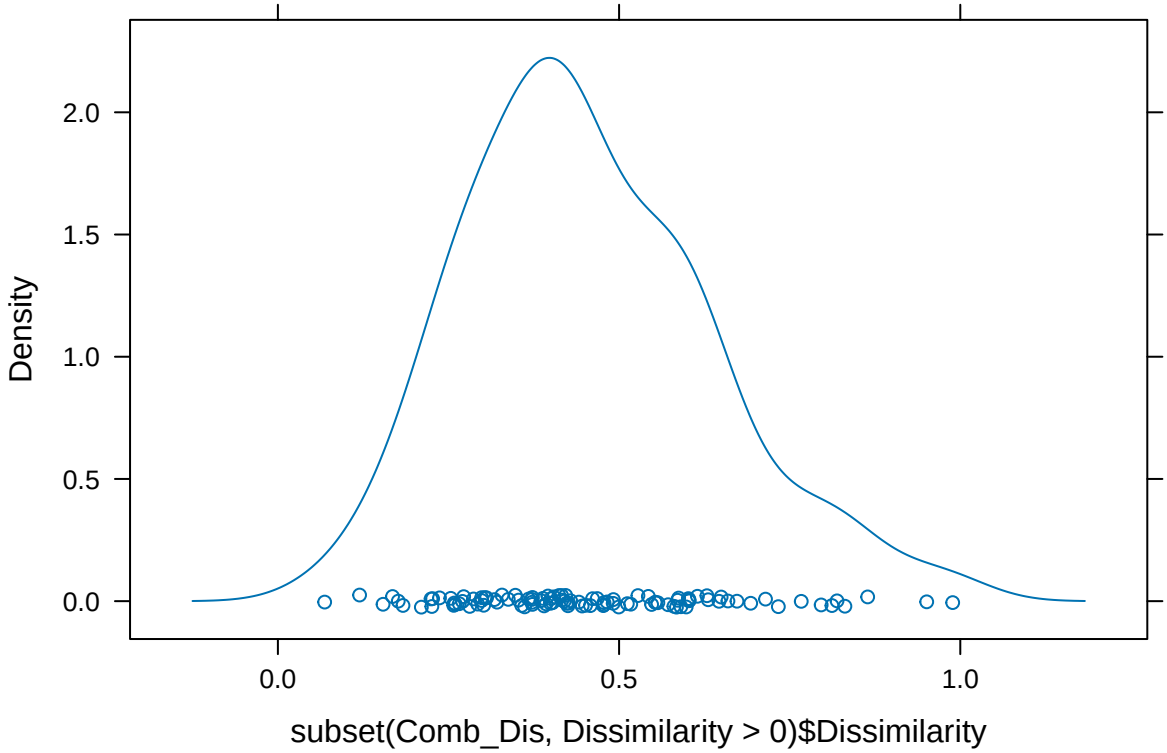
```r
corrplot( type = "upper", order = "hclust",  tl.col = "black", tl.srt = 45, p.mat = GuNi_pcor$p, diag
```




```r
densityplot(Comb_Dis$Dissimilarity)
```




```
densityplot(subset(Comb_Dis, Dissimilarity >0)$Dissimilarity)
```

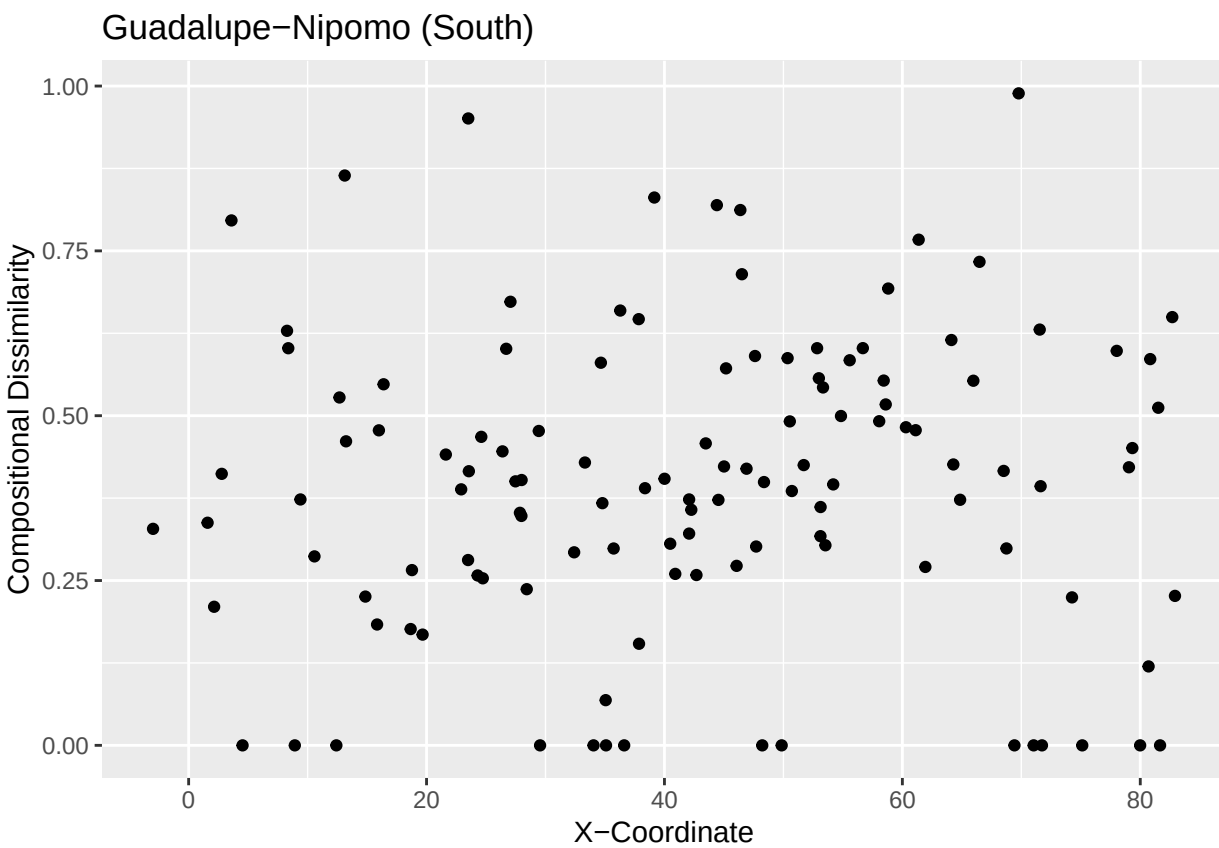



```
summary(lm(Dissimilarity ~ X, data = Comb_Dis)) # Distance from waterline not sig
```

```
##
## Call:
```

```
## lm(formula = Dissimilarity ~ X, data = Comb_Dis)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.41530 -0.13080  0.00722  0.15282  0.57889
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.3788594  0.0450447   8.411 9.89e-14 ***
## X           0.0004462  0.0009386   0.475    0.635
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2279 on 120 degrees of freedom
## Multiple R-squared:  0.00188,    Adjusted R-squared:  -0.006438
## F-statistic: 0.226 on 1 and 120 DF,  p-value: 0.6354
```

```
ggplot(aes(x = X, y = Dissimilarity), data = Comb_Dis)+
  geom_point()+
  labs(x = "X-Coordinate", y = "Compositional Dissimilarity", title = "Guadalupe-Nipomo (South)")
```

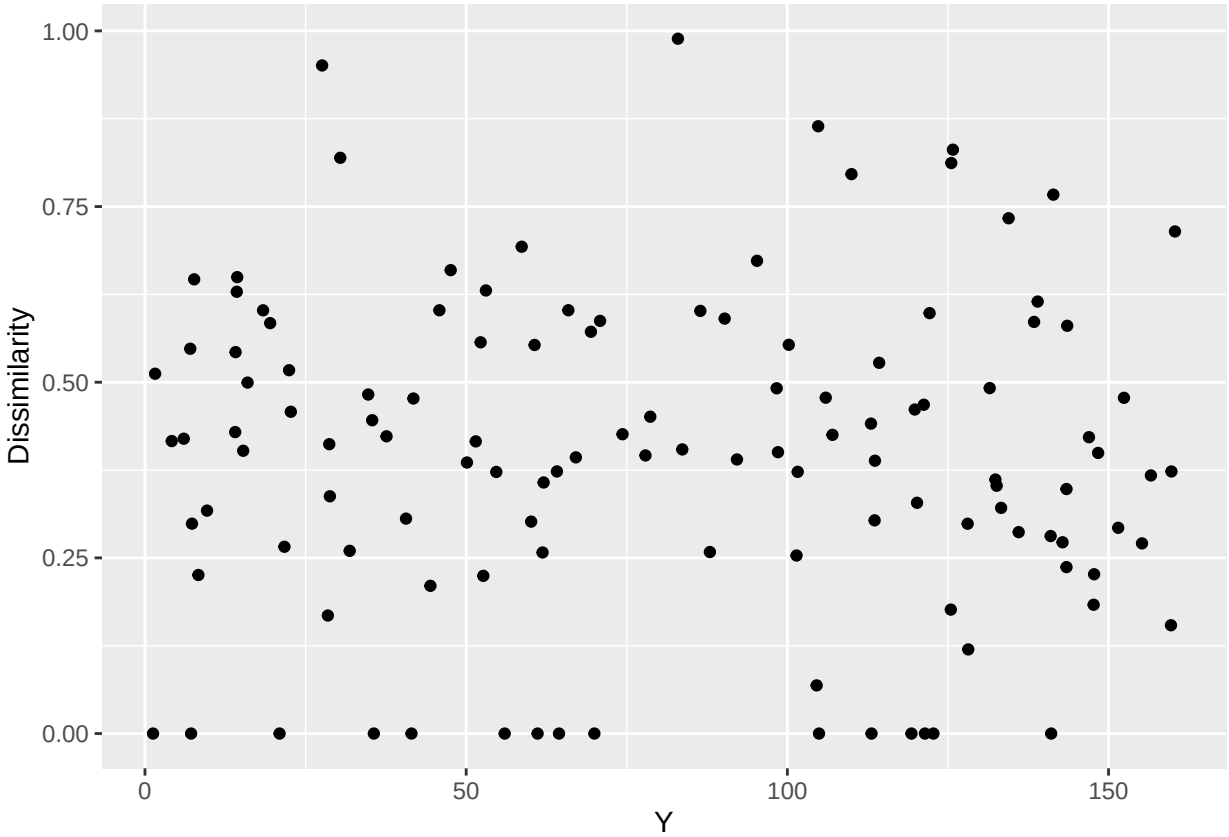



```
summary(lm(Dissimilarity ~ Y, data = Comb_Dis))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ Y, data = Comb_Dis)
##
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max
## -0.41155 -0.13251  0.00509  0.15344  0.59137
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.4117684  0.0401077  10.267   <2e-16 ***
## Y           -0.0001717  0.0004257  -0.403    0.687
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2279 on 120 degrees of freedom
## Multiple R-squared:  0.001354,   Adjusted R-squared:  -0.006968
## F-statistic: 0.1627 on 1 and 120 DF,  p-value: 0.6874
```

```r
ggplot(aes(x = Y, y = Dissimilarity), data = Comb_Dis)+
  geom_point()
```

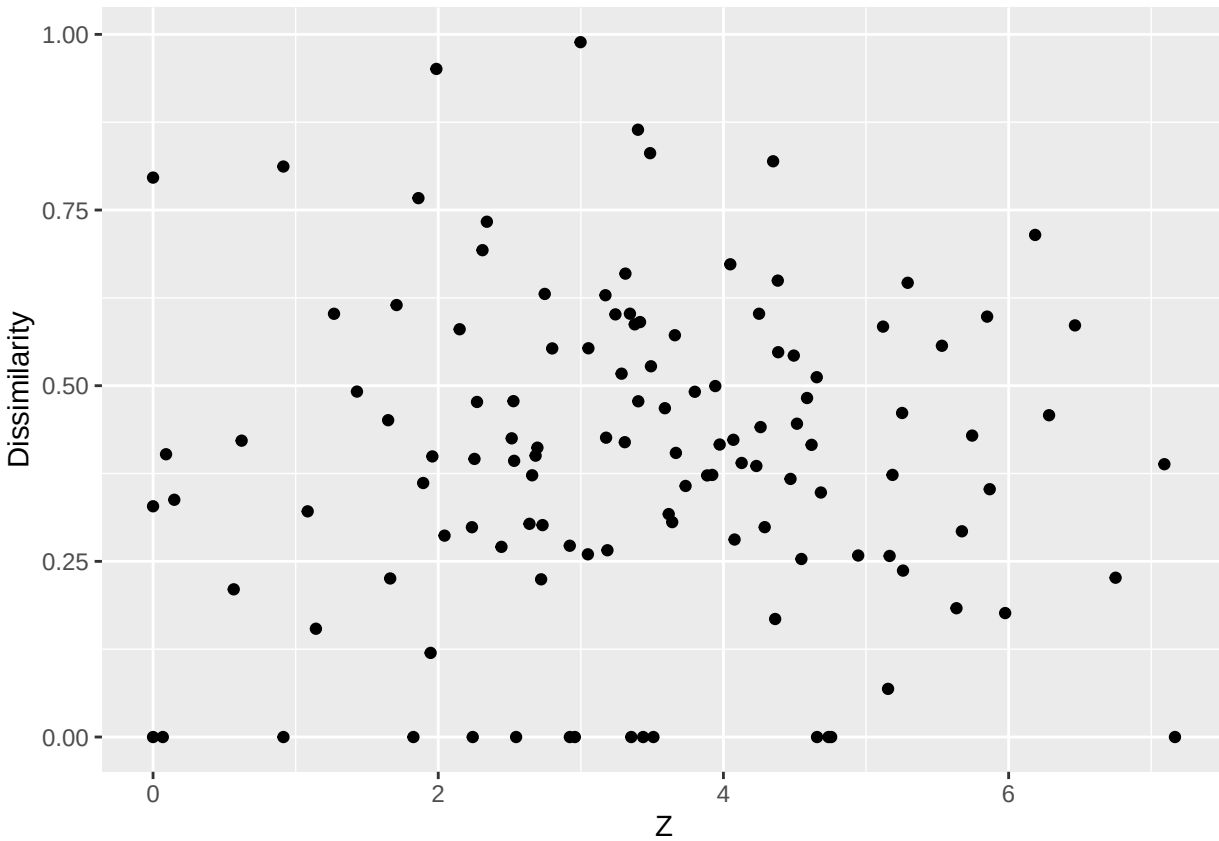



```r
summary(lm(Dissimilarity ~ Z, data = Comb_Dis))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ Z, data = Comb_Dis)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.39937 -0.13071  0.00435  0.15548  0.59116
##
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.3965541  0.0486737   8.147 4.04e-13 ***
## Z           0.0003933  0.0129198   0.030    0.976
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2281 on 120 degrees of freedom
## Multiple R-squared:  7.722e-06,  Adjusted R-squared:  -0.008326
## F-statistic: 0.0009266 on 1 and 120 DF,  p-value: 0.9758
```

```
ggplot(aes(x = Z, y = Dissimilarity), data = Comb_Dis)+
  geom_point()
```




```
summary(lm(Dissimilarity ~ X+Y+Z,  data = Comb_Dis))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ X + Y + Z, data = Comb_Dis)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.42111 -0.12966  0.00747  0.15092  0.57887
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.3951627  0.0696989   5.670 1.03e-07 ***
## X            0.0004479  0.0009601   0.467    0.642
## Y           -0.0001680  0.0004292  -0.391    0.696
```

```
## Z           -0.0008227  0.0132063  -0.062    0.950
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2296 on 118 degrees of freedom
## Multiple R-squared:  0.003195,   Adjusted R-squared:  -0.02215
## F-statistic: 0.1261 on 3 and 118 DF,  p-value: 0.9445
```

```r
summary(lm(Dissimilarity ~ SoilTemp+SoilMoist+Slope+Aspect+pH+salinity_ppt, data = Comb_Dis))
```

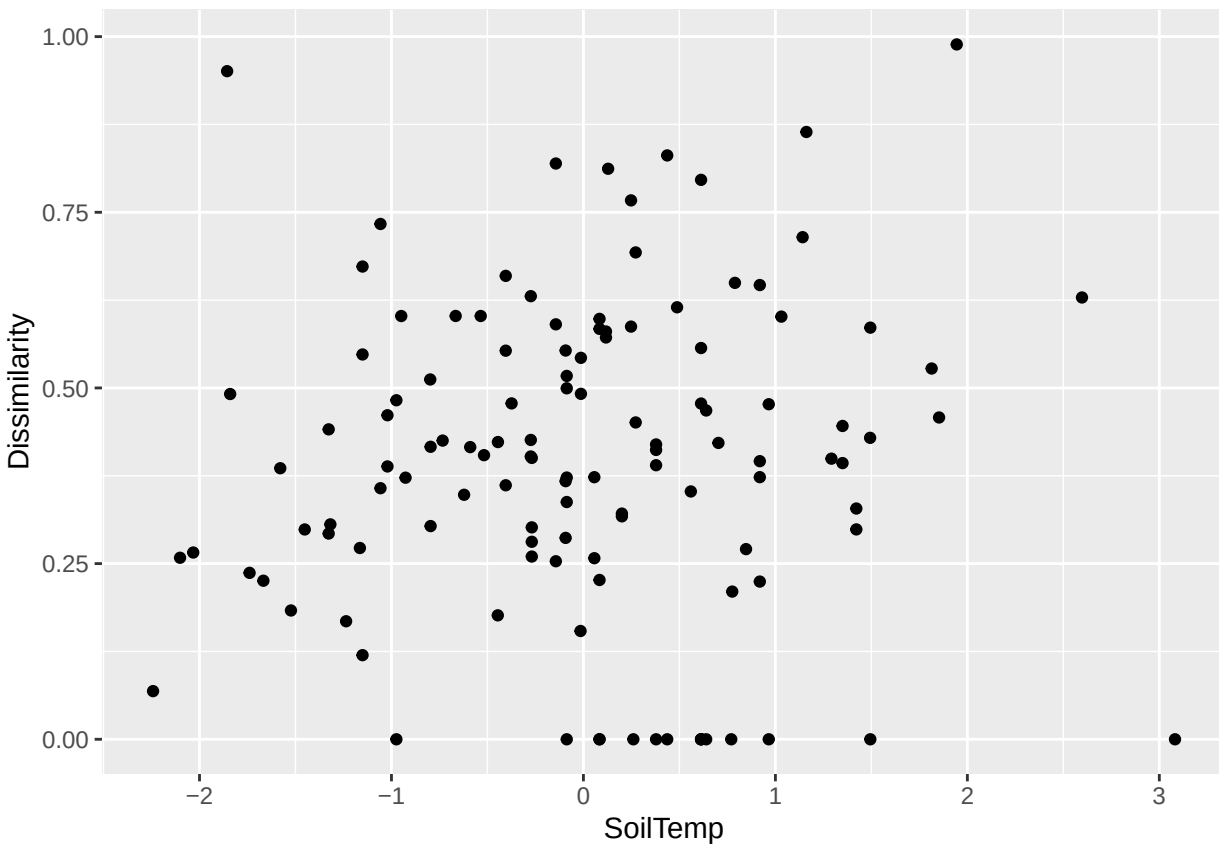
```
##
## Call:
## lm(formula = Dissimilarity ~ SoilTemp + SoilMoist + Slope + Aspect +
##     pH + salinity_ppt, data = Comb_Dis)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.48073 -0.10194 -0.01187  0.16941  0.57506
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.677326   1.062319   0.638   0.5260
## SoilTemp      0.001262   0.031186   0.040   0.9678
## SoilMoist     0.068672   0.033323   2.061   0.0433 *
## Slope         0.004704   0.003292   1.429   0.1578
## AspectN       0.051532   0.129092   0.399   0.6911
## AspectNE      0.089855   0.145730   0.617   0.5397
## AspectNW      0.075717   0.172209   0.440   0.6616
## AspectS       0.060224   0.132487   0.455   0.6509
## AspectSE      0.227947   0.151545   1.504   0.1374
## AspectSW      0.091828   0.143077   0.642   0.5233
## AspectW      -0.058450   0.139659  -0.419   0.6769
## pH           -0.027614   0.111065  -0.249   0.8044
## salinity_ppt -0.299929   0.146806  -2.043   0.0451 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.248 on 65 degrees of freedom
##   (44 observations deleted due to missingness)
## Multiple R-squared:  0.2067, Adjusted R-squared:  0.0603
## F-statistic: 1.412 on 12 and 65 DF,  p-value: 0.1835
```

```r
summary(lm(Dissimilarity ~ SoilTemp, data = Comb_Dis))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ SoilTemp, data = Comb_Dis)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.4359 -0.1161  0.0075  0.1591  0.5756
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.39800    0.02062  19.300   <2e-16 ***
```

```
## SoilTemp     0.01229    0.02089   0.588    0.557
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2278 on 120 degrees of freedom
## Multiple R-squared:  0.002877,   Adjusted R-squared:  -0.005432
## F-statistic: 0.3463 on 1 and 120 DF,  p-value: 0.5573
```

```
ggplot(aes(x = SoilTemp, y = Dissimilarity), data = Comb_Dis)+
  geom_point()
```

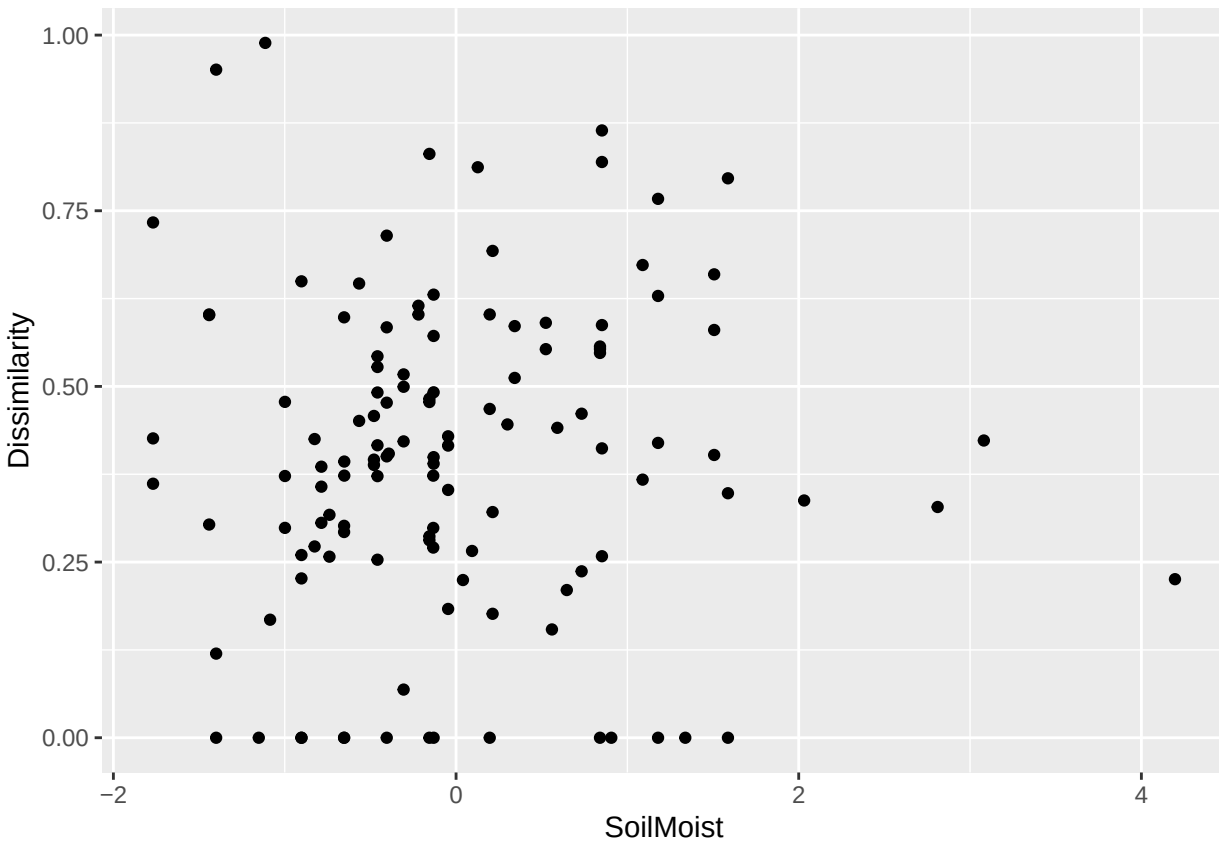



```
summary(lm(Dissimilarity ~ SoilMoist, data = Comb_Dis))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ SoilMoist, data = Comb_Dis)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.40663 -0.13114  0.00339  0.15184  0.59700
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.397963   0.020646   19.27   <2e-16 ***
## SoilMoist   0.005459   0.021017    0.26    0.796
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 0.228 on 120 degrees of freedom
## Multiple R-squared:  0.0005619,  Adjusted R-squared:  -0.007767
## F-statistic: 0.06746 on 1 and 120 DF,  p-value: 0.7955
```

```r
ggplot(aes(x = SoilMoist, y = Dissimilarity), data = Comb_Dis)+
  geom_point()
```




```r
summary(lm(Dissimilarity ~ SoilTemp*SoilMoist, data = Comb_Dis))
```

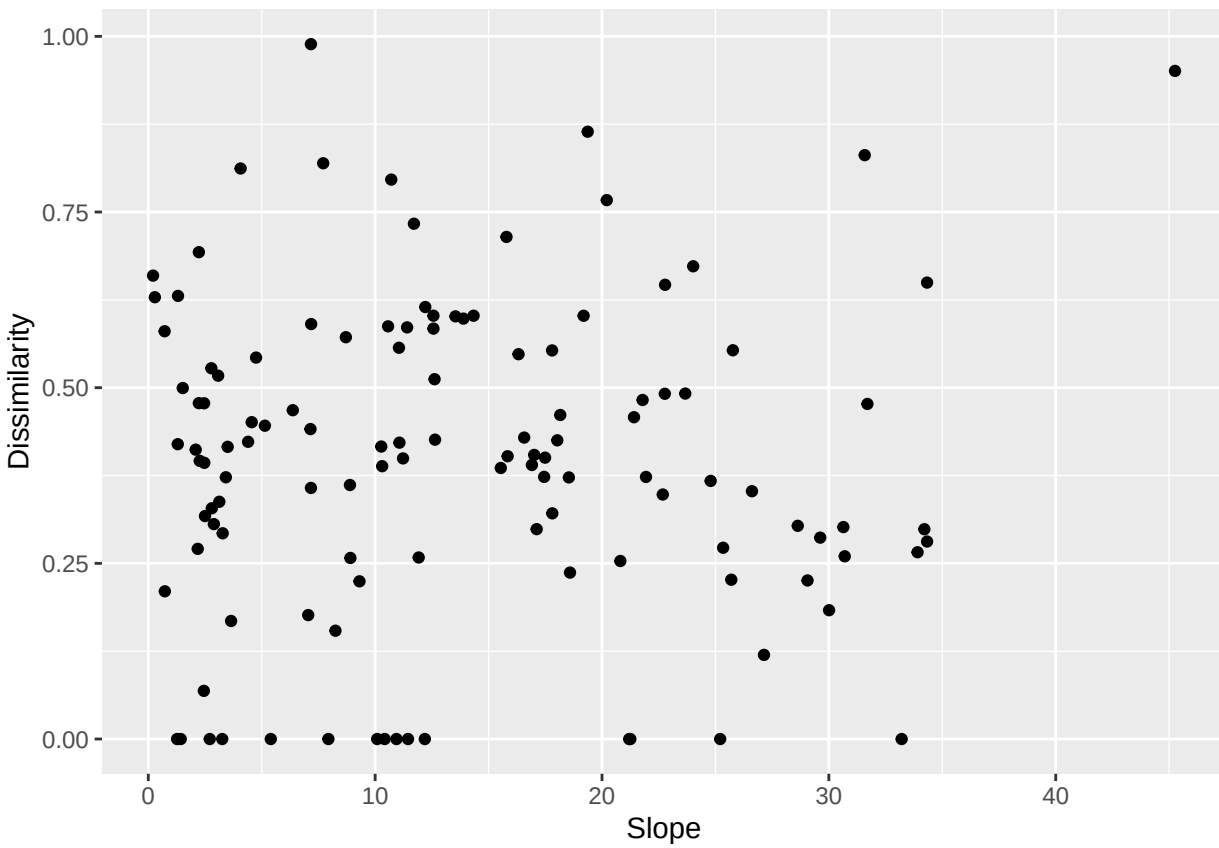
```
##
## Call:
## lm(formula = Dissimilarity ~ SoilTemp * SoilMoist, data = Comb_Dis)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.44357 -0.11279  0.00071  0.14782  0.63461
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)         0.39864    0.02064  19.316   <2e-16 ***
## SoilTemp            0.01417    0.02094   0.677    0.500
## SoilMoist           0.01278    0.02166   0.590    0.556
## SoilTemp:SoilMoist  0.02664    0.01986   1.342    0.182
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2279 on 118 degrees of freedom
## Multiple R-squared:  0.01846,    Adjusted R-squared:  -0.00649
```

```
## F-statistic: 0.7399 on 3 and 118 DF,  p-value: 0.5303
```

```r
summary(lm(Dissimilarity ~ Slope, data = Comb_Dis))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ Slope, data = Comb_Dis)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.40913 -0.13613  0.00395  0.15216  0.59481
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.3899360  0.0350171  11.136   <2e-16 ***
## Slope       0.0005781  0.0020543   0.281    0.779
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.228 on 120 degrees of freedom
## Multiple R-squared:  0.0006595,  Adjusted R-squared:  -0.007668
## F-statistic: 0.07919 on 1 and 120 DF,  p-value: 0.7789
```

```r
ggplot(aes(x = Slope, y = Dissimilarity), data = Comb_Dis)+
  geom_point()
```

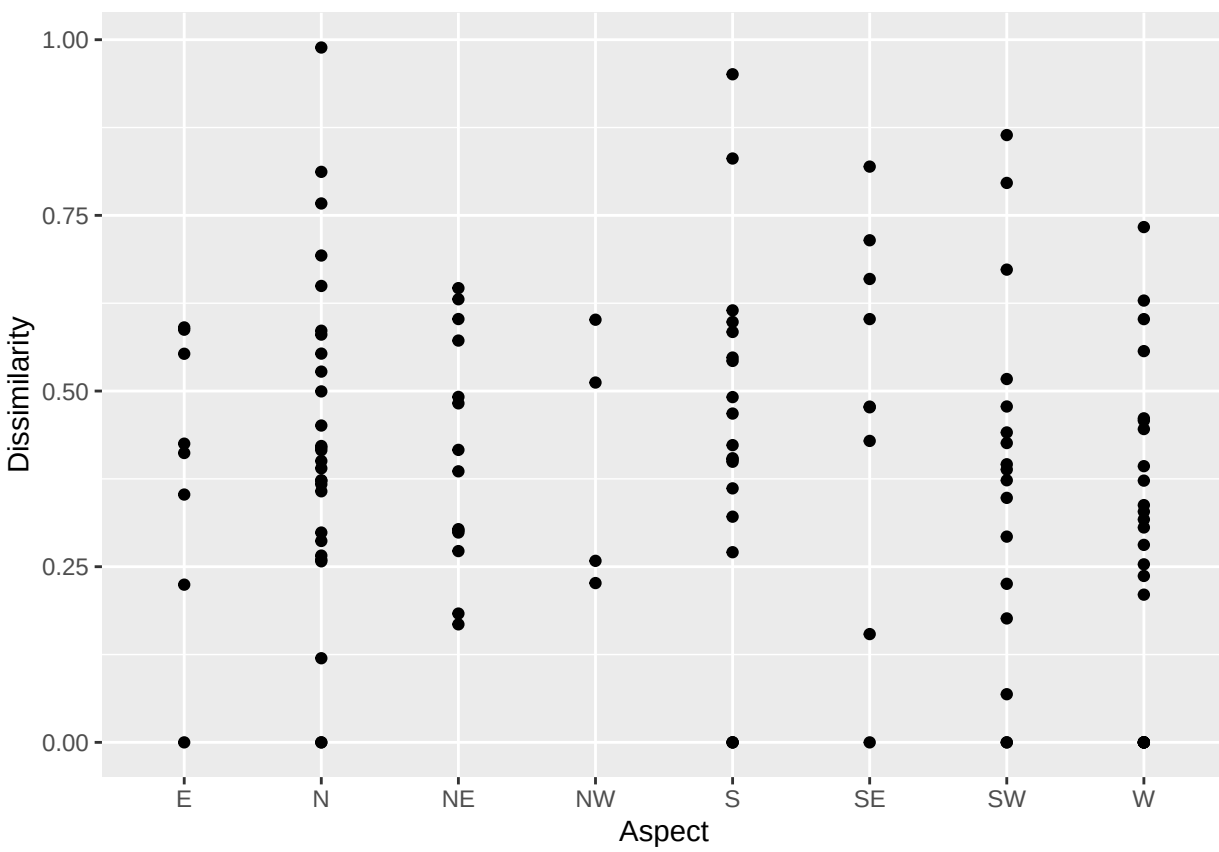



```r
summary(lm(Dissimilarity ~ Aspect, data = Comb_Dis))
```

```
##
```

```
## Call:
## lm(formula = Dissimilarity ~ Aspect, data = Comb_Dis)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.48152 -0.13762 -0.00419  0.15086  0.55625
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.393118   0.080432   4.888 3.37e-06 ***
## AspectN      0.039519   0.091201   0.433    0.666
## AspectNE     0.017931   0.100826   0.178    0.859
## AspectNW     0.006533   0.139312   0.047    0.963
## AspectS      0.039052   0.095881   0.407    0.685
## AspectSE     0.088399   0.110543   0.800    0.426
## AspectSW    -0.012893   0.097538  -0.132    0.895
## AspectW     -0.092139   0.093378  -0.987    0.326
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2275 on 114 degrees of freedom
## Multiple R-squared:  0.05494,    Adjusted R-squared:  -0.003086
## F-statistic: 0.9468 on 7 and 114 DF,  p-value: 0.4737
```

```r
ggplot(aes(x = Aspect, y = Dissimilarity), data = Comb_Dis)+
  geom_point()
```

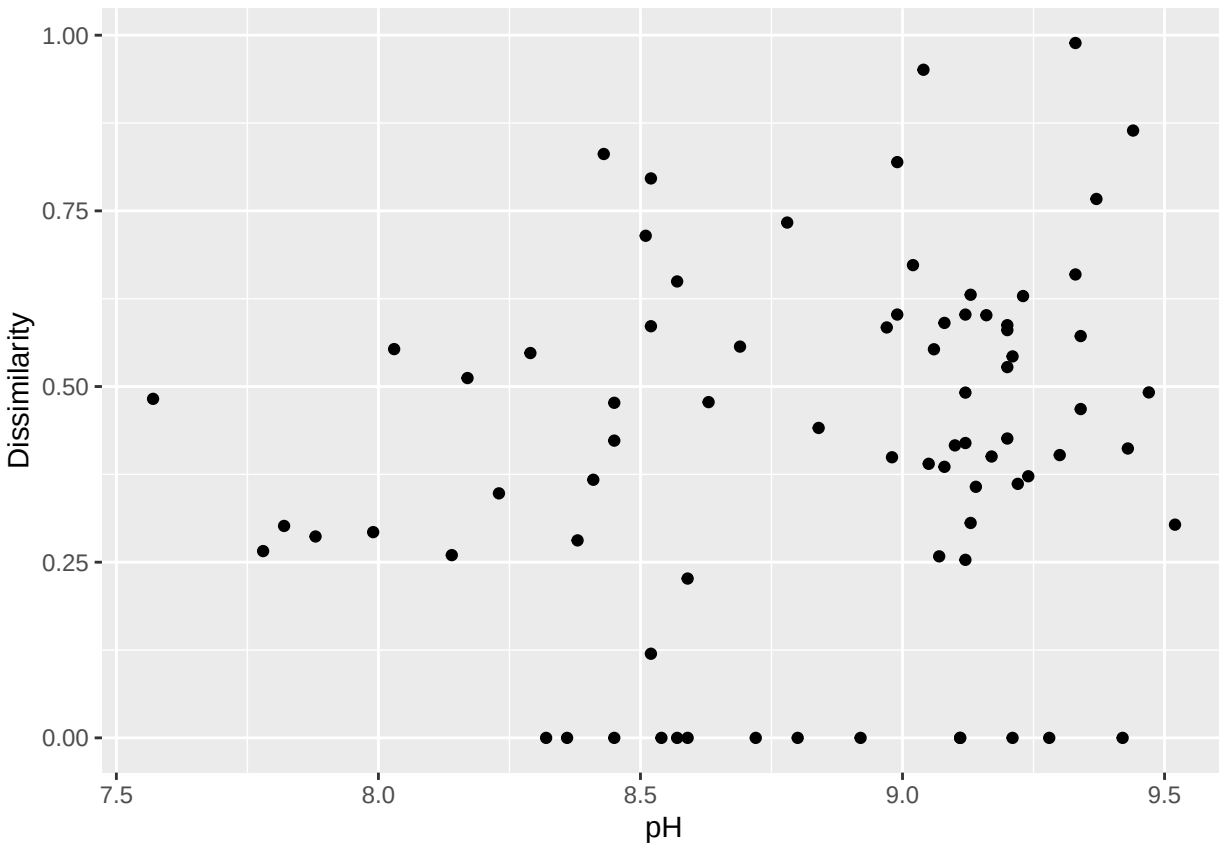


```r
summary(lm(Dissimilarity ~ pH, data = Comb_Dis))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ pH, data = Comb_Dis)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.47145 -0.12432 -0.00122  0.16683  0.52665
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.49260    0.55497  -0.888    0.378
## pH           0.10234    0.06267   1.633    0.107
##
## Residual standard error: 0.2531 on 76 degrees of freedom
##   (44 observations deleted due to missingness)
## Multiple R-squared:  0.0339, Adjusted R-squared:  0.02119
## F-statistic: 2.667 on 1 and 76 DF,  p-value: 0.1066
```

```r
ggplot(aes(x = pH, y = Dissimilarity), data = Comb_Dis)+
  geom_point()
```

```
## Warning: Removed 44 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

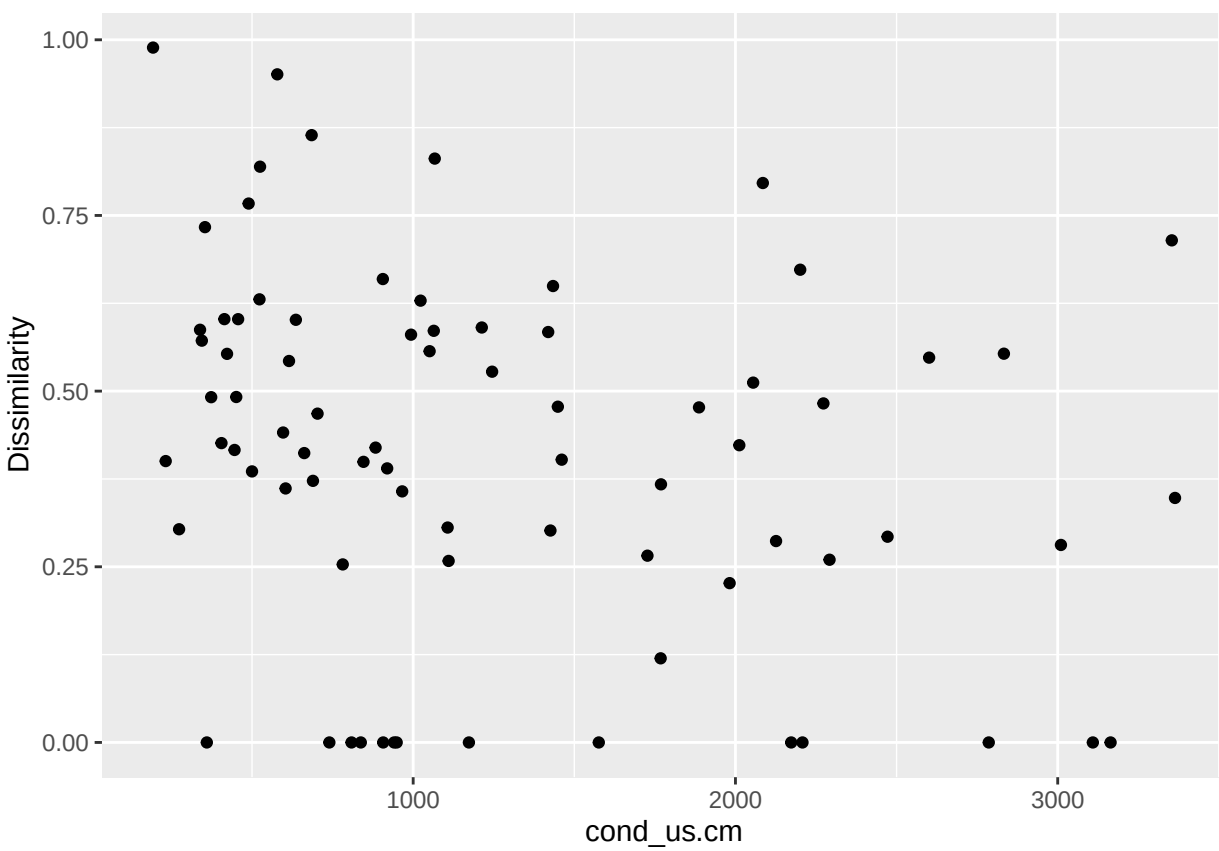



```r
summary(lm(Dissimilarity ~ cond_us.cm, data = Comb_Dis))
```

```
##
## Call:
## lm(formula = Dissimilarity ~ cond_us.cm, data = Comb_Dis)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.48378 -0.11661  0.00629  0.15580  0.49192
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.122e-01  5.063e-02   10.12 9.98e-16 ***
## cond_us.cm  -7.906e-05  3.336e-05   -2.37   0.0203 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2485 on 76 degrees of freedom
##   (44 observations deleted due to missingness)
## Multiple R-squared:  0.06882,    Adjusted R-squared:  0.05657
## F-statistic: 5.617 on 1 and 76 DF,  p-value: 0.02033
```

```r
ggplot(aes(x = cond_us.cm, y = Dissimilarity), data = Comb_Dis)+
  geom_point()
```

```
## Warning: Removed 44 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



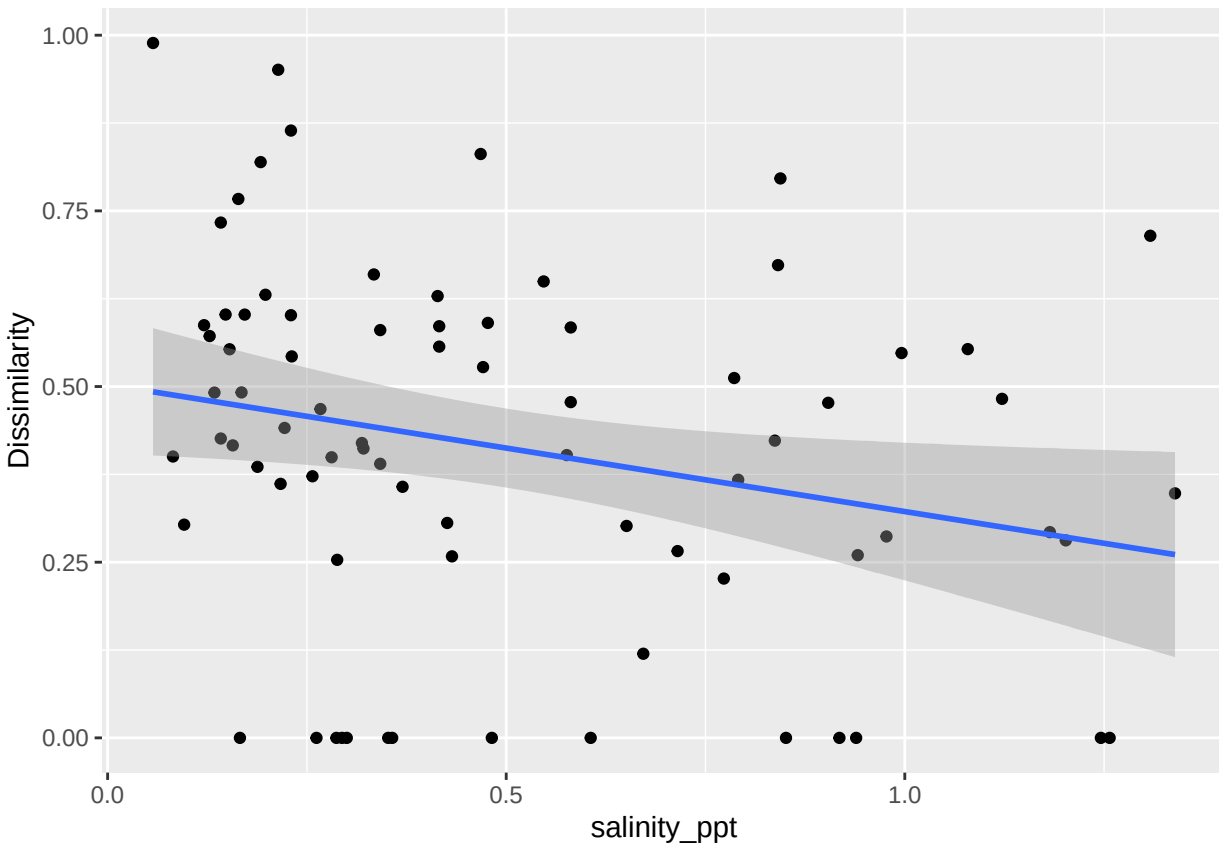
```r
summary(lm(Dissimilarity ~ salinity_ppt, data = Comb_Dis)) #***
```

```
##
## Call:
## lm(formula = Dissimilarity ~ salinity_ppt, data = Comb_Dis)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.47278 -0.11692  0.00562  0.15649  0.49641
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.50278    0.04918  10.224 6.26e-16 ***
## salinity_ppt -0.18068    0.08054  -2.243   0.0278 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2494 on 76 degrees of freedom
##   (44 observations deleted due to missingness)
## Multiple R-squared:  0.0621, Adjusted R-squared:  0.04976
## F-statistic: 5.032 on 1 and 76 DF,  p-value: 0.02779
```

```r
ggplot(aes(x = salinity_ppt, y = Dissimilarity), data = Comb_Dis)+
  geom_point()+
  geom_smooth(method = "lm")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 44 rows containing non-finite outside the scale range
## (`stat_smooth()`).
## Removed 44 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Negligible Effects

```
Mixed.mod.1 <- lmer(Dissimilarity ~ SoilTemp + SoilMoist + Slope + Aspect +  X +  Z  + (1|Site), data =
summary(Mixed.mod.1)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: Dissimilarity ~ SoilTemp + SoilMoist + Slope + Aspect + X + Z +
##     (1 | Site)
##    Data: subset(Comb_Dis, Dissimilarity > 0)
##
## REML criterion at convergence: -12.7
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.0790 -0.6293 -0.1038  0.5131  3.1597
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  Site     (Intercept) 0.005482 0.07404
##  Residual             0.027736 0.16654
## Number of obs: 107, groups:  Site, 3
##
## Fixed effects:
##               Estimate Std. Error t value
## (Intercept)  0.3951190  0.1046243   3.777
## SoilTemp     0.0573273  0.0180195   3.181
## SoilMoist    0.0086430  0.0189710   0.456
```

```
## Slope        0.0010563  0.0018899   0.559
## AspectN      0.0221338  0.0726586   0.305
## AspectNE    -0.0067168  0.0795584  -0.084
## AspectNW    -0.0549615  0.1078929  -0.509
## AspectS      0.0706362  0.0773286   0.913
## AspectSE     0.0840467  0.0882713   0.952
## AspectSW     0.0680103  0.0804657   0.845
## AspectW      0.0075406  0.0775801   0.097
## X            0.0011937  0.0008982   1.329
## Z           -0.0074262  0.0111288  -0.667
```

```
##
## Correlation matrix not shown by default, as p = 13 > 12.
## Use print(x, correlation=TRUE)  or
##     vcov(x)        if you need it
```

```
Mixed.Null.1 <- lmer(Dissimilarity ~ 1  + (1|Site), data = subset(Comb_Dis, Dissimilarity >0.0))
Mixed.ST<- lmer(Dissimilarity ~ SoilTemp  + (1|Site), data = subset(Comb_Dis, Dissimilarity >0.0))
Mixed.X<- lmer(Dissimilarity ~ X  + (1|Site), data = subset(Comb_Dis, Dissimilarity >0.0))

anova(Mixed.mod.1, Mixed.Null.1)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: subset(Comb_Dis, Dissimilarity > 0)
## Models:
## Mixed.Null.1: Dissimilarity ~ 1 + (1 | Site)
## Mixed.mod.1: Dissimilarity ~ SoilTemp + SoilMoist + Slope + Aspect + X + Z + (1 | Site)
##              npar     AIC     BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## Mixed.Null.1    3 -62.006 -53.987 34.003   -68.006
## Mixed.mod.1    15 -57.711 -17.619 43.855   -87.711 19.705 12    0.07288 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(Mixed.ST, Mixed.Null.1)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: subset(Comb_Dis, Dissimilarity > 0)
## Models:
## Mixed.Null.1: Dissimilarity ~ 1 + (1 | Site)
## Mixed.ST: Dissimilarity ~ SoilTemp + (1 | Site)
##              npar     AIC     BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## Mixed.Null.1    3 -62.006 -53.987 34.003   -68.006
## Mixed.ST        4 -72.459 -61.767 40.229   -80.459 12.453  1  0.0004174 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(Mixed.X, Mixed.Null.1)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: subset(Comb_Dis, Dissimilarity > 0)
## Models:
## Mixed.Null.1: Dissimilarity ~ 1 + (1 | Site)
## Mixed.X: Dissimilarity ~ X + (1 | Site)
##              npar     AIC     BIC logLik -2*log(L) Chisq Df Pr(>Chisq)
## Mixed.Null.1    3 -62.006 -53.987 34.003   -68.006
```

```
## Mixed.X         4 -61.416 -50.725 34.708   -69.416  1.41  1     0.2351
```

```r
anova(Mixed.ST, Mixed.X)
```

```
## refitting model(s) with ML (instead of REML)
```

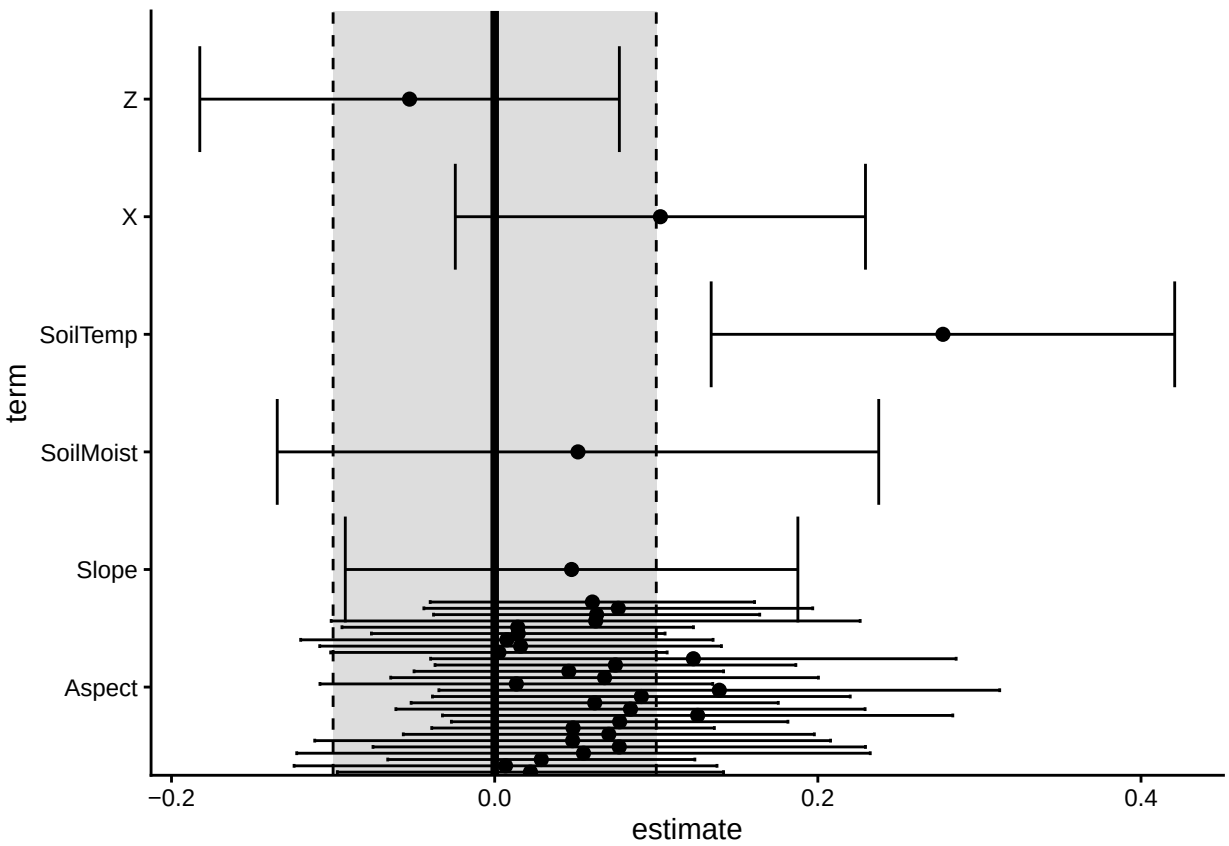
```
## Data: subset(Comb_Dis, Dissimilarity > 0)
## Models:
## Mixed.ST: Dissimilarity ~ SoilTemp + (1 | Site)
## Mixed.X: Dissimilarity ~ X + (1 | Site)
##          npar     AIC     BIC logLik -2*log(L) Chisq Df Pr(>Chisq)
## Mixed.ST    4 -72.459 -61.767 40.229   -80.459
## Mixed.X     4 -61.416 -50.725 34.708   -69.416     0  0
```

```r
Mixed.cmp<- avg_comparisons(Mixed.mod.1, variables = list(SoilTemp = "minmax", SoilMoist = "minmax", Sl
```

```
## Warning: For this model type, `marginaleffects` only takes into account the
## uncertainty in fixed-effect parameters. This is often appropriate when
## `re.form=NA`, but may be surprising to users who set `re.form=NULL` (default)
## or to some other value. Call `options(marginaleffects_safe = FALSE)` to silence
## this warning.
```

```r
Mixed.cmp <- as.data.frame(Mixed.cmp)


for(j in 1:dim(Mixed.cmp)[1]){
  if(Mixed.cmp$term[j] == "Aspect"){
    if(Mixed.cmp$estimate[j] <= 0){
   Mixed.cmp$estimate[j] <- (Mixed.cmp$estimate[j] * -1)
   NewLow <- (Mixed.cmp$conf.high[j] * -1)
   Mixed.cmp$conf.high[j] <- (Mixed.cmp$conf.low[j] * -1)
   Mixed.cmp$conf.low[j] <- NewLow
  }
}
}



ggplot(Mixed.cmp, aes(x = estimate,
                  xmin = conf.low,
                  xmax = conf.high,
                  y = term,
                  group = contrast)) +
     annotate("rect", xmin = -0.1, xmax = 0.1, ymin = -Inf, ymax = Inf, alpha = .2)+
  geom_vline(xintercept = c(-0.1, .1), linetype = "dashed") +
  geom_vline(xintercept = 0, linetype = "solid", linewidth = 1.5)+
  geom_errorbarh(aes(xmin = conf.low,  xmax = conf.high,),position = position_dodgev(height = 1.5)) +
  geom_point(size = 2, position = position_dodgev(height = 1.5)) +
  theme_classic()
```

```
## Warning: position_dodgev requires non-overlapping y intervals
```

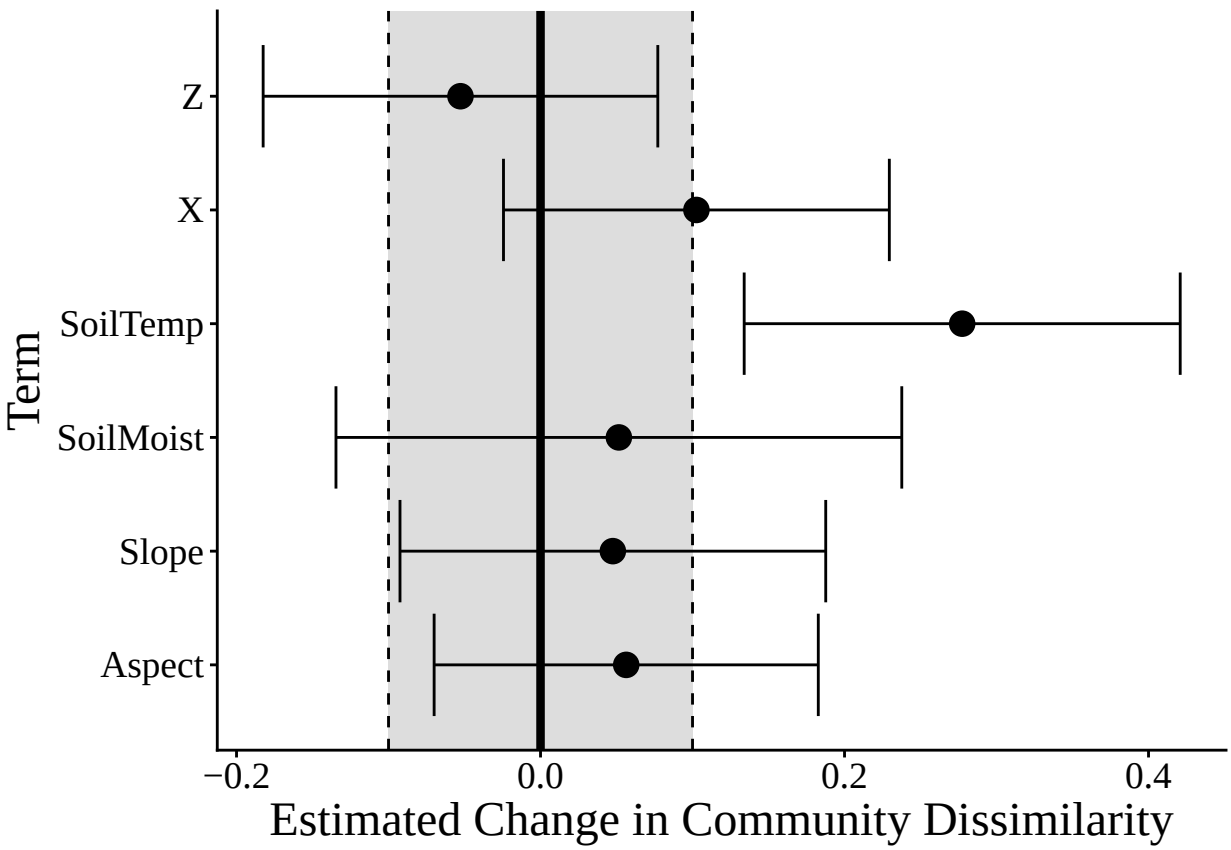



```
Mixed.cmp<-Mixed.cmp %>%
  group_by(term) %>%
  summarise_all(.funs = mean)
```

```
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.
```

```
ggplot(Mixed.cmp, aes(x = estimate,
                  xmin = conf.low,
                  xmax = conf.high,
                  y = term,
                  group = contrast)) +
     annotate("rect", xmin = -0.1, xmax = 0.1, ymin = -Inf, ymax = Inf, alpha = .2)+
  geom_vline(xintercept = c(-0.1, .1), linetype = "dashed") +
  geom_vline(xintercept = 0, linetype = "solid", linewidth = 1.5)+
  geom_errorbarh(aes(xmin = conf.low,  xmax = conf.high,),position = position_dodgev(height = 1.5)) +
  geom_point(size = 4, position = position_dodgev(height = 1.5)) +
  theme_classic()+
  labs(x = "Estimated Change in Community Dissimilarity", y = "Term")+
  theme(text = element_text(family = "serif", size = 18))
```

```
## Warning: position_dodgev requires non-overlapping y intervals
```

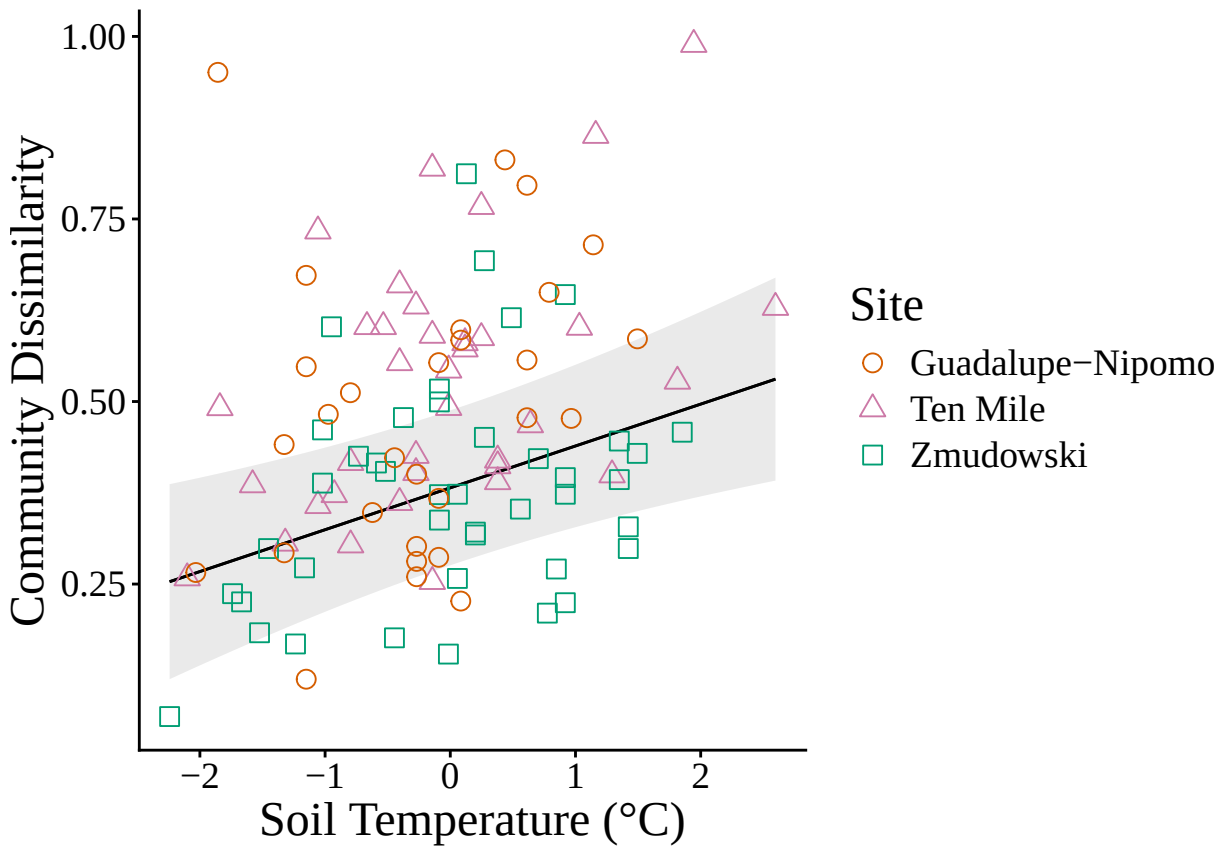



```
ggsave("Within-Marginal.png", dpi = 600, height = 7, width = 10)
```

```
## Warning: position_dodgev requires non-overlapping y intervals
```

```
plot_predictions(Mixed.mod.1, condition = "SoilTemp")+
  geom_point(data = subset(Comb_Dis, Dissimilarity >0), size = 3,aes(x = SoilTemp, y = Dissimilarity, c
  labs(x = "Soil Temperature (°C)", y =  "Community Dissimilarity")+
  scale_color_manual(labels = c("Guadalupe-Nipomo", "Ten Mile", "Zmudowski"), values = c("#d55e00", "#c
  scale_shape_manual(labels = c("Guadalupe-Nipomo", "Ten Mile", "Zmudowski"), values = c(1,2,0), name =
  theme_classic()+
  theme(text = element_text(family = "serif", size = 18))
```

```
## Warning: For this model type, `marginaleffects` only takes into account the
## uncertainty in fixed-effect parameters. This is often appropriate when
## `re.form=NA`, but may be surprising to users who set `re.form=NULL` (default)
## or to some other value. Call `options(marginaleffects_safe = FALSE)` to silence
## this warning.
## Warning: For this model type, `marginaleffects` only takes into account the
## uncertainty in fixed-effect parameters. This is often appropriate when
## `re.form=NA`, but may be surprising to users who set `re.form=NULL` (default)
## or to some other value. Call `options(marginaleffects_safe = FALSE)` to silence
## this warning.
```




```
  #geom_smooth(data = subset(Comb_Dis, Dissimilarity >0), aes(x = SoilTemp, y = Dissimilarity), method =

ggsave("Within-SoilTemp.png", dpi = 600, height = 7, width = 10)
```

```
Tracker<- data.frame()
SingleTrack<-0

for (i in 1:100) {

  Rand.1<- transform(Comb_Dis, Dissimilarity = sample(Dissimilarity) )
  Rand.mod.1 <- lmer(Dissimilarity ~ SoilTemp + SoilMoist + Slope + Aspect + X +  Z  + (1|Site), data =


  while(isSingular(Rand.mod.1) == TRUE){
    SingleTrack <- SingleTrack + 1
   Rand.1<- transform(Comb_Dis, Dissimilarity = sample(Dissimilarity) )
  Rand.mod.1 <- lmer(Dissimilarity ~ SoilTemp + SoilMoist + Slope + Aspect + X +  Z  + (1|Site), data =
    }

    Rand.cmp<- avg_comparisons(Rand.mod.1, variables = list(SoilTemp = "minmax", SoilMoist = "minmax",

  for(j in 1:dim(Rand.cmp)[1]){
  if(Rand.cmp$term[j] == "Aspect"){
    if(Rand.cmp$estimate[j] <= 0){
   Rand.cmp$estimate[j] <- (Rand.cmp$estimate[j] * -1)
```

```r
    NewLow <- (Rand.cmp$conf.high[j] * -1)
    Rand.cmp$conf.high[j] <- (Rand.cmp$conf.low[j] * -1)
    Rand.cmp$conf.low[j] <- NewLow
  }
}
}

  Rand.cmp<-Rand.cmp %>%
    group_by(term) %>%
    summarise_all(.funs = mean)

   Run <- rep(i, dim(Rand.cmp)[1])
    Rand.cmp<- cbind(Rand.cmp, Run)



    Tracker <- rbind(Tracker, Rand.cmp)
    print(i)


}
```

Randomization Test

```
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
```

```
## Warning: For this model type, `marginaleffects` only takes into account the
## uncertainty in fixed-effect parameters. This is often appropriate when
## `re.form=NA`, but may be surprising to users who set `re.form=NULL` (default)
## or to some other value. Call `options(marginaleffects_safe = FALSE)` to silence
## this warning.
```

```
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.
```

```
## [1] 1
```

```
## boundary (singular) fit: see help('isSingular')
```

```
## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.
```

```
## [1] 2

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 3

## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 4

## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 5

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 6

## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
```

```
## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 7

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 8

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 9

## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 10

## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
```

```
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 11

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 12

## boundary (singular) fit: see help('isSingular')
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## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
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##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 13

## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 14

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
```

```
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 15

## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 16

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 17

## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 18

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.
```

```
## [1] 19

## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 20

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 21

## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
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## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 22

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 23

## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
```

```
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
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##   ...`.
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## Caused by warning in `mean.default()`:
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## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 24

## boundary (singular) fit: see help('isSingular')
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## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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## The first warning was:
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##   ...`.
## i In group 1: `term = "Aspect"`.
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## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 25

## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
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##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 26

## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
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## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
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```

```
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 27

## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 28

## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
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##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 29

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
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## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 30

## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
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##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 31
```

```
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
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## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 32

## boundary (singular) fit: see help('isSingular')
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##   ...`.
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## Caused by warning in `mean.default()`:
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## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 33

## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 34

## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
```

```
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 35

## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
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##   ...`.
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## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 36

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 37

## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 38

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 39

## boundary (singular) fit: see help('isSingular')
```

```
## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.
```

```
## [1] 40
```

```
## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.
```

```
## [1] 41
```

```
## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.
```

```
## [1] 42
```

```
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
```

```
## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.
```

```
## [1] 43
```

```
## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
```

```
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 44

## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 45

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 46

## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
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## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 47

## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
```

```
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 48

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 49

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 50

## boundary (singular) fit: see help('isSingular')
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## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 51

## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
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## boundary (singular) fit: see help('isSingular')
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## boundary (singular) fit: see help('isSingular')
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## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 52

## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
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## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 53

## boundary (singular) fit: see help('isSingular')
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## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 54

## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
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## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
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## [1] 55

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
```

```
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
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## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 56

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 57

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 58

## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 59

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.
```

```
## [1] 60

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 61

## boundary (singular) fit: see help('isSingular')
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## [1] 62

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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## The first warning was:
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##   ...`.
## i In group 1: `term = "Aspect"`.
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## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 63

## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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## The first warning was:
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## [1] 64

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
```

```
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
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## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 65

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
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## [1] 66

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
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## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
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## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 67

## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
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## [1] 68

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
##   ...`.
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## Caused by warning in `mean.default()`:
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## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.
```

```
## [1] 69

## boundary (singular) fit: see help('isSingular')
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## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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##   ...`.
## i In group 1: `term = "Aspect"`.
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## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 70

## boundary (singular) fit: see help('isSingular')
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## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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## [1] 71

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
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## [1] 72

## boundary (singular) fit: see help('isSingular')
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```

```
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 73

## boundary (singular) fit: see help('isSingular')
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## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 74

## boundary (singular) fit: see help('isSingular')

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## [1] 75

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
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## [1] 76

## boundary (singular) fit: see help('isSingular')
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## [1] 77

## boundary (singular) fit: see help('isSingular')
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## [1] 78

## boundary (singular) fit: see help('isSingular')

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## [1] 79

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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## [1] 80

## boundary (singular) fit: see help('isSingular')
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```
## [1] 81

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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## The first warning was:
## i In argument: `contrast = (new("standardGeneric", .Data = function (x, ...)
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## [1] 82

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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## i In group 1: `term = "Aspect"`.
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## [1] 83

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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## i In group 1: `term = "Aspect"`.
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## ! argument is not numeric or logical: returning NA
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## [1] 84

## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
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##   ...`.
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
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## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.

## [1] 85

## boundary (singular) fit: see help('isSingular')
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## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
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##   ...`.
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## Caused by warning in `mean.default()`:
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## [1] 86

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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## The first warning was:
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## [1] 87

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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## [1] 88

## boundary (singular) fit: see help('isSingular')
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## [1] 89

## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
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```

```
## [1] 90

## boundary (singular) fit: see help('isSingular')

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
## Warning: There were 6 warnings in `summarise()`.
## The first warning was:
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## [1] 91

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## [1] 92

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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##   ...`.
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## [1] 93

## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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## [1] 94

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## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.
```

```
## [1] 95
```

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## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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```

```
## [1] 96
```

```
## boundary (singular) fit: see help('isSingular')
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```

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## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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```
## [1] 97
```

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## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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```

```
## [1] 98
```

```
## boundary (singular) fit: see help('isSingular')
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## Warning: For this model type, `marginaleffects` only takes into account the uncertainty in fixed-effe
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## [1] 99
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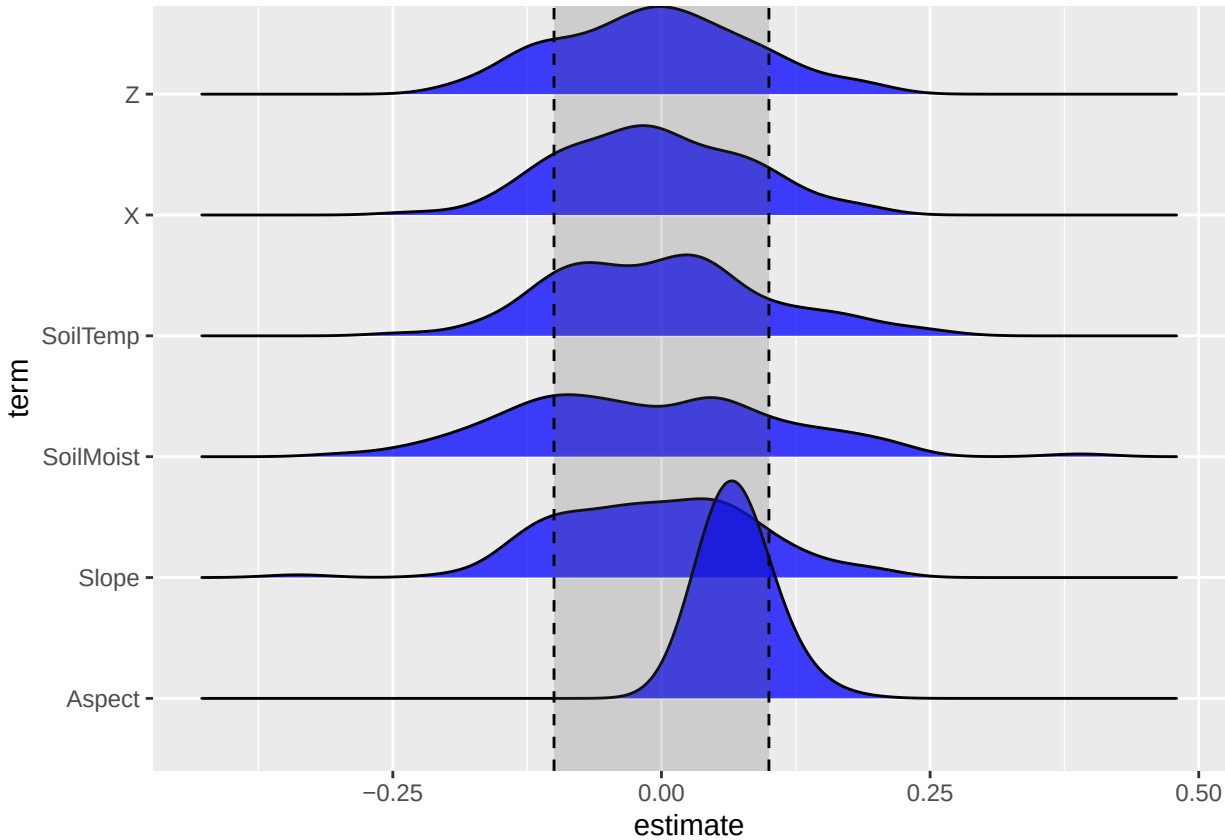
## i In group 1: `term = "Aspect"`.
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA
## i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.
```

```
## [1] 100
```

```
    # ggplot(Tracker, aes(x = estimate,
    #                xmin = conf.low,
    #                xmax = conf.high,
    #                y = term,
    #                group = contrast)) +
    #   annotate("rect", xmin = -0.1, xmax = 0.1, ymin = -Inf, ymax = Inf, alpha = .2)+
    #   geom_vline(xintercept = c(-0.1, .1), linetype = "dashed") +
    #   geom_vline(xintercept = 0, linetype = "solid", linewidth = 1.5)+
    #   geom_errorbarh(position = position_dodgev(height = 1.5)) +
    #   geom_point(size = 2, position = position_dodgev(height = 1.5)) +
    #   theme_classic()

ggplot(Tracker, aes(x = estimate,
                    y = term,
                    group = term))+
  geom_density_ridges(fill = "blue", alpha = 0.75)+
      annotate("rect", xmin = -0.1, xmax = 0.1, ymin = -Inf, ymax = Inf, alpha = .2)+
      geom_vline(xintercept = c(-0.1, .1), linetype = "dashed")
```

```
## Picking joint bandwidth of 0.0299
```




```r
length(unique(Tracker$Run))
```

```
## [1] 100
```

```r
Tracker %>% group_by(term) %>% summarize_all(mean)
```

```
## # A tibble: 6 x 10
##   term  contrast estimate std.error statistic p.value s.value conf.low conf.high
##   <chr>    <dbl>    <dbl>     <dbl>     <dbl>   <dbl>   <dbl>    <dbl>     <dbl>
## 1 Aspe~       NA  7.05e-2    0.0856   0.0172    0.494    1.52  -0.0702     0.211
## 2 Slope       NA -6.58e-3    0.0871  -0.0862    0.468    1.60  -0.150      0.137
## 3 Soil~       NA -1.28e-2    0.119   -0.107     0.462    1.56  -0.208      0.183
## 4 Soil~       NA  5.33e-4    0.104    0.00741   0.516    1.33  -0.170      0.171
## 5 X           NA -7.41e-3    0.0822  -0.0916    0.478    1.53  -0.143      0.128
## 6 Z           NA -6.30e-3    0.0846  -0.0710    0.478    1.55  -0.145      0.133
## # i 1 more variable: Run <dbl>
```

```r
save.image(file="DuneComm2.RData")
```